

General Decision Models: Benchmarking and Insights Beyond Jev

Feiyu Duan^{1,2,*}, Jiayu Lin^{1,2,*}, Jia Wang^{1,3,*}, Jun Xiang^{2,*}, Jialiang Wu², Xinnong Zhang^{1,2}, Hanqi Yan⁴, Siyuan Wang⁵, Zhongyu Wei^{1,2,†}

¹Shanghai Innovation Institute, ²Fudan University, ³Tongji University, ⁴King’s College London, ⁵Chinese University of Hong Kong

* Equal Contribution, names are listed in alphabetical order., † Corresponding author.

General decision models, such as Jev, have recently emerged as efficient alternatives to LLMs for structured judgment and selection. But what kinds of decisions can these models reliably make, and how does their behavior change when individual decisions are composed into larger systems? To study this, we introduce **JEVal**, a bilingual benchmark comprising 11,257 instances from 36 datasets across 10 application domains, and evaluate 25 model configurations spanning general decision models and generative LLMs. Our results show that (1) general decision models are most competitive when decisions can be resolved from available evidence, but weaken when they require specialist knowledge or faithful uncertainty estimation: they can often identify the most likely outcome while substantially overstating its probability. (2) In more dynamic and realistic systems involving long-horizon, multi-step interactions, the advantages of fast local decision making are offset by reliability failures at the system level. on τ -bench, faster local decisions reduce median episode time but lower task success as decision errors accumulate over long trajectories. (3) In large-scale social simulation, decision models approach strong generative LLMs on individual response prediction at substantially lower inference cost, yet remain weaker in user profiling and exhibit larger aggregate estimation errors and systematic bias. Finally, we propose **InnerJev-4B** and **InnerJev-27B**, which internalize an open-weight LLM’s own reasoning into a single-pass first-token decision through **Reasoning-to-Readout Self-Distillation**, with InnerJev-27B performing on par with Jev on JEVal while answering a typical query in about 0.1 s.

Emails: fyduan25@m.fudan.edu.cn; zywei@fudan.edu.cn

JEVal Data: <https://huggingface.co/datasets/carlosxiang/JEVal>

InnerJev Code: <https://github.com/amazingljy1206/InnerJev>

InnerJev Models: InnerJev-27B full-rank, InnerJev-27B LoRA, InnerJev-4B full-rank, InnerJev-4B LoRA

InnerJev Training Data: InnerJev-27B, InnerJev-4B

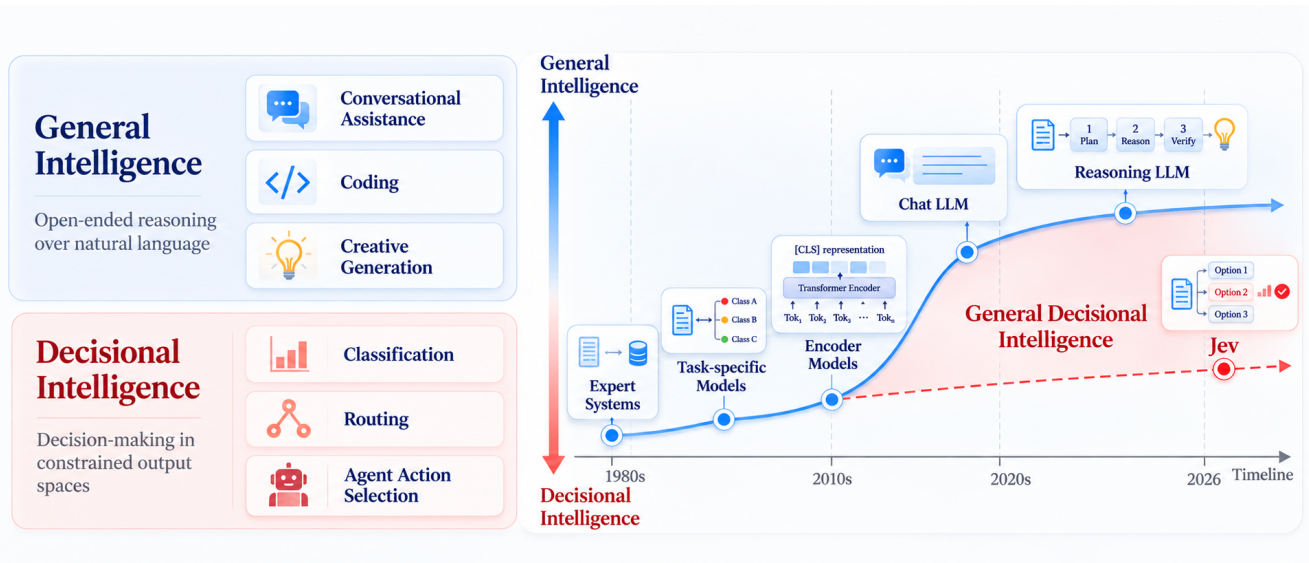


Figure 1 A conceptual view of general and decisional intelligence. The left panel contrasts open-ended language capabilities with decision-making in constrained output spaces. The right panel illustrates the evolution from expert systems and task-specific models to encoder models, chat and reasoning LLMs, and general decision models represented by Jev.

1 Introduction

Large language models (LLMs) have increasingly exhibited **general intelligence: the ability to adapt across diverse tasks and domains without task-specific retraining** (Achiam et al., 2023; Grattafiori et al., 2024). Their capabilities have expanded from language generation to reasoning, code generation, and agentic workflows (Guo et al., 2025; Hui et al., 2024; Wang et al., 2025b). However, modern AI systems also involve another broad class of computational demands: making structured judgments or selections directly from a given context, such as preference modeling, importance attribution, and action selection (Lambert et al., 2025; Zhou et al., 2024; Wang et al., 2024b). We refer to this capability as **decisional intelligence: the ability to make effective structured decisions over a constrained outcome space**.

Existing models occupy different regions along these two capability dimensions, and Figure 1 provides a conceptual summary of this evolution. Early expert systems and subsequent task-specific models were primarily designed for predefined decision problems (Shortliffe, 1977; LeCun et al., 1998). With the development of pre-trained representation learning, encoder models became capable of handling more complex and diverse natural-language judgment tasks (Devlin et al., 2019). In parallel, ChatLLMs and reasoning LLMs substantially expanded task coverage along the direction of general intelligence (Team et al., 2024, 2026). Recently, models represented by Jev (TypeSafe AI, 2026) have begun to explore a new region: rather than targeting open-ended text generation, they directly map natural-language contexts to structured decisions while attempting to retain applicability across tasks. We define **general decisional intelligence as the ability to perform structured decision making across diverse tasks and domains without task-specific retraining**. Models that exhibit this capability are referred to as **general decision models**.

However, although decision models operate over finite output spaces, the capabilities required by different tasks can vary substantially. Knowledge-intensive decisions require access to factual knowledge (Fang et al., 2025). Symbolic decision tasks involving logic and mathematics further require compositional reasoning (Yu et al., 2025), while long-context decisions require retrieving information distributed across extended inputs (Yang et al., 2025). In agentic settings, action selection may additionally depend on jointly reasoning over the current state, interaction history and future planning (Barres et al., 2025; Abhyankar et al., 2026). **We therefore aim to characterize the current frontier of general decisional intelligence¹, from the reliability of local decisions to their behavior when composed in long-horizon workflows and aggregated in large-scale systems.** Our analysis reveals several key insights:

Finding 1: General decision models perform best with explicit evidence in local decision tasks, but weaken when decisions require latent knowledge or calibrated uncertainty. We introduce **JEVal**, a bilingual benchmark comprising 11,257 instances from 36 datasets across 10 application domains (Section 3), and evaluate 20+ models spanning general decision models and LLMs. On JEVal, the strongest decision models approach leading thinking LLMs in overall accuracy and are particularly competitive when the evidence is contained in the input, but fall behind on specialist-knowledge domains such as medicine, finance, and law (Section 4.2). Controlled probability experiments reveal a parallel limitation: decision models can often identify the most likely outcome correctly while substantially overstating its probability (Section 5.1).

Finding 2: Strong local decision performance does not directly translate into reliable long-horizon agent behavior. On τ -bench (Yao et al., 2024), using Jev for next-action selection reduces median episode time, but also lowers task success and increases tail latency. We find that this gap mainly arises from accumulation of local decision errors over multi-turn trajectories (Sections 5.3).

Finding 3: General decision models make large-scale social simulation more scalable, but still face limitations in individual profiling, aggregate estimation, and systematic bias. Across user profiling, individual survey prediction, and population-level simulation, we find that current general decision models approach strong generative LLMs on individual response prediction at substantially lower inference cost and can drive thousands of simulated agents while preserving most state-level outcomes. However, they remain less effective in user-interest profiling, exhibit substantially larger errors in aggregate vote-share estimation, and retain systematic directional biases also observed in LLMs (Sections 5.4).

Jev is accessible only through a hosted API. We therefore ask: **can a general decision model like Jev be built from an open-weight LLM?** Human cognition is often described as two systems: System One decides by intuition, System Two deliberates (Kahneman, 2011), and intuition grows out of practiced deliberation (Kahneman and Klein, 2009). In these terms, Jev is a System One model, whereas an LLM has both: its next-token distribution is already an immediate decision, and it can also think before answering, repairing wrong immediate decisions. Therefore, **the model can build its intuition as humans do, teaching its System One with its own System Two.** Concretely, we restrict the

¹We examine this frontier through Jev as a representative general decision model. Unless otherwise specified, the term “general decision models” refers to Jev throughout the remainder of this paper.

logits of the first answer token to the labels of the valid options, which gives a decision head that covers all three JEval decision formats in a single forward pass. We then propose *Reasoning-to-Readout Self-Distillation*, which distills the answer distribution that the model reaches at the end of its reasoning chain into this head, without labeled answers or a stronger teacher. Applied to Qwen3.5-4B and Qwen3.8-27B, it yields two open-source decision models, **InnerJev-4B** and **InnerJev-27B**. InnerJev-27B performs comparably to Jev on JEval, and InnerJev-4B surpasses decision models of the same size, while answering a typical query in about 60 ms and 110 ms, respectively, on a single H100 GPU.

Our contributions are summarized as follows:

- **A comprehensive decision-model benchmark spanning 10 domains and over 10K bilingual instances.** We introduce JEval, which formulates diverse tasks as decision problems. JEval contains 11,257 bilingual instances from 36 datasets across 10 application domains.
- **A systematic analysis of the capability boundaries and failure modes of general decision models.** Across JEval and targeted diagnostic tasks, we find that general decision models are strongest when decisions can be resolved from available evidence, but weaken when they require specialist knowledge or faithful uncertainty estimation.
- **Evaluation in dynamic and realistic decision systems.** We move beyond static benchmarks to evaluate general decision models in interactive agent workflows and large-scale social simulations, revealing system-level behaviors that cannot be observed from isolated decision accuracy alone.
- **Open general decision models.** We open-source InnerJev, a general decision model at both 4B and 27B scales with competitive performance.

2 Related Work

From task-specific predictors to general decision models. Decision-making is a fundamental capability of intelligent systems. Early discriminative models rely on task-specific predictors, such as classification heads for pretrained encoders (Devlin et al., 2019) or candidate scoring in rerankers (Zhang et al., 2025b). Later work relaxes fixed label spaces through natural-language task and label descriptions, including entailment-based zero-shot classification (Yin et al., 2019), the Universal Discriminator (Xu et al., 2023), and GLiClass (Stepanov et al., 2025). Jev further extends this direction toward general decision models with request-time choices, binary judgments, and scores (TypeSafe AI, 2026). However, how broadly such models generalize beyond conventional classification remains unclear, motivating the systematic evaluation in JEval.

Evaluation of decision models. A broad range of existing benchmarks can be formulated as decision tasks over finite output spaces, covering capability demands such as knowledge reasoning, long-context understanding, and agentic action selection. MMLU-Pro evaluates knowledge and reasoning (Wang et al., 2024c), LongBench focuses on long-context understanding (Bai et al., 2024), and τ -bench evaluates interactive tool use (Yao et al., 2024), while HELM highlights joint evaluation of accuracy, calibration, and efficiency (Liang et al., 2022). Recent studies have also evaluated Jev in text annotation (Ibrahim and Zaki, 2026), response judging (Li et al., 2026; Rao and Callison-Burch, 2026), alignment-failure detection (Guo et al., 2026), legal-document evaluation (Zhang et al., 2026), and agent delegation (Wu and Lim, 2026), alongside studies of output validity (Sun and Xu, 2026), context sensitivity (Xu, 2026), and confidence calibration (Guo et al., 2017). However, these evaluations typically focus on individual tasks, applications, or reliability aspects, leaving the broader capabilities of general decision models underexplored. JEval therefore unifies diverse decision tasks and evaluation dimensions to systematically characterize their capability frontier.

3 JEval Benchmark

In this section, we present **JEval**, a benchmark for evaluating general decisional intelligence. Figure 2 show the overview of JEval. We first introduce the task formulation (Section 3.1), then describe how the benchmark is collected, curated, and normalized (Section 3.2), and finally we report the statistics of our benchmark (Section 3.3).

3.1 Task formulation

JEval formulates each evaluation sample as a general decision instance

$$\mathcal{D} = (x, y^*), \quad x = (s, i, c) \quad (1)$$

where s , i , and c denote the *state*, *instruction*, and *criteria*, respectively, and y^* is the gold label. The state provides the information required for the decision, the instruction specifies the decision objective, and the criteria defines

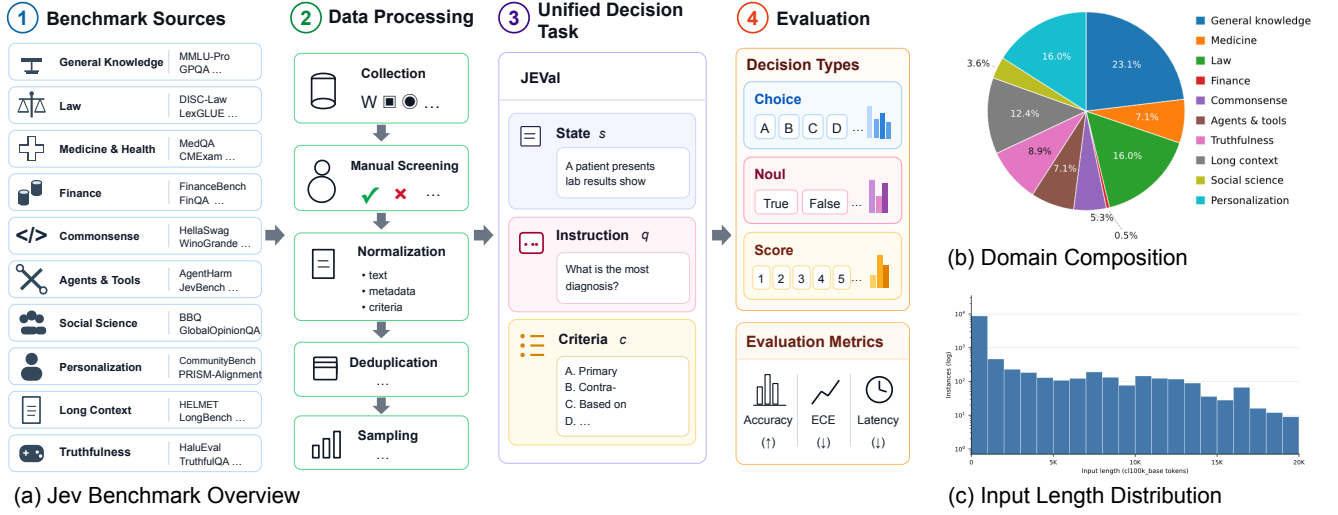


Figure 2 Overview of JEval. (a) Data processing pipeline. (b) Share of the 11,257 JEval instances contributed by each domain. (c) Input length over JEval measured over the state, instruction, and candidates in c1100k_base tokens. The frequency axis is logarithmic.

the valid decision space. Let $\mathcal{Y}(x)$ denote the finite decision space corresponding to input x , whose valid range is determined by c , with $y^* \in \mathcal{Y}(x)$. Given x , the model is required to produce a probability distribution $p_\theta(y | x)$ over $\mathcal{Y}(x)$, subject to

$$p_\theta(y | x) \geq 0, \quad \sum_{y \in \mathcal{Y}(x)} p_\theta(y | x) = 1, \quad (2)$$

and the final decision is obtained as

$$\hat{y} = \arg \max_{y \in \mathcal{Y}(x)} p_\theta(y | x) \quad (3)$$

According to the structure of the decision space, JEval categorizes tasks into three primary formats:

- **choice**: the model selects one outcome from an explicitly specified finite set of candidates;
- **noul**: the model performs a binary judgment over the decision space $\{\text{true}, \text{false}\}$;
- **score**: the model predicts one value from an ordered set of discrete rating levels.

3.2 Data Processing

Data Collection. We first define several common domains, including general knowledge reasoning, commonsense inference, agents, mathematics, medicine, among others. We collect datasets from each domain that are compatible with the decision tasks defined in Section 3.1. Specifically, we prioritize datasets with explicit decision objectives and well-defined outcome spaces. Tasks that cannot be reliably converted into a finite decision space, or whose semantics would be altered by such conversion, are excluded.

Data Curation and Normalization. For the selected datasets, two human experts further inspect the task definitions and sample content, remove instances that are unsuitable for decision-task formulation, and retain only datasets whose reference answers have been manually verified. We then normalize the samples into the unified JEval format: background text, dialogue history, or environment states required for decision making are organized as the **state**; the original questions or task descriptions are used as the **instructions**; and the meanings of candidate options in the **criteria** are supplemented based on the original dataset documentation. For tasks that do not explicitly provide candidate options, we construct the candidate space and randomly shuffle the option order to mitigate positional bias. We control input length against a 20K-token budget (20,480 tokens) and cap the number of candidate options at 255; the retained upper tail above this length budget is quantified in Appendix B.2.

Deduplication and Sampling. After normalization, we remove duplicate questions using exact string matching, resulting in more than 202K candidate samples. As the source datasets vary substantially in size, we further resample the data to prevent a small number of large datasets from dominating the overall evaluation and to

improve balance across domains and subtasks. For datasets with an official test split, we sample at most 200 instances from the test set for each subtask. For datasets without an official test split, we apply the same sampling rule to the available validation or train split.

Further dataset details are provided in Appendix B.1.

3.3 Benchmark Statistics

The resulting benchmark covers 10 application domains, 36 datasets, and 58 subtasks, comprising 11,257 instances in total. As shown in Figure 2:, general knowledge and reasoning is the largest domain (23.1%), followed by law and personalization and preference (16.0% each). JEVal contains 9,457 English and 1,800 Chinese instances. Most examples use the **choice** format (85.1%), followed by **noul** (12.0%) and **score** (2.8%). For **choice** tasks, the number of candidates ranges from 2 to 255, with a median of five. Figure 2 shows the input-length distribution up to 20K tokens, measured over the joint length of state, instruction, and candidates. Additional format, candidate-count, and domain-level length statistics are provided in Appendix B.2.

4 Experiments

4.1 Models and Evaluation Protocol

Models. We evaluate 25 configurations in three groups. Appendix C.3 lists the settings of each.

- **Decision models.** Jev (TypeSafe AI, 2026); the open-weight Kev² at 0.8B, 4B, 9B, and 27B; Intern-Decision³ at 0.8B, 2B, and 4B; Laya⁴, OpenJev-4B-v5⁵, Nimble-9B⁶, NanoJev⁷, LightJev⁸, and JevForge-0.8B⁹; and our InnerJev-4B and InnerJev-27B (Section 6).
- **Non-thinking generative LLMs.** Qwen3.5-4B (Qwen Team, 2026a), Qwen3.8-27B (Qwen Team, 2026b), DeepSeek-V4-Flash (DeepSeek-AI, 2026), GPT-6-Luna, and GPT-6-Sol (OpenAI, 2026). For the two GPT-6 models, we request no reasoning through the API.
- **Thinking generative LLMs.** Qwen3.5-4B at its default effort, Qwen3.8-27B at **xhigh** effort, and DeepSeek-V4-Flash and GLM-5.3-Flash (Z.ai, 2026) at **high** effort.

Implementation details. We deploy open-weight models locally on NVIDIA H100 GPUs using vLLM and access Jev and the GPT-6 models through hosted APIs. Generative LLMs use a sampling temperature of 1.0, with reasoning and final-answer tokens counted toward a shared output budget. Open-weight decision baselines retain their released numerical precision and calibration parameters. Appendix C.3 provides detailed hardware configurations, numerical precision settings, and token budgets.

Evaluation metrics. We evaluate predictive performance and response efficiency using the following metrics. Let $\mathcal{C}$ denote all 10,938 classification instances and $\mathcal{V}_m \subseteq \mathcal{C}$ the subset with valid responses from configuration m .

- **Accuracy.** We measure decision correctness as

$$\text{Accuracy}_m = \frac{100}{|\mathcal{C}|} \sum_{i \in \mathcal{V}_m} \mathbf{1}[\hat{y}_{mi} = y_i], \quad (4)$$

where $\hat{y}_{mi}$ and y_i are the predicted and reference labels.

- **Expected calibration error (ECE).** We partition valid responses into ten equal-width confidence bins B_{mj} and compute

$$\text{ECE}_m = \sum_{\substack{j=0 \\ |B_{mj}| > 0}}^9 \frac{|B_{mj}|}{|\mathcal{V}_m|} |\text{acc}(B_{mj}) - \text{conf}(B_{mj})|, \quad (5)$$

where acc and conf denote the fraction of correct predictions and mean confidence in each bin.

²<https://github.com/jaredpalmer/kev>

³<https://github.com/InternLM/Intern-Decision>

⁴<https://github.com/NandhaKishorM/laya>

⁵<https://huggingface.co/AlexWortega/openjev>

⁶<https://github.com/bespokelabsai/nimble>

⁷<https://github.com/TianyuCodings/NanoJev>

⁸<https://github.com/rongxinzy/LightJev>

⁹<https://github.com/zwliJay/jev-forge>

Table 1 The strongest decision models match thinking LLMs in accuracy and are better calibrated than every generative LLM. We evaluate 25 configurations on the 10,938 choice and noul questions of JEval. Accuracy counts invalid responses as incorrect, while ECE and Brier_{raw} are computed over valid responses.

Model	ACC (%) ↑	ECE ↓	Brier _{raw} ↓
<i>General Decision models</i>			
Jev	78.38	0.0532	<u>0.287</u>
Kev-0.8b	49.63	0.0432	0.617
Kev-4b	68.90	0.0843	0.435
Kev-9b	70.87	0.0579	0.404
Kev-27b	77.17	0.0356	0.304
Intern-Decision-0.8B	47.71	0.0635	0.643
Intern-Decision-2B	55.72	0.0214	0.553
Intern-Decision-4B	70.74	<u>0.0152</u>	0.399
Laya	26.30	0.2485	0.833
Openjev-4b-v5	70.20	0.1472	0.481
Nimble-9b	73.87	0.1045	0.369
Nanojev	25.34	0.1300	0.806
Lightjev	37.20	0.1145	0.742
Jevforge-0.8b	31.04	0.1164	0.794
InnerJev-4B (ours)	71.92	0.0843	0.383
InnerJev-27B (ours)	79.24	0.0149	0.279
<i>Non-thinking generative LLMs</i>			
Qwen3.5-4B	61.37	0.2526	0.781
Qwen3.8-27B	73.81	0.1697	0.446
DeepSeek-V4-Flash	72.83	0.1923	0.494
GPT-6-Luna	80.93	0.1159	0.297
GPT-6-Sol	79.92	0.1309	0.322
<i>Thinking generative LLMs</i>			
Qwen3.5-4B default think effort	70.73	0.1716	0.489
Qwen3.8-27B xhigh	79.49	0.1005	0.319
DeepSeek-V4-Flash high	79.69	0.1449	0.382
GLM-5.3-Flash high	<u>80.37</u>	0.0784	0.316

- **Raw probability-vector squared error (Brier_{raw}).** We evaluate candidate probabilities against one-hot labels:

$$\text{Brier}_{\text{raw},m} = \frac{1}{|\mathcal{V}_m|} \sum_{i \in \mathcal{V}_m} \sum_{k=1}^{K_i} (p_{mik} - y_{ik})^2, \quad (6)$$

where K_i is the candidate count, $p_{mik} \in [0, 1]$ is the reported probability, and y_{ik} is the one-hot reference.

- **Response latency.** We report mean latency over valid responses on a separate 200-question test, stratified into six input-length bins up to 16K Jev-reported tokens.

MAE on score tasks and benchmark-run latencies from a separate setup are reported in Appendix D.

4.2 Main Results

Finding 1. The strongest decision models **match thinking LLMs in accuracy** with **a single forward pass**.

Overall accuracy. The seven most accurate configurations in Table 1 fall within three points of one another. They are the two GPT-6 models, the thinking modes of GLM-5.3-Flash, DeepSeek-V4-Flash, and Qwen3.8-27B, and two decision models, InnerJev-27B and Jev. Open-weight generative LLMs reach this group only with thinking, which adds six to nine points to their accuracy. Without thinking, the best of them is more than four points below Jev. InnerJev-27B matches the thinking mode of its backbone, Qwen3.8-27B, and InnerJev-4B is more accurate than thinking Qwen3.5-4B and every other 4B decision model (Section 6). Among open decision models, accuracy grows with model size: Kev rises from about 50% at 0.8B to 77% at 27B. Laya, NanoJev, LightJev, and JevForge-0.8B stay below 40%.

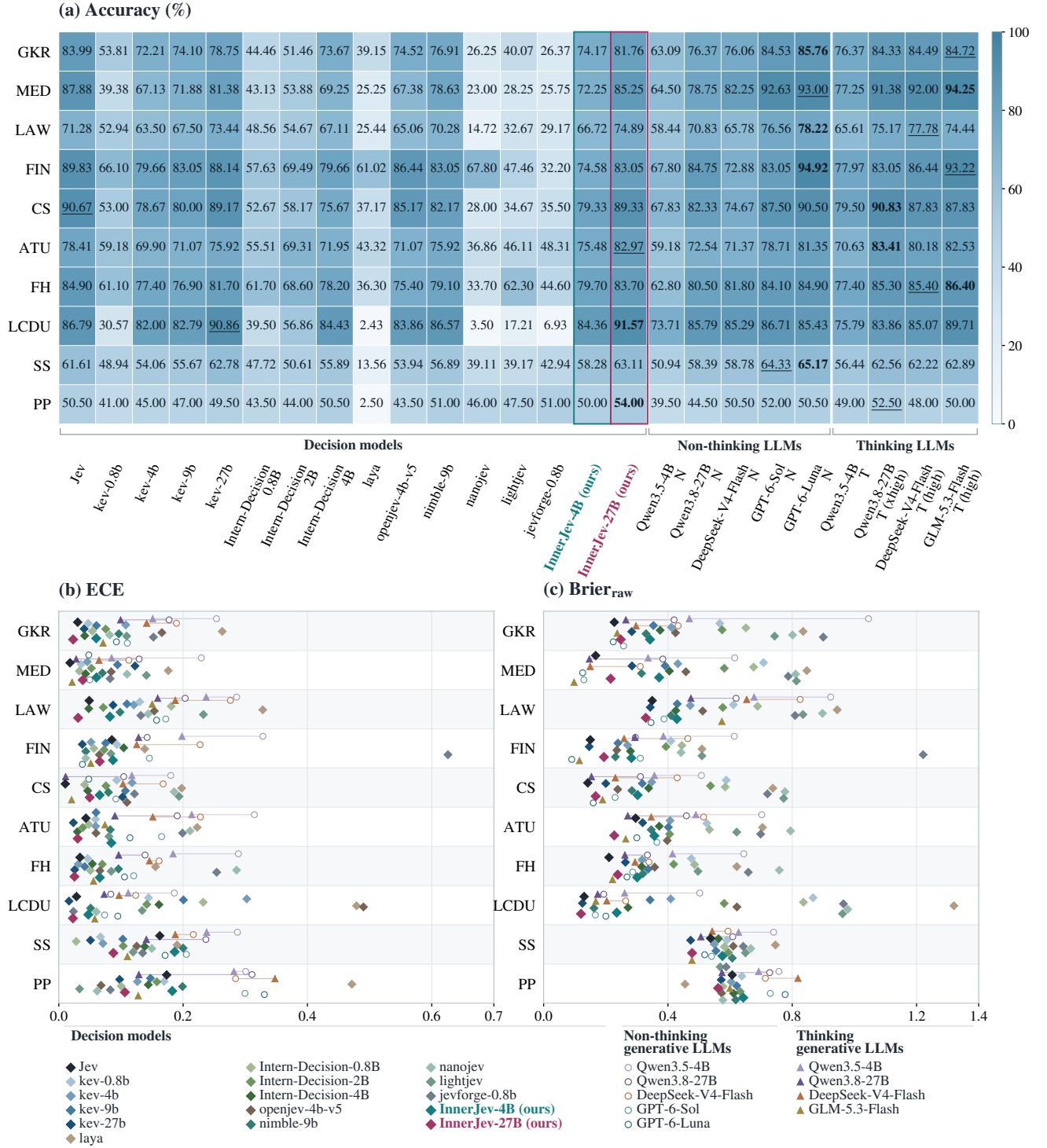


Figure 3 Decision models lead when the evidence is in the input and trail generative LLMs in medicine, finance, and law. We break the JEval results of all 25 configurations down by the ten application domains. (a) Accuracy, (b) ECE, and (c) Brier_{raw}; N and T denote non-thinking and thinking. GKR: general knowledge and reasoning; MED: medicine; LAW: law; FIN: finance; CS: commonsense; ATU: agent and tool use; FH: factuality and hallucination; LCDU: long-context and document understanding; SS: social science; PP: personalization and preferences.

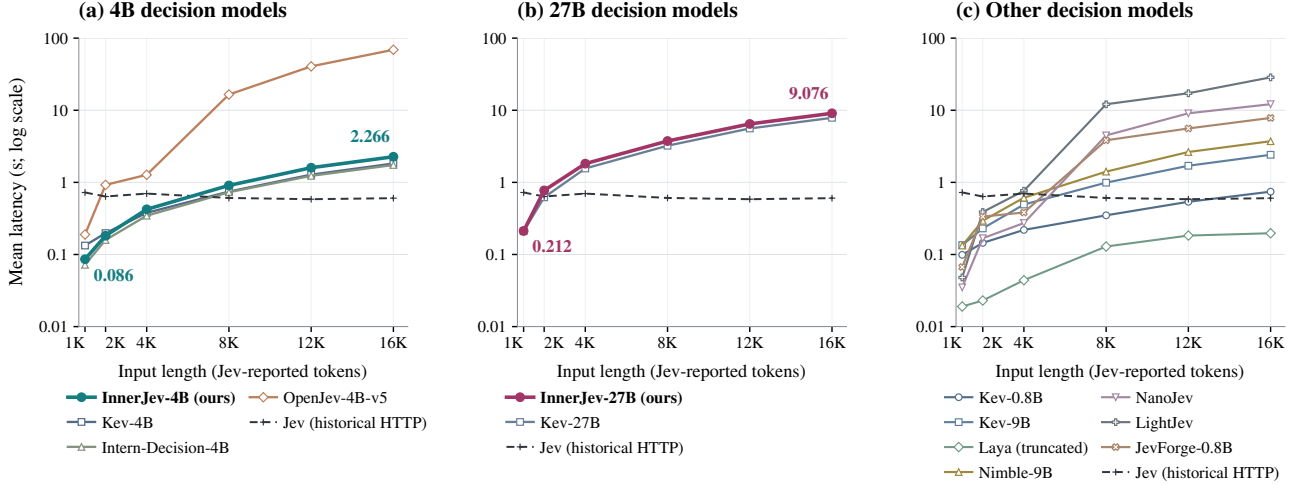


Figure 4 Local decision models are faster than hosted Jev on short inputs and slower on long ones. We measure mean latency on a 200-question test stratified into six input-length bins of up to 16K tokens. (a) 4B decision models, (b) 27B decision models, and (c) the other decision models. Jev’s latency is that of its hosted API and includes network transport, whereas local models run on RTX 4090 GPUs and exclude queueing and transport.

Finding 2. Decision models **lead when the evidence is in the input** and **trail when the decision requires specialist knowledge**.

Accuracy by domain. Figure 3(a) reports accuracy in each of the ten domains. On long-context and document understanding, InnerJev-27B and Kev-27B are the two most accurate of all 25 configurations, and thinking gives the generative LLMs almost no gain. On agent and tool use, InnerJev-27B is within half a point of the best configuration. The order reverses in medicine, finance, and law, where the best decision model is three to six points below the best generative LLM. Thinking does not explain this gap, since GPT-6-Luna ranks first or second in all three domains without thinking. Among decision models, Jev leads in general knowledge, medicine, finance, commonsense, and factuality, and InnerJev-27B leads in the other five domains. Social science and personalization are the hardest domains for every configuration, and no configuration exceeds 54% on personalization. Appendix D reports every benchmark.

Finding 3. Decision models give **confidence that tracks their accuracy** more closely than the probabilities generative LLMs write out.

Calibration. Jev, Kev-27B, Intern-Decision-4B, and InnerJev-27B all have lower ECE than every generative configuration, including GPT-6-Luna, the most accurate one. The ECE of InnerJev-27B is the lowest in Table 1, about one fifth of that of the best-calibrated generative LLM. Thinking lowers the ECE of every LLM run in both modes, but not to the level of these decision models. InnerJev-27B and Jev also have the lowest Brier_{raw}. ECE covers only the confidence of the returned option: Kev-0.8B is well calibrated while answering half of the questions correctly. Section 5.1 examines the full probability distribution, where Jev overstates the probability of its top choice.

Finding 4. Local decision models are **faster than hosted Jev on short inputs** and **slower on long ones**.

Latency. Figure 4 reports mean latency by input length. Jev’s hosted latency, network transport included, stays below one second at every length, while the latency of local models grows with input length. On inputs of up to 1K tokens, which make up most of JEval, every local model is faster than Jev: InnerJev-4B answers in under 0.1s and InnerJev-27B in about 0.2s. InnerJev-27B is as fast as Kev-27B and two points more accurate. The 27B models become slower than Jev at about 2K tokens, and most 4B models between 4K and 8K tokens. On the longest inputs, the 27B models take several seconds per question.

5 Analysis: From Local Decisions to Systems

This section goes beyond overall accuracy to examine the capability boundaries and system-level behavior of decision models. We first probe **local decision boundaries**: exact numerical and probabilistic reasoning (Section 5.1) and long

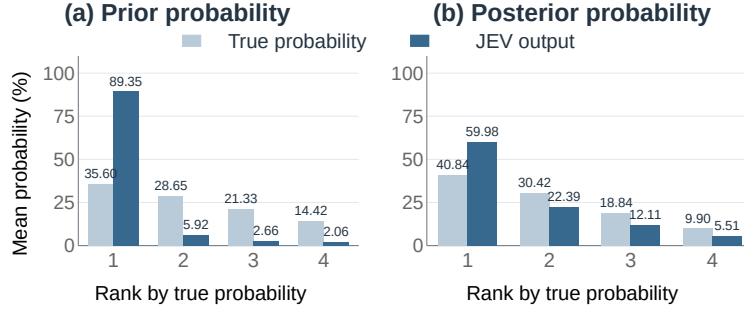


Figure 5 Jev ranks the options correctly but concentrates the probability on its top choice. We compare Jev’s output probabilities with analytically determined distributions over four options, using 100 items per task. Options are ordered by true probability. (a) Prior items, where no clue is observed. (b) Posterior items, where the probabilities follow from an observed clue.

inputs (Section 5.2). Choosing the most likely outcome is not the same as estimating its probability. We then examine **long-horizon decision making** in a multi-step tool-using agent (Section 5.3), where faster action selection reduces median episode time but does not improve task success. Finally, we examine **large-scale decision systems** through social simulation from individuals to populations (Section 5.4). Low-cost inference supports scalable simulation with competitive individual predictions, while profiling limitations, aggregate errors, and systematic bias remain.

5.1 Struggles with Exact Computation and Probability Estimation

Finding 5. Decision models **choose well but estimate poorly**: they identify the most likely outcome while **overstating its probability**, and they land **near exact numerical answers** but rarely on them.

To assess decision models’ mathematical ability and decision-making under uncertainty, we construct four tasks with 100 items each. Examples of these tasks are shown in Figure 13 in Appendix E.1. Every mathematical item has a unique correct answer; every probability item has an analytically determined distribution. We compare Jev with four LLMs under different inference modes, following the input/output format requirements of Section 4; prompts are also provided in Appendix E.1.

Tasks and metrics. For *repeating-decimal division*, the target quantity is $z = a/b \in [0, 100]$, with a non-terminating decimal expansion near a rounding boundary. For *area inversion*, $z = \sqrt{A}$ for 50 square sides and $z = \sqrt{A/\pi}$ for 50 circle radii. Both use $\mathcal{C} = \{0, \dots, 100\}$ and gold answer $y^* = \arg \min_{n \in \mathcal{C}} |z - n|$, excluding $z \in \mathbb{Z} + \frac{1}{2}$. With returned choice $\hat{y}_i$ and $N = 100$, we report

$$\text{Acc}_{\pm d} = \frac{100}{N} \sum_i \mathbf{1}\{|\hat{y}_i - y_i^*| \leq d\}, \quad (7)$$

for $d \in \{0, 1\}$; $d = 0$ gives Acc, and missing choices score zero.

For *prior* and *posterior probability*, let $Y \in \{A, B, C, D\}$ be a hidden label, with prior $\pi_k = n_k / \sum_j n_j$ and likelihood $\ell_k = P(X | Y = k)$. Prior items provide no observation, giving $p_k = \pi_k$. Posterior items observe $e = X$ in 50 cases

Table 2 Jev lands near the exact answer and picks the most likely label, while thinking LLMs recover both the exact value and the probabilities. We construct four tasks of 100 items each, two with a unique numerical answer and two with an analytically determined distribution, and compare Jev with four LLMs, three of them in both inference modes. Acc and Acc_{±1} are the share of answers that are exact and within one integer of the gold value; E_{mode} and E_{all} are the mean absolute errors, in percentage points, of the probability of the gold option and of all four options.

Model	Inference mode	Repeating division		Area inversion		Prior probability			Posterior probability		
		Acc (%)↑	Acc _{±1} (%)↑	Acc (%)↑	Acc _{±1} (%)↑	Acc (%)↑	E_{mode} (pp)↓	E_{all} (pp)↓	Acc (%)↑	E_{mode} (pp)↓	E_{all} (pp)↓
Jev	Native	44	97	39	73	100	53.75	26.91	67	28.12	18.13
Qwen3.5-4B	Non-thinking	34	68	24	45	48	18.66	13.12	38	21.84	17.40
Qwen3.8-27B	Non-thinking	53	99	45	95	98	2.85	2.06	46	12.37	9.87
DeepSeek-V4-Flash	Non-thinking	60	100	69	98	66	9.90	6.54	43	18.67	14.99
Qwen3.5-4B	default think	72	73	82	84	81	0.41	0.38	82	0.74	0.63
Qwen3.8-27B	Thinking (xhigh)	98	100	99	100	100	0.10	0.06	100	0.79	0.48
DeepSeek-V4-Flash	Thinking (high)	100	100	96	97	98	0.00	0.00	93	0.0011	0.0014
GLM-5.3-Flash	Thinking (high)	100	100	99	100	97	0.00	0.00	99	0.15	0.23

and $e = \bar{X}$ in the other 50, giving

$$p_k = P(Y = k | e) = \frac{\pi_k L_k(e)}{\sum_j \pi_j L_j(e)}, \quad (8)$$

where $L_k(X) = \ell_k$ and $L_k(\bar{X}) = 1 - \ell_k$. The gold choice is $k_i^* = \arg \max_k p_{ik}$. We report choice Acc and probability errors in percentage points (pp). Let $\mathcal{V}$ denote the valid responses and $N_{\text{valid}} = |\mathcal{V}|$. For raw returned probabilities q_{ik} ,

$$E_{\text{mode}} = \frac{100}{N_{\text{valid}}} \sum_{i \in \mathcal{V}} |q_{ik_i^*} - p_{ik_i^*}|, \quad (9)$$

$$E_{\text{all}} = \frac{100}{4N_{\text{valid}}} \sum_{i \in \mathcal{V}} \sum_k |q_{ik} - p_{ik}|. \quad (10)$$

Jev lands near the exact answer but rarely on it. Jev answers 44% of division items and 39% of area items exactly, above only non-thinking Qwen3.5-4B among the evaluated configurations, yet 97% and 73% of its answers fall within one integer of the gold value (Table 2). Because every item lies near a rounding boundary, this pattern shows that Jev recovers the magnitude of the quotient or root but not the final digit that decides the rounding. Thinking supplies exactly this step: Qwen3.8-27B rises from 53% to 98% on division and from 45% to 99% on area inversion once it reasons.

Jev selects the most likely label but overstates its probability. Jev picks the mode on all 100 prior items and on 67 posterior items, ahead of every evaluated non-thinking LLM on both tasks. Its probabilities, however, carry the largest errors of all configurations. On prior items, it assigns the true mode 89.35% of the mass on average, against 35.60% in the true distribution (Figure 5); on posterior items, its own two highest-probability options receive 90.30% of the mass. Jev therefore ranks the options well but compresses the distribution toward its top choice. Bayesian updating adds a second limitation: with thinking enabled, every evaluated LLM surpasses Jev in posterior accuracy, reaching 82 to 100% against its 67%.

Accuracy hides this gap wherever probabilities are consumed. A decision that only needs the argmax is unaffected by a compressed distribution, but any downstream use that averages or thresholds probabilities inherits the distortion. Section 5.4.3 shows the population-level consequence: aggregated voter decisions call the right states while misestimating their vote shares.

5.2 Long Contexts Still Degrade Constrained Decisions

Finding 6. A **constrained output space** does not make the input easier to read: decision models **degrade with context length** and **lose evidence in the middle**, just as generative LLMs do.

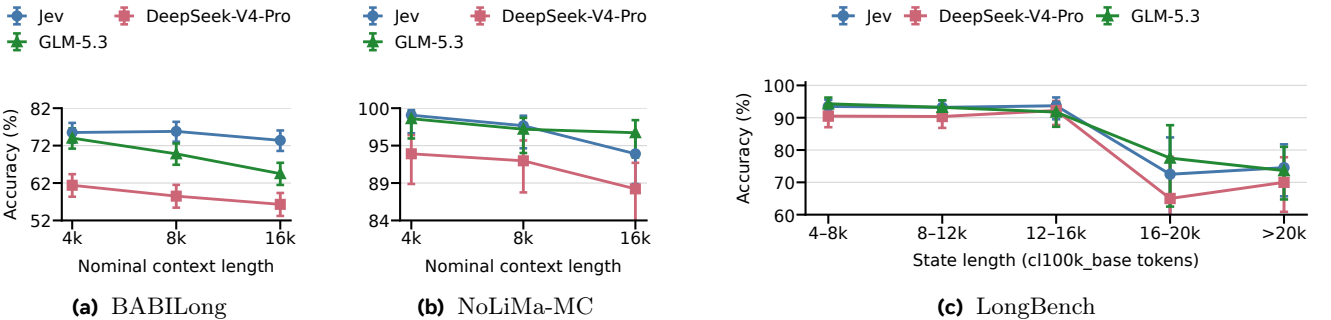


Figure 6 Accuracy falls as the context grows, for Jev and for generative LLMs alike. We evaluate Jev, DeepSeek-V4-Pro, and GLM-5.3 on three long-context benchmarks. (a) BABILong and (b) NoLiMa-MC at context lengths of 4k, 8k, and 16k. (c) LongBench, binned by the length of the `state` field. Error bars are 95% Wilson intervals.

We evaluate Jev, DeepSeek-V4-Pro, and GLM-5.3 on LongBench (Bai et al., 2024), BABILong (Kuratov et al., 2024), and NoLiMa-MC (Modarressi et al., 2025). LongBench examples are stratified by the `c1100k_base` token length of the `state` field, while BABILong and NoLiMa-MC are evaluated under 4k, 8k, and 16k context-length settings.

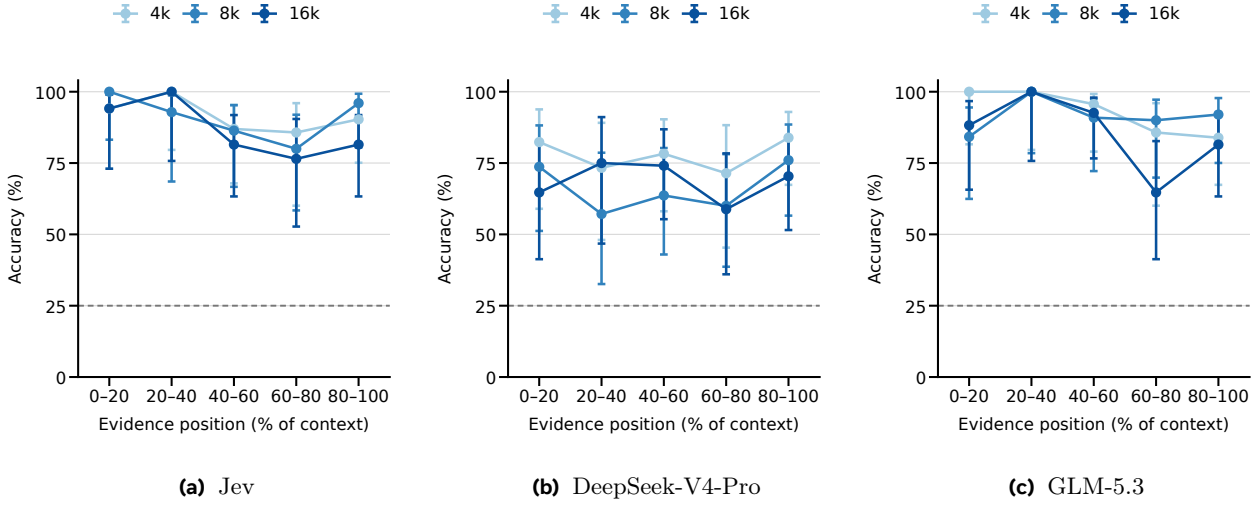


Figure 7 All three models use evidence in the middle of the context least. We locate the supporting sentence of each BABILong QA1 example and group accuracy by its relative position in the context, at context lengths of 4k, 8k, and 16k. Error bars are 95% Wilson intervals.

Accuracy falls as the input grows, for decision models and LLMs alike. Decision accuracy declines with context length for all three models (Figure 6). On LongBench, Jev’s accuracy falls from 93.5% for examples in the 4–8k token range to 74.5% for contexts around 20k tokens, and all three models lose accuracy beyond 16k tokens. A constrained output space fixes what the model must produce, not what it must read, so the bottleneck shifts entirely to locating the relevant evidence.

Evidence in the middle of the context is used least. Motivated by the lost-in-the-middle phenomenon (Liu et al., 2024), we identify the sentence containing the gold supporting evidence in each BABILong QA1 example and partition its normalized character position into five equal-width intervals. After aggregating over the three context-length settings, all three models show the U-shaped pattern previously reported for generative LLMs: accuracy reaches its minimum when the supporting evidence appears near the middle of the context (Figure 7). Jev follows the same pattern, with its accuracy dropping to 80.4% in the middle-position bin.

5.3 Fast Local Decisions in Agentic Workflows

Finding 7. A **faster decision step** does not make a **faster agent**: errors in action selection **compound over an episode**, lowering success and lengthening the slowest runs.

Table 3 With Jev selecting actions, the typical episode is faster, but fewer episodes succeed and the slowest ones take longer. We replace the action-selection step of the τ -bench tool-calling agent with Jev, keep the same base LLM for arguments and responses, and run each test task once. Pass is the episode success rate, Turns is the mean number of agent turns, and P50 and P95 are percentiles of episode wall-clock time.

Env.	System	Pass (%)	Turns	P50 (s)	P95 (s)
Retail	Base (tool call)	86.1	13.22	257.3	661.9
	Jev + LLM	73.9	13.90	147.1	702.3
Airline	Base (tool call)	58.0	13.68	277.8	795.2
	Jev + LLM	48.0	12.35	242.6	1147.7
Overall	Base (tool call)	77.6	13.36	267.1	695.5
	Jev + LLM	66.1	13.44	160.6	816.0

To evaluate Jev in realistic agent workflows, we integrate it into the open-source τ -bench tool-calling agent (Yao et al., 2024) and test it in the Retail and Airline environments. The baseline uses DeepSeek-V4-Flash to jointly select tools and generate arguments. In the Jev condition, Jev selects an available tool or the **respond** action, while the same base model generates the corresponding arguments or user-facing response. Both systems use the same user simulator, temperature 0, environments, and reward checker. We report episode success (Pass¹), mean agent turns, and P50 and P95 wall-clock time. Appendix E.2 details the matched protocol and per-turn workflows.

Jev shortens typical episodes but lowers task success. Jev + LLM succeeds on 64.2% of episodes against 77.6% for the baseline, and the drop is similar in Retail, 73.0% against 86.1%, and in Airline, 44.0% against 58.0% (Table 3). Median episode time falls from 267.1s to 156.1s while mean turns stay nearly unchanged at 13.30 against 13.36, so the saving comes from cheaper turns rather than shorter trajectories. The tail moves the other way: P95 rises from 695.5s to 1,009.5s.

Single-step errors compound into the slow tail. The long-running episodes are concentrated among failures. Some failures begin when Jev selects an inappropriate next tool or action in a multi-step task, causing the trajectory to deviate from the goal; others begin after action selection, when the base LLM generates incorrect tool arguments. Both kinds of error trigger repeated lookups, recovery attempts, and retries, and some trajectories eventually time out. A faster decision step therefore speeds up successful trajectories, but each unreliable choice is paid for over the remaining turns of the episode.

5.4 From Individual Decisions to Population Simulation

Social simulation is a natural application for decision models: every simulated person repeatedly makes a constrained choice, and a simulation needs many such choices at low cost. We test decision models at three levels of increasing scope. We first ask whether they can annotate people (Section 5.4.1), then whether they can simulate an individual’s survey responses (Section 5.4.2), and finally whether the decisions of many simulated individuals aggregate into population-level outcomes that match reality (Section 5.4.3). Jev is evaluated at the annotation and individual levels, and an earlier checkpoint of InnerJev-27B, the open decision model built in Section 6, at the individual and population levels.

5.4.1 Agreement in User Interest Profiling

Finding 8. Jev falls short of general-purpose LLMs in user-interest profiling, so applying it directly to social-science annotation calls for caution.

Annotating people, for example by labeling the interests expressed in their social-media activity, is one of the most basic tasks in computational social science and the first step in building simulated agents. Each annotation is a choice among predefined categories, which makes it a natural fit for a decision model. We therefore first test whether Jev can serve as such an annotator by comparing the interest profiles it assigns to users with those of a general-purpose LLM and of a reference profiler.

Experimental Setup. We profile a held-out cohort of social-media users as multi-label sets over 20 predefined top-level interest categories, using Jev 1.13 and Qwen3.8-27B in non-thinking and thinking modes, and compare each against a frozen Qwen3-Max production profile on the 898 accounts for which all systems return valid outputs. We report the mean per-user Jaccard similarity between interest sets and the rate of exact set matches. None of the outputs is treated as ground truth.

Table 4 Jev agrees less with the reference profiler than Qwen3.8-27B does. We profile 898 social-media accounts as multi-label sets over 20 interest categories and compare each system with a frozen Qwen3-Max reference. Jaccard similarity is computed per user and averaged, and an exact match requires identical sets of categories.

Compared System	Mean Jaccard ↑	Exact Match ↑
Jev 1.13	53.88%	15.03%
Qwen3.8-27B non-thinking	65.24%	25.06%
Qwen3.8-27B thinking	65.86%	28.84%

Jev agrees less with a reference profiler than a general-purpose LLM does. Jev’s interest profiles reach a mean Jaccard similarity of 53.88% with the reference, against 65.24% and 65.86% for non-thinking and thinking Qwen3.8-27B, and match it exactly for 15.03% of users, against 25.06% and 28.84% (Table 4). Thinking adds less than one Jaccard point, and no system reproduces the reference exactly for most users. Because Qwen3.8-27B and the reference belong to the same model family, part of its advantage may reflect shared preferences rather than more accurate profiles.

5.4.2 Individual-Level Social Survey Prediction

Finding 9. Decision models predict individual survey responses on par with generative LLMs at a much lower cost.

Table 5 Decision models predict individual survey responses about as accurately as generative LLMs at a fraction of their latency. We predict responses from demographic profiles on SocioBench, using the same 277,500 respondent–question pairs across ten domains for all systems. InnerJev-27B is close to thinking DeepSeek-V4-Flash, and Jev to its non-thinking mode. Latency is the mean time in seconds per item.

Domain	Jev		Qwen3.5-4B				DeepSeek-V4-Flash-0731				Ours: InnerJev-27B	
	Native		Non-thinking		Thinking		Non-thinking		Thinking (high)		Native	
	ACC ↑	Latency ↓	ACC ↑	Latency ↓	ACC ↑	Latency ↓	ACC ↑	Latency ↓	ACC ↑	Latency ↓	ACC ↑	Latency ↓
Citizenship	43.00	0.478	39.46	6.999	41.84	102.333	43.60	268.818	45.41	302.180	<u>45.09</u>	8.412
Environment	34.85	0.480	30.30	6.832	33.62	101.452	35.51	234.424	38.29	283.856	<u>36.21</u>	8.919
Family	37.23	0.493	29.17	6.728	33.55	106.760	40.31	260.248	42.94	328.831	<u>40.77</u>	8.490
Health	35.44	0.494	30.46	6.816	35.03	97.687	34.83	250.956	<u>37.44</u>	292.322	39.99	8.623
National Identity	37.70	0.504	31.69	7.473	34.30	98.672	35.93	249.620	<u>37.92</u>	182.197	38.86	8.618
Religion	39.69	0.494	34.77	8.319	38.39	101.512	39.90	81.061	42.42	178.482	<u>40.65</u>	6.131
Role of Government	<u>38.66</u>	0.519	34.97	7.513	36.17	97.840	38.02	74.523	39.86	189.612	38.39	8.321
Social Inequality	36.03	0.486	29.05	7.115	31.34	103.303	34.83	104.599	<u>36.75</u>	147.013	36.82	8.711
Social Networks	36.82	0.497	34.53	7.117	37.72	110.469	37.54	112.630	39.96	115.594	<u>38.51</u>	8.898
Work Orientations	36.27	0.506	32.36	6.164	36.70	113.625	37.33	113.121	<u>41.93</u>	111.529	42.39	8.668
Overall	37.69	0.496	32.87	7.124	36.04	103.394	37.90	173.999	40.41	211.649	<u>39.87</u>	8.357

Annotating people is not yet simulating them. We next ask whether a decision model can simulate an individual: given a respondent’s demographic profile, it must predict how that person answers a sociological survey question, a standard testbed for LLM-based simulation of individuals. Each prediction is a choice among the answer options, so the task directly measures how well a decision model represents individual opinions.

Experimental Setup. We predict individual survey responses from respondents’ demographic profiles on SocioBench (Wang et al., 2025a), using the same 277,500 respondent–question pairs across ten domains for all systems. We compare Jev 1.13, InnerJev-27B, and two generative LLMs, Qwen3.5-4B and DeepSeek-V4-Flash-0731, each in non-thinking and thinking modes. The LLMs are served with vLLM on eight H100 GPUs at temperature 0.5, with an 8,192-token budget for reasoning and answer, and InnerJev-27B uses its native option-probability readout.

At the individual level, decision models match LLMs at a fraction of their latency. InnerJev-27B reaches 39.87% overall accuracy, 0.54 points below the 40.41% of the best configuration, thinking DeepSeek-V4-Flash-0731, at a recorded mean latency of 8.36s against 211.65s (Table 5). It ranks among the top two in nine of the ten domains and leads four of them. Jev performs on par with non-thinking DeepSeek-V4-Flash-0731, at 37.69% against 37.90%, with the lowest latency of all systems at 0.50s. Thinking improves both LLMs in every domain, but only by 2.5–3.2 points overall, while the thinking configurations take 103–212s per item, a cost that a simulation pays once for every member of the population.

5.4.3 Population-Level Election Simulation

Finding 10. Decision models make **population-scale simulation affordable** and call **state-level outcomes** as well as generative voters, but they still exhibit **LLM-like biases**, such as a partisan lean.

Finally, we move from individuals to a population. In macro-level simulations such as ElectionSim (Zhang et al., 2024), the quantity of interest is not any single decision but the aggregate of many: hundreds of thousands of simulated voters whose choices add up to state-level outcomes. We test whether the individual decisions of a decision model, aggregated in this way, agree with the real outcome of the 2024 U.S. presidential election.

Experimental Setup. We use InnerJev-27B to drive the voter agents of ElectionSim, which simulates the election state by state with agents built from real social-media users. We sample about 330,000 voters, a ratio of 1/1,000, and describe each by state, five demographic attributes, and recent social-media posts; ElectionSim’s area attribute is unavailable to us. Each voter chooses between Harris and Trump under both option orders, and a state’s predicted Democratic two-party share is the voter-weighted mean of $P(\text{Harris})$. We compare with the archived output of the original pipeline at the same ratio, in which GPT-4o-mini writes each vote. Following ElectionSim, we count correctly called states and battleground states and measure the error of the predicted vote share; Appendix E.3 defines these metrics and reports them in full.

Aggregated decisions call the states but misestimate the shares. InnerJev-27B calls 47 of 51 states and 12 of 15 battleground states, one more of each than the original pipeline (Figure 8), but its vote-share errors are about

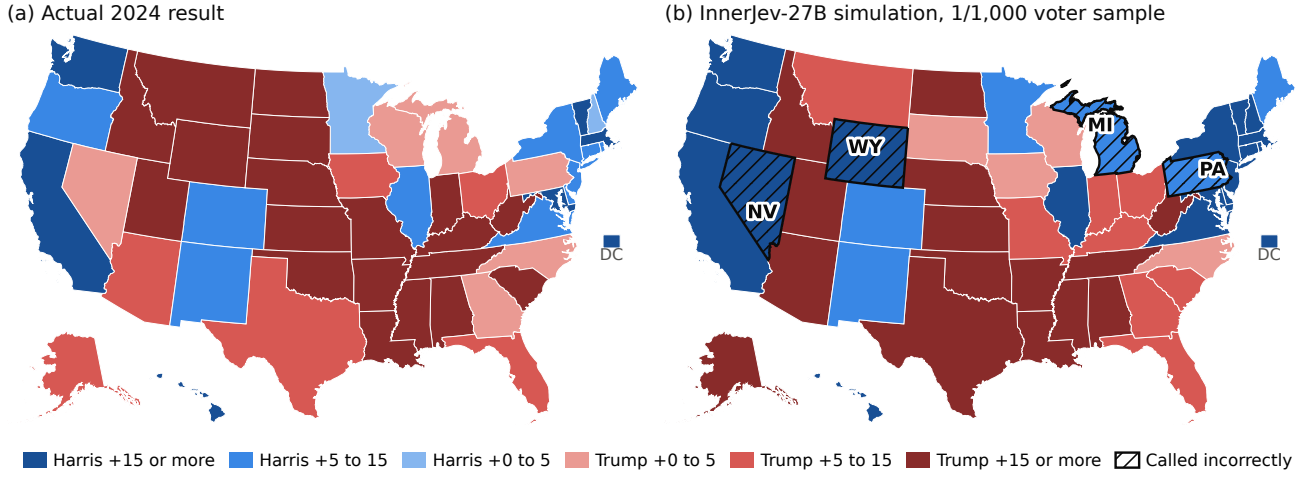


Figure 8 InnerJev-27B calls 47 of 51 states correctly, and all four misses are Trump-won states called for Harris. We simulate the 2024 U.S. presidential election in ElectionSim, with InnerJev-27B driving about 330,000 voter agents sampled at a ratio of 1/1,000. (a) Actual two-party margin. (b) Simulated margin. Appendix E.3 lists the per-state results.

twice as large (Table 40 in Appendix E.3). A state call depends only on which side of 50% the averaged $P(\text{Harris})$ falls, whereas the vote share depends on the level of every voter’s $P(\text{Harris})$, so the two can diverge in the same way that choosing and estimating diverge in Section 5.1. The largest errors occur in sparsely sampled states such as Wyoming and Alaska, and all four misses are Trump-won states called for Harris.

Decision models inherit the partisan lean of LLM voters. Every run overestimates the Democratic share. InnerJev-27B, the original pipeline, and two preliminary runs of Jev through its hosted API all overestimate Harris’s share, by two to six points on average; of the Jev runs, one uses the ElectionSim setting at a ratio of 1/100,000, and the other uses demographics only on the SocioVerse voter pool (Zhang et al., 2025a). In the demographics-only run, every misclassified state is a Trump-won state called for Harris despite thousands of simulated voters per state, and liberal Republicans defect to Harris about twice as often as conservative Democrats defect to Trump. ElectionSim reports the same overestimation for generative agents (Zhang et al., 2024), so moving from generation to decision preserves the lean.

Decisions make population-scale simulation cheap. Each voter costs two forward passes and no generated tokens, and $P(\text{Harris})$ yields expected vote shares directly. Reducing the vote-share errors and the inherited lean is what remains before decision models can replace generative voter agents.

6 Distilling Reasoning into Decision

Finding 11. An open-weight LLM becomes a general decision model through its **first-token readout**, and distilling its **own post-reasoning distribution** into that readout brings it to Jev’s level in a single forward pass.

Jev is available only through a hosted API, and its training recipe is not public. The preceding sections therefore study general decision models from the outside. This section asks the complementary question:

Can such a model be rebuilt from an open-weight LLM?

We answer in two steps. First, an LLM already contains a decision model, and its decisions can be read from the first answer token: the next-token distribution at the answer position, restricted to the labels of the valid options, yields choice, noul, and score decisions from a single forward pass. We call this the **First-Token Decision Readout** (Section 6.1). Second, we distill the answer distribution that the model reaches at the end of its reasoning chain into its first answer token, and call this the **Reasoning-to-Readout Self-Distillation** (Section 6.2). Figure 9 illustrates both steps. Applied to Qwen3.5-4B and Qwen3.8-27B, it yields the open models InnerJev-4B and InnerJev-27B, and InnerJev-27B performs on par with Jev on JEval at a fraction of its latency (Section 6.3).

6.1 Reading Decision from the First Token

To rebuild a decision model from an LLM, we first need the LLM to produce what a decision model produces: a probability for every valid option, from a single forward pass. An LLM normally gives its decision by writing it out, as the generative baselines of Section 4 do, which costs decoding steps and yields probabilities only as numbers

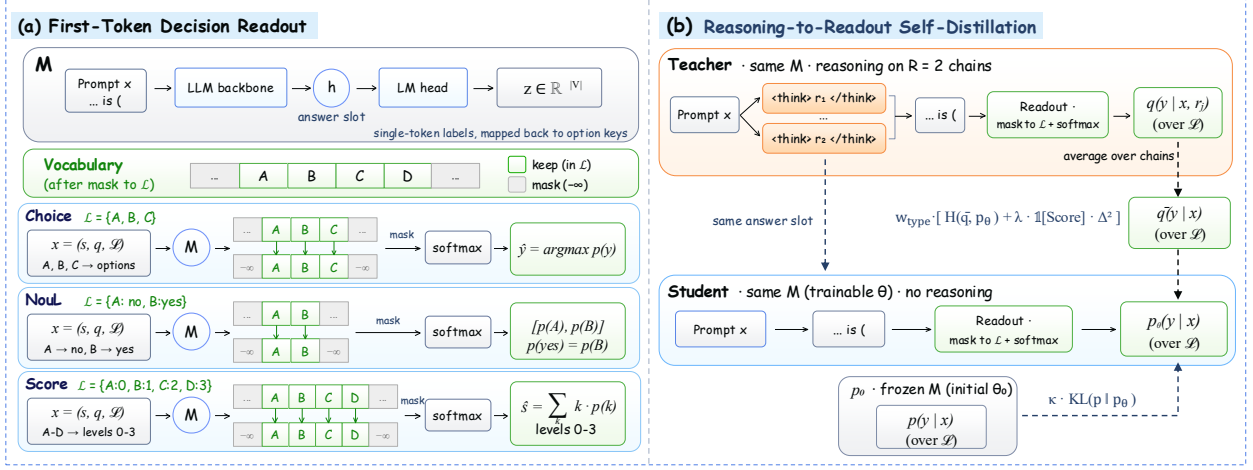


Figure 9 Overview of how we build a decision model from an open-weight LLM. (a) First-Token Decision Readout: the next-token distribution at the answer position is restricted to the labels of the valid options. One forward pass gives the selected option for **choice**, the probability of true for **noul**, and the expected level for **score**. (b) Reasoning-to-Readout Self-Distillation: the model first reasons, and its readouts at the end of the reasoning chains are averaged into a target distribution. The same model without reasoning is trained to match this target at its first answer token, with a KL term that keeps it close to the readout of the frozen original model.

the model writes. The decision is available earlier, however. Once the prompt ends where the answer is due, the model’s next-token distribution already assigns a probability to the label of each option. The First-Token Decision Readout takes this distribution as the decision: we read the next-token distribution at the first answer position and restrict it to the labels of the valid options (Figure 9a).

Concretely, we render each instance $x = (s, i, c)$ as plain text that lists the state, the instruction, and the candidate options labeled (A), (B), ..., and ends with the answer prefix “The correct answer is (”, without a chat template (Appendix F.1). Options are labeled A to Z and then AA, AB, and so on, keeping only labels that the tokenizer encodes as a single token; the first 255 such labels cover every candidate set in JEval (Section 3.2) and are single tokens for both backbones, so each option corresponds to one row of the output layer. Let $h_\theta(x)$ denote the final-layer hidden state at the last prefix position, w_ℓ the output embedding of token ℓ , and $\ell(y)$ the label token of option y . The readout is the next-token softmax restricted to the labels of the valid options:

$$p_\theta(y | x) = \frac{\exp(w_{\ell(y)}^\top h_\theta(x))}{\sum_{y' \in \mathcal{Y}(x)} \exp(w_{\ell(y')}^\top h_\theta(x))}, \quad y \in \mathcal{Y}(x). \quad (11)$$

The output-layer rows of the option labels thus act as a decision head over $\mathcal{Y}(x)$, with no new parameters and no generated tokens. All three JEval formats share this head. For **choice** questions, the most probable label is mapped back to its option key. For **noul** questions, the two outcomes are presented as options and the model returns the probability of **true**. For **score** questions, every rating level is presented as an option and the model returns the expected level $\hat{s} = \sum_k v_k p_\theta(k | x)$, where v_k is the value of level k . These are the same typed outputs that Jev returns, so the readout can replace Jev wherever Jev is used, and because it needs one prefill pass and no decoding, its latency depends only on the backbone and the input length.

Even without training, this readout of Qwen3.8-27B comes within a few points of Jev. The pretrained output layer is also the right decision head to use: in preliminary experiments, training only the label rows of the output layer barely changed accuracy, and replacing them with a newly initialized decision head lowered it by several points. The readout therefore turns an open-weight LLM into a general decision model as it is, with its pretrained head intact.

6.2 Learning from the Model’s Own Reasoning

The readout gives the model’s decision before it has done any reasoning. The same model can also reason first and then answer, and we observe that this changes its decisions considerably, and almost always for the better. The change is large: on JEval, thinking adds five to ten points to each open LLM we run in both modes (Section 4.2), and on the exact-computation and probability tasks of Section 5.1 it lifts Qwen3.8-27B from about half of the items correct to nearly all of them. The change also has a clear direction. Reasoning repairs the readout rather than disrupting it: on the public JevBench questions, thinking fixes all but one readout error of Qwen3.8-27B without

changing any correct answer, and on our other development sets it corrects far more answers than it breaks in all three formats.

What the readout lacks is therefore not knowledge but the reasoning that brings it out: after reasoning, the model reaches decisions that its readout misses, and these decisions are its own. The model can therefore act as its own teacher, with no need for labeled answers or a stronger teacher. Reasoning-to-Readout Self-Distillation builds on this: it distills the answer distribution that the model reaches at the end of its reasoning chain into its first answer token, so that a single forward pass arrives where the chain would have led (Figure 9b).

Teacher: the distribution at the end of reasoning. For each training question, the model generates two independent reasoning chains with thinking enabled. For each chain r , we append the reasoning segment and the answer prefix to the prompt, discard the answer text written after the reasoning, and apply Eq. (11) at the end of the chain to obtain $q(y | x, r)$; for Qwen3.8-27B, the top option of this readout agrees with the model’s written answer in every chain we checked. The teacher distribution $\bar{q}(y | x)$ averages q over the two chains. The target is the full distribution rather than the final answer, because downstream uses such as the population simulation of Section 5.4.3 consume probabilities, not only argmax decisions. Gold labels never enter the target: we neither filter questions by whether the teacher is correct nor replace its distribution with an annotation, and a chain without a valid readout falls back to the model’s readout without thinking.

Student: the same model without reasoning. Teacher and student share the model, the readout, and the label space, and differ only in whether a reasoning chain precedes the answer slot. The student sees only the question and learns to match, at its first answer token, the teacher’s distribution at the reasoning endpoint:

$$\mathcal{L}(\theta) = \sum_i w_{t_i} \left[H(\bar{q}_i, p_\theta(\cdot | x_i)) + \lambda \mathbf{1}[t_i = \text{score}] \Delta_i^2 \right] + \sum_i \kappa_{t_i} \text{KL}(p_0(\cdot | x_i) \| p_\theta(\cdot | x_i)), \quad (12)$$

where H is the cross-entropy, t_i is the question format, and w_t are inverse-frequency weights that balance the formats. For **score** questions, Δ_i is the difference between the student’s and the teacher’s expected levels divided by the width of the rating scale, with $\lambda = 0.5$. The KL term anchors the student to the frozen readout p_0 of the original model to limit forgetting, with $\kappa = 0.05$, raised to 0.2 for hard-labeled **noul** questions. Because the student must reach the post-reasoning decision without the chain, the computation of the chain has to be absorbed into a single forward pass; this is how the model internalizes its reasoning. The first term is knowledge distillation (Hinton et al., 2015) restricted to the valid options; unlike prior work that distills explicit reasoning into direct answers (Deng et al., 2023; Yu et al., 2024), the target is only the decision distribution at the reasoning endpoint, matched at the first answer token.

We update the attention and MLP projections of all language layers at full rank, with LoRA adapters (Hu et al., 2022) on the same modules as a lighter alternative; the embeddings and the output layer stay frozen, so the label rows used by the readout are unchanged and the trained model has the inference cost of the untrained readout. Training runs for a single epoch, with the checkpoint selected on a frozen development set (Appendix F.2). Qwen3.5-4B and Qwen3.8-27B follow exactly the same recipe, each learning from its own reasoning with its own readout as the anchor p_0 .

Training data: any question with options. Since no gold label enters the target, any question with a well-defined option set can serve as training data, and the design question becomes which questions to use. We select them by the capabilities a general decision model is expected to cover. The base pool draws on public decision datasets in eight groups: language understanding, general knowledge, commonsense, logic, English and Chinese medicine, rule-based decisions and computation, and safety and subjective judgment. Development performance rises steadily with the number of such questions, so for InnerJev-27B we extend the pool with science, quantitative reasoning, professional, and factual-boundary questions from further public sources, for 39,279 questions in total. Beyond this point the gain stalls, because additional questions bring little to learn: questions that the model writes itself tend to be easy for it, and it cannot reliably check whether harder ones are correct (Appendix F.4).

6.3 Results

Several open decision models have been released to reproduce Jev. We compare InnerJev-4B and InnerJev-27B with these reproductions and with Jev itself (Table 1). Accuracy is the pooled JEVal accuracy of Section 4.2, and latency is the mean over inputs of up to 1K tokens, which cover most of JEVal, measured on a single H100 GPU for the open models; Jev’s latency is that of its hosted API and includes network transport.

InnerJev sits on the accuracy–latency frontier. Both models lie on the Pareto frontier of Figure 10, and InnerJev-27B sets its upper end: it is the most accurate decision model on JEVal, nearly two points above Jev, and answers in a fraction of Jev’s time on typical inputs. InnerJev-4B is the most accurate of the models that answer within a

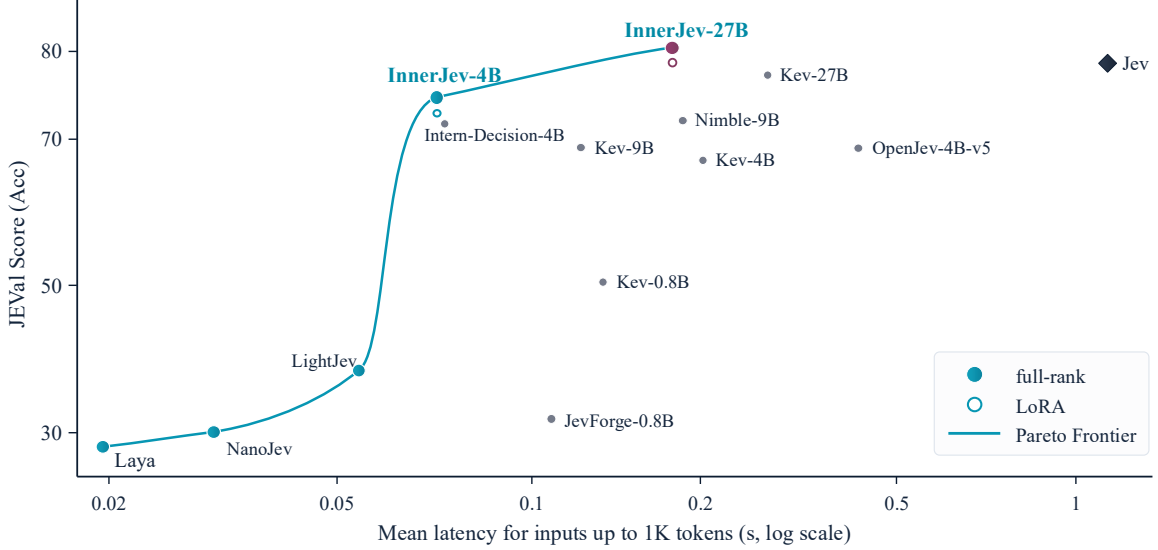


Figure 10 InnerJev-4B and InnerJev-27B lie on the accuracy–latency Pareto frontier, and InnerJev-27B exceeds Jev in accuracy at a fraction of its latency. We compare open reproductions of Jev, and Jev itself, on pooled JEval accuracy and on mean latency for inputs of up to 1K tokens. Open models are measured on a single H100 GPU, and Jev through its hosted API, including network transport.

tenth of a second, ahead of Intern-Decision-4B at a similar latency. On the same GPU, their median latency is about 60 ms for InnerJev-4B and 110 ms for InnerJev-27B. The full-rank and LoRA versions land within a point of each other at both scales, so the gain comes from what the model learns, its own post-reasoning distribution, rather than from how its weights are updated. The gain carries over to newly constructed questions that played no role in training or model selection, where InnerJev-27B improves on the untrained readout by about seven points. Calibration follows the strength of the teacher: InnerJev-27B has the lowest ECE and Brier_{raw} of all systems in Table 1, whereas InnerJev-4B is less calibrated than Intern-Decision-4B.

Distillation transfers the weighing of options, not intermediate computation. Relative to its untrained readout, InnerJev-27B gains most on label-selection benchmarks such as intent recognition, harmful-request judgment, and emotion classification, and it matches or exceeds Jev on more than half of the benchmarks. On NLU-Eval-68, CLINC150, AgentHarm, and GoEmotions, the distilled readout recovers most of the gain that thinking brings to Qwen3.8-27B. In these tasks, reasoning mainly weighs the evidence for each option, and the result of that weighing fits into the first-token distribution. On reasoning-intensive benchmarks the picture reverses: on GPQA and MMLU-Pro, thinking lifts Qwen3.8-27B beyond Jev’s level, yet distillation recovers a quarter of that gain or less. Such reasoning computes intermediate results, as in multi-step arithmetic, solution verification, or aggregation over many records, and a single forward pass does not reproduce that computation even when trained on its outcome. This matches Section 5.1, where Jev lands near exact numerical answers but rarely on them. The teacher sets a second limit: on TruthfulQA, thinking Qwen3.8-27B stays well below Jev, so there is little to transfer, and distillation lowers accuracy.

7 Future Directions

Our results show that current decision models offer broad capabilities and substantial efficiency gains, while their limits, scalability, and integration with generative models remain open questions. Therefore, we further discuss several open questions:

Scaling Toward Collective Intelligence. General intelligence has largely been provided by generative LLMs, whose high inference latency and computational cost limit large-scale parallel deployment, particularly in settings that require continuous interaction among many agents, such as large-scale social simulation, multi-agent coordination, and collective behavior modeling. Fast general decision models make it possible to compose broadly capable decision-making components at substantially lower cost, opening a new avenue for studying collective behavior and emergent intelligence at scale.

The Capability Frontier of General Decision Models. Our results show that finite output spaces do not restrict decision tasks to simple pattern matching; general decision models can already handle knowledge-intensive, reasoning,

long-context, and agentic tasks. However, we also observe a remaining capability gap relative to state-of-the-art LLMs, and it remains unclear how far this can be extended. Characterizing this progression is an important direction for understanding the limits of general decisional intelligence.

System-Level Composition of Heterogeneous Intelligence. Future AI systems may benefit from combining components with different capability rather than relying on a single model for all computation. Generative LLMs are well suited for open-ended reasoning, planning, and complex generation, while fast decision models can handle large volumes of frequent and structured decisions. Combining such complementary components offers a natural path toward balancing system capability, inference latency, and computational cost.

8 Conclusion

We presented a systematic study of **general decisional intelligence**, from isolated decisions to their behavior when composed in larger systems. We introduced **JEVal**, a bilingual benchmark comprising 11,257 instances from 36 datasets across 10 domains, and found that current general decision models can approach strong thinking LLMs on broad decision tasks, especially when decisions can be resolved from available evidence, while remaining weaker in specialist knowledge, faithful uncertainty estimation, exact computation, and long-context evidence use. Moving beyond static benchmarks, we further showed that strong local decision performance does not directly translate into reliable system behavior: in interactive agent workflows, faster local decisions coincided with lower task success, while the trajectory audit identified action errors and integration-level empty responses without isolating a dominant cause. In large-scale social simulation, efficient decision inference enables scalable population modeling but remains vulnerable to profiling errors, inaccurate aggregate estimates, and systematic bias. Finally, we showed that general decision models can be constructed from open-weight LLMs through first-token decision readout and **Reasoning-to-Readout Self-Distillation**, yielding **InnerJev-4B** and **InnerJev-27B** with competitive single-pass performance.

9 Limitations

Broader and more complex agentic evaluation. Our current analysis evaluates general decision models in two representative agentic settings: tool-using agents and social simulation. While these experiments provide initial evidence of how decision models behave within larger agent systems, they cover only a limited range of workflows and interaction patterns. Future work should extend evaluation to broader and more complex agentic settings, including richer tool environments and multi-agent coordination, to better understand how general decision models perform at the system level.

Broader model coverage. Our current evaluation covers only a limited set of general decision models, with Jev serving as the primary representative. Future work should evaluate a broader range of open- and closed-source decision models and LLMs across different model families, scales, and training paradigms to examine whether the observed capability and efficiency patterns generalize more broadly.

Training and construction of general decision models. Beyond evaluating existing models, we explore an open-source training approach based on Reasoning-to-Readout Self-Distillation as an initial attempt to construct general decision models. This represents only one possible training paradigm. Future work should investigate dedicated training objectives, alternative model architectures, broader training data, and more scalable training strategies for improving general decisional intelligence.

References

- Reyna Abhyankar, Qi Qi, and Yiyang Zhang. Osworld-human: Benchmarking the efficiency of computer-use agents. *Proceedings of Machine Learning and Systems*, 8:482–494, 2026.
- Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Alvenschmidt, Sam Altman, Shyamal Anadkat, et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.
- Maksym Andriushchenko, Alexandra Souly, Mateusz Dziemian, Derek Duenas, Maxwell Lin, Justin Wang, Dan Hendrycks, Andy Zou, Zico Kolter, Matt Fredrikson, et al. Agentharm: A benchmark for measuring harmfulness of llm agents. In *International Conference on Learning Representations*, volume 2025, pages 79185–79220, 2025.
- Yushi Bai, Xin Lv, Jiajie Zhang, Hongchang Lyu, Jiankai Tang, Zhidian Huang, Zhengxiao Du, Xiao Liu, Aohan Zeng, Lei Hou, et al. Longbench: A bilingual, multitask benchmark for long context understanding. In *Proceedings of the 62nd annual meeting of the association for computational linguistics (volume 1: Long papers)*, pages 3119–3137, 2024.

- Victor Barres, Honghua Dong, Soham Ray, Xujie Si, and Karthik Narasimhan. τ^2 -bench: Evaluating conversational agents in a dual-control environment, 2025.
- Ilias Chalkidis, Abhik Jana, Dirk Hartung, Michael Bommarito, Ion Androutsopoulos, Daniel Katz, and Nikolaos Aletras. Lexglue: A benchmark dataset for legal language understanding in english. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 4310–4330, 2022.
- Zhiyu Chen, Wenhui Chen, Charese Smiley, Sameena Shah, Iana Borova, Dylan Langdon, Reema Moussa, Matt Beane, Ting-Hao Huang, Bryan R Routledge, et al. Finqa: A dataset of numerical reasoning over financial data. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 3697–3711, 2021.
- Christopher Clark, Kenton Lee, Ming-Wei Chang, Tom Kwiatkowski, Michael Collins, and Kristina Toutanova. Boolq: Exploring the surprising difficulty of natural yes/no questions. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 2924–2936, 2019.
- Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. Think you have solved question answering? try arc, the ai2 reasoning challenge. *arXiv preprint arXiv:1803.05457*, 2018.
- DeepSeek-AI. DeepSeek-V4: Towards highly efficient million-token context intelligence. *arXiv preprint arXiv:2606.19348*, 2026. <https://arxiv.org/abs/2606.19348>.
- Dorottya Demszky, Dana Movshovitz-Attias, Jeongwoo Ko, Alan Cowen, Gaurav Nemade, and Sujith Ravi. Goemotions: A dataset of fine-grained emotions. In *Proceedings of the 58th annual meeting of the association for computational linguistics*, pages 4040–4054, 2020.
- Yuntian Deng, Kiran Prasad, Roland Fernandez, Paul Smolensky, Vishrav Chaudhary, and Stuart Shieber. Implicit chain of thought reasoning via knowledge distillation. *arXiv preprint arXiv:2311.01460*, 2023.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)*, pages 4171–4186, 2019.
- Esin Durmus, Karina Nguyen, Thomas I Liao, Nicholas Schiefer, Amanda Aspell, Anton Bakhtin, Carol Chen, Zac Hatfield-Dodds, Danny Hernandez, Nicholas Joseph, et al. Towards measuring the representation of subjective global opinions in language models. *arXiv preprint arXiv:2306.16388*, 2023.
- Jinyuan Fang, Zaiqiao Meng, and Craig Macdonald. Kirag: Knowledge-driven iterative retriever for enhancing retrieval-augmented generation. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 18969–18985, 2025.
- Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.
- Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q Weinberger. On calibration of modern neural networks. In *International conference on machine learning*, pages 1321–1330. PMLR, 2017.
- Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Peiyi Wang, Qihao Zhu, Runxin Xu, Ruoyu Zhang, Shirong Ma, Xiao Bi, et al. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.
- Ruoqi Guo, Yi Liu, Gelei Deng, Yuekang Li, Lida Zhao, Yutao Wu, Simin Chen, Ying Zhang, and Leo Yu Zhang. Just ask jev: Reinforcement learning for calibrated decisions as a zero-shot detector of ai alignment failures. *arXiv preprint arXiv:2609.29429*, 2026.
- Dan Hendrycks, Collin Burns, Anya Chen, and Spencer Ball. Cuad: An expert-annotated nlp dataset for legal contract review. *arXiv preprint arXiv:2103.06268*, 2021.
- Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015.
- Johannes Hoffart, Mohamed Amir Yosef, Ilaria Bordino, Hagen Fürstenau, Manfred Pinkal, Marc Spaniol, Bilyana Taneva, Stefan Thater, and Gerhard Weikum. Robust disambiguation of named entities in text. In *Proceedings of the 2011 conference on empirical methods in natural language processing*, pages 782–792, 2011.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.
- Yuzhen Huang, Yuzhuo Bai, Zhihao Zhu, Junlei Zhang, Jinghan Zhang, Tangjun Su, Junteng Liu, Chuancheng Lv, Yikai Zhang, Yao Fu, et al. C-eval: A multi-level multi-discipline chinese evaluation suite for foundation models. *Advances in neural information processing systems*, 36:62991–63010, 2023.
- Binyuan Hui, Jian Yang, Zeyu Cui, Jiayi Yang, Dayiheng Liu, Lei Zhang, Tianyu Liu, Jiajun Zhang, Bowen Yu, Keming Lu, et al. Qwen2. 5-coder technical report. *arXiv preprint arXiv:2409.12186*, 2024.

- Hazem Ibrahim and Yasir Zaki. Evaluating decision models for text annotation in computational social science. *arXiv preprint arXiv:2609.24574*, 2026.
- Pranab Islam, Anand Kannappan, Douwe Kiela, Rebecca Qian, Nino Scherrer, and Bertie Vidgen. Financebench: A new benchmark for financial question answering. *arXiv preprint arXiv:2311.11944*, 2023.
- Di Jin, Eileen Pan, Nassim Oufattole, Wei-Hung Weng, Hanyi Fang, and Peter Szolovits. What disease does this patient have? a large-scale open domain question answering dataset from medical exams. *Applied Sciences*, 11(14):6421, 2021.
- Daniel Kahneman. *Thinking, Fast and Slow*. Farrar, Straus and Giroux, New York, 2011.
- Daniel Kahneman and Gary Klein. Conditions for intuitive expertise: A failure to disagree. *American Psychologist*, 64(6): 515–526, 2009.
- Hannah R Kirk, Alexander Whitefield, Paul Röttger, Andrew Bean, Katerina Margatina, Juan Ciro, Rafael Mosquera, Max Bartolo, Adina Williams, He He, et al. The prism alignment dataset: What participatory, representative and individualised human feedback reveals about the subjective and multicultural alignment of large language models. *Advances in Neural Information Processing Systems*, 37:105236–105344, 2024.
- Yuri Kuratov, Aydar Bulatov, Petr Anokhin, Ivan Rodkin, Dmitry Sorokin, Artyom Sorokin, and Mikhail Burtsev. Babilong: Testing the limits of llms with long context reasoning-in-a-haystack. *Advances in Neural Information Processing Systems*, 37:106519–106554, 2024.
- Nathan Lambert, Valentina Pyatkin, Jacob Morrison, LJ Miranda, Bill Yuchen Lin, Khyathi Chandu, Nouha Dziri, Sachin Kumar, Tom Zick, Yejin Choi, et al. Rewardbench: Evaluating reward models for language modeling. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 1755–1797, 2025.
- Stefan Larson, Anish Mahendran, Joseph J Peper, Christopher Clarke, Andrew Lee, Parker Hill, Jonathan K Kummerfeld, Kevin Leach, Michael A Laurenzano, Lingjia Tang, et al. An evaluation dataset for intent classification and out-of-scope prediction. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 1311–1316, 2019.
- Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998.
- Haonan Li, Yixuan Zhang, Fajri Koto, Yifei Yang, Hai Zhao, Yeyun Gong, Nan Duan, and Timothy Baldwin. Cmmlu: Measuring massive multitask language understanding in chinese. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 11260–11285, 2024.
- Junyi Li, Xiaoxue Cheng, Xin Zhao, Jian-Yun Nie, and Ji-Rong Wen. Halueval: A large-scale hallucination evaluation benchmark for large language models. In *Proceedings of the 2023 conference on empirical methods in natural language processing*, pages 6449–6464, 2023.
- Yubo Li, Yidi Miao, Ramayya Krishnan, and Rema Padman. Jev-as-a-judge: Accept when confident, escalate when unsure. *arXiv preprint arXiv:2609.26550*, 2026.
- Percy Liang, Rishi Bommasani, Tony Lee, Dimitris Tsipras, Dilara Soylu, Michihiro Yasunaga, Yian Zhang, Deepak Narayanan, Yuhuai Wu, Ananya Kumar, et al. Holistic evaluation of language models. *arXiv preprint arXiv:2211.09110*, 2022.
- Jiayu Lin and Zhongyu Wei. Communitybench: Benchmarking community-level alignment across diverse groups and tasks. *arXiv preprint arXiv:2601.13669*, 2026.
- Stephanie Lin, Jacob Hilton, and Owain Evans. Truthfulqa: Measuring how models mimic human falsehoods. In *Proceedings of the 60th annual meeting of the association for computational linguistics (volume 1: long papers)*, pages 3214–3252, 2022.
- Junling Liu, Peilin Zhou, Yining Hua, Dading Chong, Zhongyu Tian, Andrew Liu, Helin Wang, Chenyu You, Zhenhua Guo, Lei Zhu, et al. Benchmarking large language models on cmexam-a comprehensive chinese medical exam dataset. *Advances in Neural Information Processing Systems*, 36:52430–52452, 2023.
- Nelson F Liu, Kevin Lin, John Hewitt, Ashwin Paranajape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the association for computational linguistics*, 12:157–173, 2024.
- Xingkun Liu, Arash Eshghi, Pawel Swietojanski, and Verena Rieser. Benchmarking natural language understanding services for building conversational agents. In *Increasing naturalness and flexibility in spoken dialogue interaction: 10th international workshop on spoken dialogue systems*, pages 165–183. Springer, 2021.
- Ali Modarressi, Hanieh Deilamsalehy, Franck Dernoncourt, Trung Bui, Ryan A Rossi, Seunghyun Yoon, and Hinrich Schütze. Nolima: Long-context evaluation beyond literal matching. *arXiv preprint arXiv:2502.05167*, 2025.
- OpenAI. Introducing GPT-6 sol and luna. OpenAI blog, September 2026. <https://openai.com/index/introducing-gpt-6-sol-and-luna/>.
- Alicia Parrish, Angelica Chen, Nikita Nangia, Vishakh Padmakumar, Jason Phang, Jana Thompson, Phu Mon Htut, and Samuel R Bowman. Bbq: A hand-built bias benchmark for question answering. In *Findings of the Association for Computational Linguistics: ACL 2022*, pages 2086–2105, 2022.

- Qwen Team. Qwen3.5: Towards native multimodal agents. Qwen blog, February 2026a. <https://qwen.ai/blog?id=qwen3.5>.
- Qwen Team. Qwen3.8-Max: A new bar for coding and cowork. Qwen blog, August 2026b. <https://qwen.ai/blog?id=qwen3.8>.
- Delip Rao and Chris Callison-Burch. Jev vs. llms as rubric judges: Cheaper, faster, and wrong in the same places. *arXiv preprint arXiv:2609.29769*, 2026.
- David Rein, Betty Li Hou, Asa Cooper Stickland, Jackson Petty, Richard Yuanzhe Pang, Julien Dirani, Julian Michael, and Samuel R Bowman. Gpqa: A graduate-level google-proof q&a benchmark. *arXiv preprint arXiv:2311.12022*, 2023.
- Keisuke Sakaguchi, Ronan Le Bras, Chandra Bhagavatula, and Yejin Choi. Winogrande: An adversarial winograd schema challenge at scale. *Communications of the ACM*, 64(9):99–106, 2021.
- Edward H Shortliffe. Mycin: A knowledge-based computer program applied to infectious diseases. In *Proceedings of the annual symposium on computer application in medical care*, page 66, 1977.
- Ihor Stepanov, Mykhailo Shtopko, Dmytro Vodianytskyi, Oleksandr Lukashov, Alexander Yavorskyi, and Mykyta Yaroshenko. Gliclass: Generalist lightweight model for sequence classification tasks. *arXiv preprint arXiv:2508.07662*, 2025.
- Yu Sun and Junhao Xu. Type-safe is not error-free: A constrained decision head follows the option name, not the rubric bound to it. *arXiv preprint arXiv:2609.26758*, 2026.
- Alon Talmor, Jonathan Herzig, Nicholas Lourie, and Jonathan Berant. Commonsenseqa: A question answering challenge targeting commonsense knowledge. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4149–4158, 2019.
- Gemini Team, Petko Georgiev, Ving Ian Lei, Ryan Burnell, Libin Bai, Anmol Gulati, Garrett Tanzer, Damien Vincent, Zhufeng Pan, Shibo Wang, et al. Gemini 1.5: Unlocking multimodal understanding across millions of tokens of context. *arXiv preprint arXiv:2403.05530*, 2024.
- Kimi Team, Tongtong Bai, Yifan Bai, Yiping Bao, Jianfeng Cai, Xinyuan Cai, Peizhou Cao, Yuxuan Cao, Ziwei Chai, Y Charles, et al. Kimi k3: Open frontier intelligence. *arXiv preprint arXiv:2607.24653*, 2026.
- James Thorne, Andreas Vlachos, Christos Christodoulopoulos, and Arpit Mittal. Fever: a large-scale dataset for fact extraction and verification. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers)*, pages 809–819, 2018.
- TypeSafe AI. Introducing system one models and Jev, September 2026. <https://typesafe.ai/blog/introducing-system-one-models-and-jev>.
- David Wadden, Shanchuan Lin, Kyle Lo, Lucy Lu Wang, Madeleine van Zuylen, Arman Cohan, and Hannaneh Hajishirzi. Fact or fiction: Verifying scientific claims. In *Proceedings of the 2020 conference on empirical methods in natural language processing (EMNLP)*, pages 7534–7550, 2020.
- Jia Wang, Ziyu Zhao, Tingjuntao Ni, and Zhongyu Wei. Sociobench: Modeling human behavior in sociological surveys with large language models. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 26268–26300, 2025a.
- Xidong Wang, Hardy Chen, Dingjie Song, Zhiyi Zhang, Zhihong Chen, Qingying Xiao, Feng Jiang, Jianquan Li, Xiang Wan, Benyou Wang, et al. Cmb: A comprehensive medical benchmark in chinese. In *Proceedings of the 2024 conference of the north american chapter of the association for computational linguistics: Human language technologies (volume 1: Long papers)*, pages 6184–6205, 2024a.
- Xingyao Wang, Yangyi Chen, Lifan Yuan, Yizhe Zhang, Yunzhu Li, Hao Peng, and Heng Ji. Executable code actions elicit better llm agents. *arXiv preprint arXiv:2402.01030*, 2024b.
- Xingyao Wang, Boxuan Li, Yufan Song, Frank F Xu, Xiangru Tang, Mingchen Zhuge, Jiayi Pan, Yueqi Song, Bowen Li, Jaskirat Singh, et al. Openhands: An open platform for ai software developers as generalist agents. In *International Conference on Learning Representations*, volume 2025, pages 65882–65919, 2025b.
- Yubo Wang, Xueguang Ma, Ge Zhang, Yuansheng Ni, Abhranil Chandra, Shiguang Guo, Weiming Ren, Aaran Arulraj, Xuan He, Ziyang Jiang, et al. Mmlu-pro: A more robust and challenging multi-task language understanding benchmark. *Advances in Neural Information Processing Systems*, 37:95266–95290, 2024c.
- Tiantong Wu and Wei Yang Bryan Lim. Reflex with jev for efficient selective control in llm agents. *arXiv preprint arXiv:2609.26532*, 2026.
- Haike Xu, Zongyu Lin, Jing Zhou, Yanan Zheng, and Zhilin Yang. A universal discriminator for zero-shot generalization. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 10559–10575, 2023.
- Zixiang Xu. Jevout: Natural context can flip decision models. *arXiv preprint arXiv:2609.30243*, 2026.
- Van Yang, Hongye Jin, Shaochen Zhong, Song Jiang, Qifan Wang, Vipin Chaudhary, and Xiaotian Han. 100-longbench: Are de facto long-context benchmarks literally evaluating long-context ability? In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 17560–17576, 2025.

- Shunyu Yao, Noah Shinn, Pedram Razavi, and Karthik Narasimhan. τ -bench: A benchmark for tool-agent-user interaction in real-world domains, 2024.
- Howard Yen, Tianyu Gao, Minmin Hou, Ke Ding, Daniel Fleischer, Peter Izsak, Moshe Wasserblat, and Danqi Chen. Helmet: How to evaluate long-context language models effectively and thoroughly, 2025.
- Wenpeng Yin, Jamaal Hay, and Dan Roth. Benchmarking zero-shot text classification: Datasets, evaluation and entailment approach. In *Proceedings of the 2019 conference on empirical methods in natural language processing and the 9th international joint conference on natural language processing (EMNLP-IJCNLP)*, pages 3914–3923, 2019.
- Ping Yu, Jing Xu, Jason Weston, and Ilia Kulikov. Distilling system 2 into system 1. *arXiv preprint arXiv:2407.06023*, 2024.
- Zhouliang Yu, Ruotian Peng, Keyi Ding, Yizhe Li, Zhongyuan Peng, Minghao Liu, Yifan Zhang, Zheng Yuan, Huajian Xin, Wenhao Huang, et al. Formalmath: Benchmarking formal mathematical reasoning of large language models. *arXiv preprint arXiv:2505.02735*, 2025.
- Shengbin Yue, Wei Chen, Siyuan Wang, Bingxuan Li, Chenchen Shen, Shujun Liu, Yuxuan Zhou, Yao Xiao, Song Yun, Xuanjing Huang, et al. Disc-lawllm: Fine-tuning large language models for intelligent legal services. *arXiv preprint arXiv:2309.11325*, 2023.
- Z.ai. GLM-5.3-Flash. Z.ai blog, August 2026. <https://z.ai/blog/glm-5.3-flash>.
- Rowan Zellers, Ari Holtzman, Yonatan Bisk, Ali Farhadi, and Yejin Choi. Hellaswag: Can a machine really finish your sentence? In *Proceedings of the 57th annual meeting of the association for computational linguistics*, pages 4791–4800, 2019.
- Fan Zhang, Yankai Chen, Zhuohan Xie, Yixi Zhou, Sijia Peng, Lei Fan, Xinhua Ji, Cunyuan Zheng, Huangyong Shan, Philip S Yu, et al. Same scores, different decisions: Evaluating jev and language models for legal document understanding. *arXiv preprint arXiv:2609.27678*, 2026.
- Xinnong Zhang, Jiayu Lin, Libo Sun, Weihong Qi, Yihang Yang, Yue Chen, Hanjia Lyu, Xinyi Mou, Siming Chen, Jiebo Luo, Xuanjing Huang, Shiping Tang, and Zhongyu Wei. ElectionSim: Massive population election simulation powered by large language model driven agents. *arXiv preprint arXiv:2410.20746*, 2024.
- Xinnong Zhang, Jiayu Lin, Xinyi Mou, Shiyue Yang, Xiawei Liu, Libo Sun, Hanjia Lyu, Yihang Yang, Weihong Qi, Yue Chen, Guanying Li, Ling Yan, Yao Hu, Siming Chen, Yu Wang, Xuanjing Huang, Jiebo Luo, Shiping Tang, Libo Wu, Baohua Zhou, and Zhongyu Wei. SocioVerse: A world model for social simulation powered by LLM agents and a pool of 10 million real-world users. *arXiv preprint arXiv:2504.10157*, 2025a.
- Yanzhao Zhang, Mingxin Li, Dingkun Long, Xin Zhang, Huan Lin, Baosong Yang, Pengjun Xie, An Yang, Dayiheng Liu, Junyang Lin, et al. Qwen3 embedding: Advancing text embedding and reranking through foundation models. *arXiv preprint arXiv:2506.05176*, 2025b.
- Wei Zhou, Heike Adel, Hendrik Schuff, and Ngoc Thang Vu. Explaining pre-trained language models with attribution scores: An analysis in low-resource settings. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 6867–6875, 2024.

A Ethics

JEVal includes tasks that may involve personal preferences, demographic attributes, harmful requests, or stereotyped scenarios, raising concerns about privacy, harmful content, and inherited dataset bias. To mitigate these risks, we do not reproduce individual participant profiles or demographic records, and sensitive examples are included solely for evaluating model behavior rather than endorsing the assumptions or labels in the source data. We also acknowledge that biases may persist through data collection and annotation, and therefore interpret results only within the scope of the sampled tasks rather than as representative evidence about real-world populations or settings.

B More Benchmark Details

B.1 Datasets by Domain

We organize the source datasets by application domain. The final evaluation set contains 11,257 examples from 36 source datasets in ten domains; datasets with multiple retained task configurations contribute a total of 58 evaluation subtasks. We describe the source task, its conversion to the JEVal decision interface, and the split and sample count used in each case. Split names below refer to the labeled source partitions used to construct the evaluation set. Each item states its resulting JEVal decision format or formats.

General knowledge and reasoning.

- **A12 ARC** contains science examination questions collected in Challenge and Easy subsets (Clark et al., 2018). We place the question in the state, ask the model to select an answer, and use the original options and answer key as the criteria and label of a **choice** task. We sample 200 examples from the test split of each subset.
- **AIDA-TestC** consists of news articles with mentions linked to knowledge-base entities (Hoffart et al., 2011). The article and marked mention form the state, while entity linking becomes a **choice** decision among the gold entity and sampled distractors. We take 200 examples from Test-C.
- **BoolQ** pairs naturally occurring yes/no questions with supporting passages (Clark et al., 2019). We retain the passage and question and map the answer to a **noul** judgment. The evaluation set contains 200 examples from the labeled train split.
- **C-Eval** evaluates Chinese knowledge across academic and professional subjects using multiple-choice questions (Huang et al., 2023). We preserve the question stem, options, and verified answer as a **choice** task and sample 200 examples from its test split.
- **CLINC150** contains user utterances labeled with 150 in-scope intents and an out-of-scope class (Larson et al., 2019). We present the utterance as state and the 151 intent labels as the decision space of a **choice** task, with 200 examples sampled from the test split.
- **CMMLU** provides Chinese multiple-choice questions spanning many knowledge domains (Li et al., 2024). Each question becomes a state with its answer options as criteria and the supplied answer as the label of a **choice** task; we sample 200 test examples.
- **FEVER** contains claims and Wikipedia evidence annotated for factual verification (Thorne et al., 2018). We use the evidence as state and ask whether it supports the claim, retaining the supported and refuted cases that admit a **noul** judgment. We sample 200 examples from the test split.
- **GoEmotions** annotates Reddit comments with fine-grained emotion labels (Demszky et al., 2020). We retain single-label examples and ask the model to make a **choice** decision over the 28-label emotion inventory. We sample 200 examples from the test split.
- **GPQA** contains expert-written questions designed to distinguish domain expertise from plausible incorrect answers (Rein et al., 2023). We convert the Diamond subset to four-option **choice** decisions, retaining its question, candidate answers, and gold answer. All 198 available test examples are included.
- **MMLU-Pro** extends multidisciplinary knowledge evaluation with more challenging, often reasoning-intensive multiple-choice questions (Wang et al., 2024c). We retain the question and options in the **choice** format and sample 200 examples from the labeled train split.
- **NLU Evaluation Data** collects utterances across 68 intents in 18 scenarios (Liu et al., 2021). We use each utterance as state and the intent inventory as criteria for a **choice** decision, selecting 200 examples from the source collection, which is represented as a test split in our converted data.

- **SciFact** pairs scientific claims with evidence abstracts and veracity annotations (Wadden et al., 2020). The abstract is the state and the claim is assessed through a **noul** support judgment; we retain examples with support or contradiction labels. We sample 200 labeled dev examples because the released test questions do not provide gold labels.

Medicine.

- **CMB** collects Chinese medical examination questions across clinical specialties (Wang et al., 2024a). We retain single-answer questions, map their stems and answer options to the **choice** interface, and join the released answer key where needed. We sample 200 examples from the labeled test split.
- **CMExam** consists of Chinese medical licensing and professional examination questions (Liu et al., 2023). We retain questions with one unambiguous answer among the supplied options and convert them to **choice** decisions. We sample 200 test examples.
- **MedQA** draws medical board-style questions from Chinese and English examination sources (Jin et al., 2021). We preserve the original question and four candidates as a **choice** task, constructing separate Chinese and English subtasks. Each contributes 200 test examples.

Law.

- **CUAD** annotates commercial contracts with clause types that legal reviewers may need to identify (Hendrycks et al., 2021). We ask whether a contract contains a specified review-relevant clause, using the contract as state and clause presence as the label of a **noul** decision. We sample 200 test examples.
- **DISC-Law** provides Chinese legal examination questions, including single-answer and multiple-answer items (Yue et al., 2023). We keep the stem and options and represent a multiple-answer selection as one **choice** decision over the valid nonempty option combinations. We sample 200 test examples for each of the single-answer and multiple-answer subtasks.
- **LexGLUE** brings together legal understanding tasks from case law, human-rights decisions, contracts, and terms of service (Chalkidis et al., 2022). We convert CaseHOLD, ECtHR-A, ECtHR-B, LEDGAR, SCOTUS, and Unfair-ToS to **choice** decisions, retaining only items with a single valid class where required. Each of the six test subtasks contributes 200 examples.

Finance.

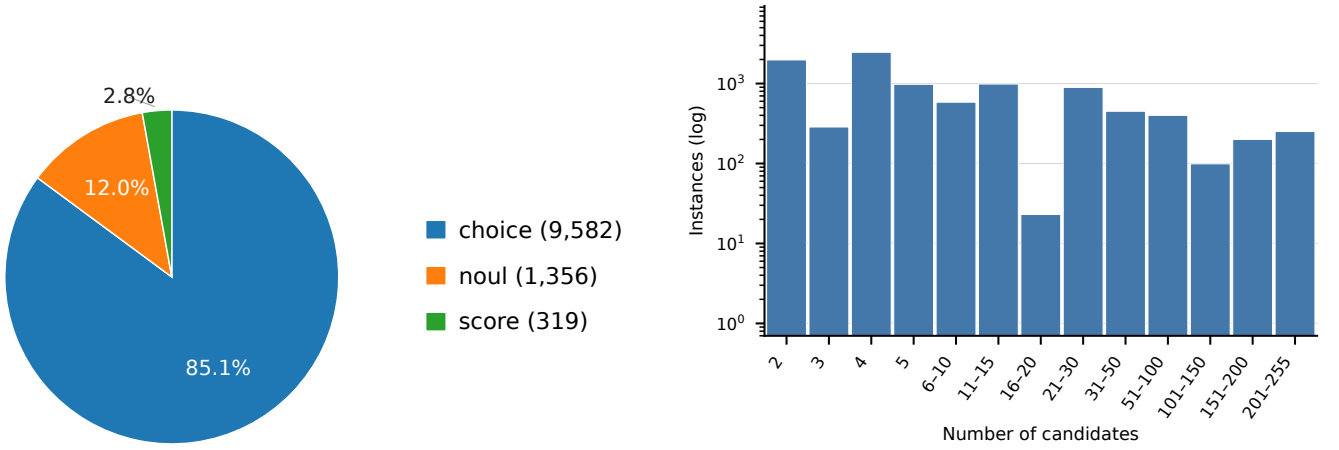
- **FinanceBench** asks questions grounded in company filings and supplies document evidence and reference answers (Islam et al., 2023). We retain questions whose reference answer supports an unambiguous **noul** judgment, placing the evidence in state and the question in instruction. The test subset contains 37 such examples.
- **FinQA** pairs financial-report passages and tables with questions and annotated reasoning programs (Chen et al., 2021). We retain questions with a verifiable yes/no answer and convert the report context and question to a **noul** decision. The test subset contains 22 examples.

Commonsense.

- **CommonsenseQA** contains multiple-choice questions whose distractors reflect related commonsense concepts (Talmor et al., 2019). We use the question as state and its five answers as the criteria of a **choice** task, sampling 200 examples from the labeled dev split.
- **HellaSwag** asks a model to select the plausible continuation of an everyday or instructional scenario (Zellers et al., 2019). The context becomes state and four endings define the criteria of a **choice** task; we sample 200 dev examples.
- **WinoGrande** tests commonsense coreference through paired sentence-completion alternatives (Sakaguchi et al., 2021). We preserve the sentence and its two fillers as a **choice** decision and sample 200 labeled dev examples.

Agents and tools.

- **AgentHarm** evaluates agent behavior on harmful and benign requests in tool-use settings (Andriushchenko et al., 2025). We turn its behavior annotations into **noul** harmfulness judgments and **score** severity ratings over the same request context. We sample 200 test records across these decision formats.



(a) Decision formats across all 11,257 instances. Slice labels show percentages; the legend gives counts.

(b) Candidate counts among 9,582 **choice** instances. The vertical axis is logarithmic.

Figure 11 Decision-format and candidate-space composition of the final JEval evaluation set.

Table 6 Domain-level composition of the final evaluation set. Language follows each source subtask’s designation, rather than a separate label for every input field. Empty planned domains are omitted.

Domain	Datasets	Subtasks	Instances	Share (%)	English	Chinese
General knowledge	12	13	2,598	23.1	2,198	400
Medicine	3	4	800	7.1	200	600
Law	3	9	1,800	16.0	1,400	400
Finance	2	2	59	0.5	59	0
Commonsense	3	3	600	5.3	600	0
Agents and tools	4	4	800	7.1	800	0
Truthfulness	2	5	1,000	8.9	1,000	0
Long context	2	7	1,400	12.4	1,000	400
Social science	2	2	400	3.6	400	0
Personalization	3	9	1,800	16.0	1,800	0
Total	36	58	11,257	100.0	9,457	1,800

- **Decider**¹⁰ contains decision prompts for structured judgments. We retain their scenario text as state and encode candidate outcomes as criteria for **choice** tasks and binary outcomes for **noul** tasks. We sample 200 records from the labeled train split.
- **JevBench**¹¹ collects requests and responses with judgments about correctness, constraint satisfaction, and related decision properties. We preserve each request–response context and formulate its judgment as a **noul**, **choice**, or **score** decision according to the supplied label. We sample 200 test records.
- **JevForge**¹² collects structured interaction states and next-step decisions for agent tasks. We use the task and available interface elements as state and criteria for **choice** action selection, and express binary checks as **noul** decisions. We sample 200 records from the labeled train split.

Truthfulness and hallucination.

- **HaluEval** pairs grounded and hallucinated responses across dialogue, question answering, and summarization, and includes direct hallucination judgments (Li et al., 2023). We turn the paired responses into two-option **choice** tasks and the general judgments into **noul** decisions. Dialogue, general, QA, and summarization each contribute 200 examples from the labeled train split.
- **TruthfulQA** asks questions chosen to elicit common falsehoods and supplies answers with truthfulness annotations (Lin et al., 2022). We use its MC1 candidates as the decision space of a **choice** task and their annotated correct answer as label, sampling 200 examples from the labeled dev split.

Long context and document understanding.

¹⁰<https://github.com/Mapika/decider>

¹¹<https://github.com/fstandhartinger/jevbench>

¹²<https://github.com/zwliJay/jev-forge>

Table 7 Input lengths by domain in `cl100k_base` tokens. Each instance is capped at 20,480 tokens before calculating the mean and maximum for this descriptive table. Stored inputs above the cap are not claimed to have been truncated during evaluation.

Domain	Mean	Maximum
General knowledge	348	3,447
Medicine	136	653
Law	2,831	20,480
Finance	565	1,591
Commonsense	85	276
Agents and tools	267	3,523
Truthfulness	327	2,312
Long context	10,433	20,480
Social science	71	221
Personalization	474	6,769
All instances	1,974	20,480

- **HELMET** evaluates long-context models with controlled tasks at different context lengths (Yen et al., 2025). We use its key-value retrieval instances: the long dictionary is state, the query identifies a key, and the listed values form the **choice** decision space. We sample 200 test examples.
- **LongBench** aggregates bilingual long-context tasks, including classification and passage retrieval (Bai et al., 2024). We retain LSHT, Retrieval-en, Retrieval-en-e, Retrieval-zh, TREC, and TREC-e, using the original class inventory or paragraph identifiers as criteria of **choice** tasks. Each of the six test subtasks contributes 200 examples.

Social science.

- **BBQ** probes social bias with questions whose answers depend on whether the provided context resolves an ambiguity (Parrish et al., 2022). We preserve the context, question, three candidate answers, and annotated answer in a **choice** task, sampling 200 test examples.
- **GlobalOpinionQA** organizes survey questions and response distributions across countries (Durmus et al., 2023). We combine a question with its country context and ask for a **choice** decision selecting the most common response among its original answer options, excluding tied modes. We sample 200 test examples.

Personalization and preference.

- **CommunityBench** evaluates whether a model can infer and follow the preferences of an online community (Lin and Wei, 2026). We convert community prediction and response preference identification into **choice** tasks conditioned on community descriptions or examples. Each test subtask contributes 200 examples.
- **CompRed**¹³ contains community-grounded response preferences from discussions in several topical domains. We present the discussion history as state and ask which candidate response the community prefers in a **choice** task. Finance, gender and sexuality, history, politics, and science each contribute 200 test examples.
- **PRISM-Alignment** records participant preferences and ratings of model responses in conversational settings (Kirk et al., 2024). We use the participant context to predict the preferred response as a **choice** task or an ordered rating band as a **score** task. Both the alignment and score subtasks contribute 200 examples from the labeled train split.

B.2 Benchmark Composition

Coverage. The final JEval evaluation set comprises 36 datasets, 58 subtasks, and 11,257 instances, after sampling from the larger pool of normalized candidate examples. We treat each upstream source as a single dataset, each distinct evaluation configuration as a subtask, and each retained decision record as an instance. Table 6 summarizes the number of datasets, subtasks, and instances by domain, together with the source-designated language distribution.

¹³<https://huggingface.co/datasets/allenai/compred>

Table 8 Representative JEval test cases. Each row shows one evaluation instance in the shared decision format.

Dataset and format	Evaluation instance
AI2 ARC Science question choice	State: It was once thought that living organisms could come from non-living matter. For example, people believed that flies would develop from rotting meat. Instruction: This idea was later disproved primarily because of Criteria: opt_1: the discovery of the atom; opt_2: better surgical techniques; opt_3: continued experimentation; opt_4: the invention of the microscope. Gold label: opt_3 (continued experimentation).
MedQA Medical examination choice	State: A 72-year-old anthropologist with long-standing hypertension visits your office for a routine exam. You notice an abnormality on his laboratory results caused by his regimen of captopril and triamterene. What abnormality did you most likely find? Instruction: Based on the question, choose the correct option. Criteria: A: Hyperkalemia; B: Hypernatremia; C: Thrombocytopenia; D: Anemia. Gold label: A (Hyperkalemia).
LexGLUE Unfair contract clause choice	State: We may modify this contract, our privacy policy and our cookies policies from time to time. Instruction: Which type of unfair clause does this terms-of-service excerpt contain? Choose None of the 8 if it contains none. Criteria: Limitation of liability; Unilateral termination; Unilateral change; Content removal; Contract by using; Choice of law; Jurisdiction; Arbitration; None of the 8. Gold label: Unilateral change.
HaluEval Response grounding noul	State: User query: “Sketch a logo for the company International Business Company.” Response: “Unfortunately, as an AI language model, I do not have the capability to generate visual content. Can I assist you with anything else?” Instruction: Does this response contain hallucinated content not supported by the query? Criteria: false: The response is faithful to the source and contains no hallucination; true: The response contains content not supported by the source (hallucinated). Gold label: false.
LongBench Paragraph retrieval choice	State (excerpt): Paragraph 16: “This species of lizard has a large head that is elongated and depressed, with the cheeks swollen in adult males.” Paragraph 17: “Biometrika was established in 1901 by Francis Galton, Karl Pearson, and Raphael Weldon to promote the study of biometrics.” The other 28 paragraphs are omitted here. Instruction (excerpt): Which paragraph in the passage below matches this description? The text describes the physical characteristics of a species of lizard. It focuses on details such as the large head with swollen cheeks in adult males, the length of the snout, . . . Criteria: Paragraph 1 through Paragraph 30 (30 candidate identifiers). Gold label: Paragraph 16.
JevBench Incident impact score	State: The icon is misaligned. Every function works. Instruction: Rate incident impact using only reported facts. Use the highest fully supported level. Criteria: 0: No function impaired; cosmetic only. 1: One user or a nonessential function impaired, with a workaround. 2: Many users blocked from a core function, no data loss. 3: Confirmed irreversible data loss or physical harm. Gold label: 0 (cosmetic only).

Decision formats and candidate spaces. The relative frequencies of the three decision formats are shown in Figure 11a. Choice decisions dominate the final set, while binary and ordered-score decisions provide distinct output structures. For the 9,582 choice instances, the number of candidates ranges from two to 255. Figure 11b shows the candidate-count distribution using the same bar-chart style as the input-length distribution in the main text. The bins cover the entire observed range.

Input length by domain. Table 7 reports the mean and maximum input length for each domain, measured over state, instruction, and criteria with `cl100k_base`. For this table, we cap each measured length at the Jev reporting limit of 20,480 tokens before computing either statistic. This gives the length scale within the Jev budget while retaining all 11,257 records in the denominator. In the stored evaluation data, 202 raw inputs exceed 20,480 tokens; the cap in this table is a reporting convention rather than evidence that those stored records were truncated before evaluation.

B.3 Representative Test Cases

Table 8 presents six instances drawn from the final JEval evaluation set: AI2 ARC (Clark et al., 2018), MedQA (Jin et al., 2021), LexGLUE (Chalkidis et al., 2022), HaluEval (Li et al., 2023), LongBench (Bai et al., 2024), and JevBench (TypeSafe AI, 2026). They illustrate how different source tasks appear under the common decision interface. We show the state, instruction, criteria, and gold label for each instance.

C More JEval experimental settings

C.1 Input and Output Protocol.

All configurations answer the same 11,257 instances from 58 benchmarks, and all but 319 score questions are choice or noul classification questions. Every model receives the same task content in the state-instruction-criteria form of Section 3.1, but the two kinds of model produce probabilities differently. Decision models read them from a prediction head over the candidates: Jev through its native interface, the other open decision models through their own encoders and heads with the candidate order unchanged, and InnerJev-4B and InnerJev-27B through a softmax over the candidate labels at the first answer position. Generative LLMs instead write their answer and a probability for every candidate as JSON that follows a question-specific schema, whether or not they think first. Appendix C.2 gives the prompt templates, schemas, and validation rules.

C.2 Prompt Templates and Response Protocol

Input representation and prompt construction. Jev receives the benchmark’s standardized **state** and **questions** fields through its native interface. The seven generative LLM configurations receive the same task content through a shared system message and task-specific user templates, shown in Figure 12. Each user message contains the task state, instructions, and the available options or score levels. Candidate identifiers, descriptions, order, and language are preserved. Strings remain unchanged, while objects and arrays are serialized as compact JSON with `ensure_ascii=False`. Missing or null descriptions are omitted together with the preceding colon. Score levels follow their original order and are indexed from 0 to $K - 1$, where K is the number of levels.

Response specification. Each LLM request includes a question-specific JSON Schema through `response_format`, with `type=json_schema` and `strict=true`. Responses are organized under **answers**, keyed by the original question identifier. Each answer specifies its **type**: **choice** responses return an original option identifier, **noul** responses return the probability that the answer is true, and **score** responses return a numerical value in $[0, K - 1]$. Choice and score responses additionally include **confidence** and a **probabilities** mapping containing every option or level exactly once, including zero-valued entries. Score responses also include a **legend** mapping level identifiers to their original descriptions. Confidence and probability values must lie in $[0, 1]$. All specified fields are required, and every schema object sets `additionalProperties=false`.

Output validation. Only the final **message.content** is parsed for LLM answers; reasoning text is excluded. Incomplete or malformed JSON, duplicate keys, non-finite numbers, missing or additional fields, invalid candidate identifiers, and out-of-range values are rejected without repair. The returned choice is not replaced by the probability argmax, and the returned score is not recomputed as a probability-weighted expectation. The independent **confidence** field is not required to equal the selected-option probability. Probability vectors are neither normalized nor rejected because their entries do not sum to one.

C.3 Model Configurations

Table 9 summarizes the inference configurations used in our main experiments, covering decision models and both non-thinking and thinking configurations of generative LLMs. For each configuration, we report the applicable reasoning controls, token budgets, and temperature. The reported context windows and token limits represent experimental budgets rather than the models’ native context capacities. Temperature is used for candidate-probability scaling in decision models and token sampling in generative LLMs.

Local benchmark runs use eight NVIDIA H100 80 GB GPUs with a development build of vLLM 0.29.1rc1. The generative Qwen models use BF16 weights, and DeepSeek-V4-Flash and GLM-5.3-Flash use FP8-quantized weights. LLM requests are non-streaming. Non-thinking requests use output budgets of 256 to 8,192 tokens depending on the response schema, and thinking requests use an 8,192-token output budget that includes a 4,096-token thinking budget. InnerJev runs with BF16 base weights and unmerged BF16 LoRA adapters, with tensor parallelism 1 and data parallelism 8. The other local decision models keep their checkpoint-specific precision and calibration parameters. Only unsuccessful requests are retried, and the first valid response is kept.

Table 9 Inference configurations for JEVal. **Context.** Token columns report maximum experimental budgets; input limits apply to complete encoded paths. Relative to released encoders, we raise Laya’s limit from 512 to 8,192 tokens and the limits of LightJev (256), JevForge (768), OpenJev (4,096), and NanoJev, Nimble, and Intern-Decision (8,192) to 24,576 tokens; Kev accepts an explicit encoding limit. Backbone positional configurations remain unchanged: 8,192 for Laya, 40,960 for NanoJev and LightJev, and 262,144 for the Qwen3.5/3.8-based decision models, including InnerJev. Some NanoJev and LightJev runs use shorter context/input limits of 8,192 and 256 tokens, respectively. GPT entries are client-side budgets. Laya retains its 48-token option limit; OpenJev processes all 24,000-character windows with 2,000-character overlap. **Output.** Thinking and final-answer tokens share the output budget; InnerJev’s one-token allowance supports candidate-logit readout. **Temperature.** Values specify candidate-probability scaling for decision models and token sampling for generative LLMs. Laya uses task-specific temperatures of 1.6369 (choice), 1.2514 (score), and 1.9834 (noul), with choice overrides of 1.9064, 1.7602, 1.0000, and 0.1006 for 2, 3–5, 6–10, and ≥ 11 options, respectively. — denotes an inapplicable or unreported scalar.

Model	Mode	Reasoning effort	Context window	Input budget	Output budget	Thinking budget	Temperature
<i>Decision models</i>							
Jev	—	—	—	—	—	—	—
Kev-0.8b	—	—	32,768	24,576	—	—	2.3511
Kev-4b	—	—	32,768	24,576	—	—	2.4061
Kev-9b	—	—	32,768	24,576	—	—	2.2974
Kev-27b	—	—	32,768	24,576	—	—	1.3819
Intern-Decision-0.8B	—	—	32,768	24,576	—	—	2.7478
Intern-Decision-2B	—	—	32,768	24,576	—	—	2.1005
Intern-Decision-4B	—	—	32,768	24,576	—	—	1.9924
Laya	—	—	8,192	8,192	—	—	—
Openjev-4b-v5	—	—	32,768	24,576	—	—	—
Nimble-9b	—	—	32,768	24,576	—	—	1.0
Nanojev	—	—	32,768	24,576	—	—	1.0
Lightjev	—	—	32,768	24,576	—	—	1.0
Jevforge-0.8b	—	—	32,768	24,576	—	—	0.9717
InnerJev-4B (ours)	—	—	32,768	24,576	1	—	1.0
InnerJev-27B (ours)	—	—	32,768	24,576	1	—	1.0
<i>Non-thinking generative LLMs</i>							
Qwen3.5-4B	Non-thinking	—	32,768	24,576	8,192	—	1.0
Qwen3.8-27B	Non-thinking	—	32,768	24,576	8,192	—	1.0
DeepSeek-V4-Flash	Non-thinking	—	32,768	24,576	8,192	—	1.0
GPT-6-Luna	Non-thinking	—	32,768	24,576	8,192	—	1.0
GPT-6-Sol	Non-thinking	—	32,768	24,576	8,192	—	1.0
<i>Thinking generative LLMs</i>							
Qwen3.5-4B	Thinking	default	32,768	24,576	8,192	4,096	1.0
Qwen3.8-27B	Thinking	xhigh	32,768	24,576	8,192	4,096	1.0
DeepSeek-V4-Flash	Thinking	high	32,768	24,576	8,192	4,096	1.0
GLM-5.3-Flash	Thinking	high	32,768	24,576	8,192	4,096	1.0

Prompt and response protocol

(a) System message	
Return only one JSON object conforming to the response JSON Schema. Use the original question ID under "answers". Include every field required for this question type. For choice and score questions, include every option or level exactly once in "probabilities", including entries with value 0. Output confidence and probability values as JSON numbers between 0 and 1. For noul questions, "noul" is the probability that the answer is true. Do not output Markdown, comments, explanations, or additional fields.	
(b) Choice task: user message	Response schema (schematic)
State: <pre>{state_text}</pre> {instructions_text} Options: <pre>{option_lines}</pre> Each line: - {option_id}: {description} Preserve order; omit the suffix for null descriptions.	<pre>{ "answers": { "{question_id}": { "type": "choice", "choice": "<original option ID>", "confidence": <number in [0,1]>, "probabilities": { <every option ID>: <number in [0,1]> } } } }</pre>
(c) Noul task: user message	Response schema (schematic)
State: <pre>{state_text}</pre> {instructions_text} Options: - true{true_description_suffix} - false{false_description_suffix} Missing/null suffix is empty; otherwise use ": " + description.	<pre>{ "answers": { "{question_id}": { "type": "noul", "noul": <number in [0,1]> } } }</pre> Prediction: true if and only if noul $\geq$ 0.5.
(d) Score task: user message	Response schema (schematic)
State: <pre>{state_text}</pre> {instructions_text} Levels: <pre>{level_lines}</pre> Each line: - {level_index}: {description} Indices 0 to K-1 preserve the original level order. Null descriptions omit the colon and following text. K denotes the number of levels.	<pre>{ "answers": { "{question_id}": { "type": "score", "score": <number in [0,K-1]>, "confidence": <number in [0,1]>, "legend": { <every level ID>: <original description> }, "probabilities": { <every level ID>: <number in [0,1]> } } } }</pre>

Figure 12 Prompt templates and response structures for the nine LLM configurations. Panel (a) presents the shared system message; panels (b)–(d) present the user templates and schematic responses for **choice**, **noul**, and **score**, respectively. Placeholders are instantiated per question. Formatting notes, value constraints, and the binary prediction rule are explanatory annotations, not additional prompt text.

D Benchmark-Level Results

This section presents benchmark-level results for all 25 model configurations across the 58 JEval benchmarks in ten domains, together with pooled domain summaries. We report accuracy, ECE, and Brier_{raw} for classification questions, MAE separately for score questions, and mean latency over valid responses across all question types. Accuracy counts skips and final failures as incorrect, whereas ECE and Brier_{raw} are computed only from valid classification responses. Bold and underline indicate the best and second-best distinct displayed values, respectively, across all 25 configurations.

Mean absolute error. For the valid **score** responses $\mathcal{S}_m$, we compute

$$\text{MAE}_m = \frac{1}{|\mathcal{S}_m|} \sum_{i \in \mathcal{S}_m} |\hat{s}_{mi} - s_i|, \quad (13)$$

where $\hat{s}_{mi}$ is the returned score and s_i is its reference value. Each question retains its defined rating scale, and

invalid outputs are excluded rather than imputed.

D.1 General knowledge and reasoning

Table 10 General knowledge and reasoning: model band 1/3. ACC (%) $\uparrow$; benchmark-run latency (s), ECE, and Brier_{raw} $\downarrow$. Rankings span all bands.

N: non-thinking; T: thinking. Parentheses specify reasoning effort.								
Metric	Jev	InnerJev-4B (ours)	InnerJev-27B (ours)	kev-0.8b	kev-4b	kev-9b	kev-27b	Intern- Decision 0.8B
aida-testc								
								$N = 200$
ACC $\uparrow$	95.00	86.00	97.50	38.00	80.00	96.00	98.50	6.50
Latency $\downarrow$	2.11	<u>5.80</u>	14.02	18.15	57.79	94.05	55.95	6.38
ECE $\downarrow$	0.009	0.043	0.012	0.169	0.532	0.302	0.077	0.051
Brier _{raw} $\downarrow$	0.032	0.193	0.040	0.807	0.600	0.191	0.037	0.980
ai2-arc-challenge								
								$N = 200$
ACC $\uparrow$	98.00	90.50	96.00	57.00	92.50	94.00	96.00	60.00
Latency $\downarrow$	0.87	5.59	12.41	18.10	56.66	94.98	56.51	6.38
ECE $\downarrow$	<u>0.008</u>	0.042	0.024	0.066	0.135	0.123	0.058	0.113
Brier _{raw} $\downarrow$	0.034	0.125	0.063	0.571	0.157	0.135	0.082	0.548
ai2-arc-easy								
								$N = 200$
ACC $\uparrow$	99.00	96.50	99.00	80.00	97.00	97.00	98.00	74.00
Latency $\downarrow$	0.83	5.64	12.53	18.07	57.07	95.13	56.22	6.38
ECE $\downarrow$	0.011	0.025	0.027	0.212	0.099	0.079	0.050	0.200
Brier _{raw} $\downarrow$	<u>0.008</u>	0.055	0.020	0.363	0.081	0.062	0.041	0.402
boolq								
								$N = 200$
ACC $\uparrow$	86.00	83.50	86.50	79.00	85.50	86.00	90.50	77.00
Latency $\downarrow$	1.04	5.64	12.25	17.94	56.75	95.62	55.36	6.37
ECE $\downarrow$	0.036	0.071	0.048	0.077	<u>0.026</u>	0.054	0.018	0.089
Brier _{raw} $\downarrow$	0.178	0.243	<u>0.164</u>	0.309	0.217	0.199	0.159	0.349
clinc150								
								$N = 200$
ACC $\uparrow$	86.50	82.00	89.00	49.00	72.50	75.50	75.00	11.00
Latency $\downarrow$	1.19	<u>5.86</u>	14.07	17.80	56.53	95.12	55.04	6.31
ECE $\downarrow$	0.056	<u>0.036</u>	0.046	0.138	0.318	0.257	0.062	0.080
Brier _{raw} $\downarrow$	0.196	0.263	0.162	0.647	0.519	0.443	0.340	0.947
fever								
								$N = 200$
ACC $\uparrow$	<u>96.50</u>	95.50	97.50	93.00	95.00	94.00	96.00	91.50
Latency $\downarrow$	0.73	5.72	12.57	17.11	56.27	94.34	55.11	6.39
ECE $\downarrow$	0.013	0.027	<u>0.017</u>	0.029	0.035	0.045	0.028	0.165
Brier _{raw} $\downarrow$	<u>0.058</u>	0.082	0.052	0.100	0.092	0.093	0.064	0.183
gpqa								
								$N = 198$
ACC $\uparrow$	73.74	39.90	51.52	27.27	38.38	37.37	46.46	31.82
Latency $\downarrow$	0.99	5.63	12.41	16.75	57.11	91.93	53.96	6.32
ECE $\downarrow$	<u>0.057</u>	0.112	0.084	0.086	0.072	0.099	0.071	0.091
Brier _{raw} $\downarrow$	0.367	0.711	0.598	0.752	0.685	0.703	0.618	0.757
go-emotions								
								$N = 200$
ACC $\uparrow$	31.00	33.00	37.00	31.50	30.50	31.00	<u>39.00</u>	16.00
Latency $\downarrow$	1.18	5.62	12.09	16.83	56.71	93.05	54.60	6.44
ECE $\downarrow$	0.306	0.287	0.219	0.102	0.045	0.102	0.057	<u>0.048</u>
Brier _{raw} $\downarrow$	0.951	0.902	0.828	0.851	0.827	<u>0.816</u>	0.755	0.934
mmlu-pro								
								$N = 200$
ACC $\uparrow$	82.00	46.50	65.00	23.50	52.00	51.00	63.50	30.00
Latency $\downarrow$	0.87	5.64	13.23	19.92	65.21	108.57	62.75	8.22
ECE $\downarrow$	0.096	0.110	<u>0.040</u>	0.032	0.084	0.071	0.094	0.054
Brier _{raw} $\downarrow$	0.309	0.647	0.462	0.839	0.624	0.611	0.493	0.824
nlu-eval-68								
								$N = 200$
ACC $\uparrow$	80.50	80.50	87.50	65.00	70.50	72.00	78.00	21.50
Latency $\downarrow$	0.79	5.89	12.67	19.86	64.93	106.50	62.35	8.16
ECE $\downarrow$	0.069	<u>0.032</u>	0.041	0.174	0.296	0.166	0.082	0.036
Brier _{raw} $\downarrow$	0.270	0.284	0.184	0.511	0.521	0.432	0.317	0.870
scifact								
								$N = 200$
ACC $\uparrow$	90.00	87.00	91.50	81.00	<u>92.50</u>	90.00	93.00	76.00
Latency $\downarrow$	0.66	5.73	13.01	19.46	62.24	104.88	61.06	7.42
ECE $\downarrow$	0.054	0.070	0.036	0.097	0.039	0.045	<u>0.025</u>	0.089
Brier _{raw} $\downarrow$	0.156	0.202	0.153	0.244	<u>0.130</u>	0.138	0.101	0.327
c-eval								
								$N = 200$
ACC $\uparrow$	84.50	69.00	76.50	33.00	65.00	66.50	71.00	39.00
Latency $\downarrow$	0.87	<u>5.60</u>	12.51	18.78	60.90	101.54	58.53	7.27
ECE $\downarrow$	<u>0.040</u>	0.078	0.051	0.127	0.068	0.084	0.038	0.082
Brier _{raw} $\downarrow$	0.234	0.418	0.314	0.708	0.470	0.396	0.372	0.689

Continued on next page

Table 10 continued; model band 1/3

N: non-thinking; T: thinking. Parentheses specify reasoning effort.

Metric	Jev	InnerJev-4B (ours)	InnerJev-27B (ours)	kev-0.8b	kev-4b	kev-9b	kev-27b	Intern- Decision 0.8B
cmmlu $N = 200$								
ACC $\uparrow$	89.00	74.00	88.00	42.00	67.00	72.50	78.50	43.50
Latency $\downarrow$	0.85	5.59	12.55	18.74	59.42	100.10	58.53	7.21
ECE $\downarrow$	<u>0.032</u>	0.058	0.059	0.107	0.087	0.074	0.125	0.094
Brier _{raw} $\downarrow$	0.173	0.331	0.189	0.669	0.426	0.380	0.313	0.654
Domain summary $N = 2,598$								
ACC $\uparrow$	83.99	74.17	81.76	53.81	72.21	74.10	78.75	44.46
Latency $\downarrow$	1.00	<u>5.69</u>	12.79	18.27	59.05	98.14	57.38	6.87
ECE $\downarrow$	<u>0.030</u>	0.060	0.023	0.047	0.108	0.061	0.041	0.054
Brier _{raw} $\downarrow$	<u>0.228</u>	0.342	0.248	0.567	0.411	0.353	0.284	0.651

Table 11 General knowledge and reasoning: model band 2/3. ACC (%) $\uparrow$; benchmark-run latency (s), ECE, and Brier_{raw} $\downarrow$. Rankings span all bands.

N: non-thinking; T: thinking. Parentheses specify reasoning effort.

Metric	Intern- Decision 2B	Intern- Decision 4B	laya	openjev-4b-v5	nimble-9b	nanojev	lightjev	jevforge-0.8b
aida-testc $N = 200$								
ACC $\uparrow$	13.00	83.50	19.00	87.50	94.00	1.50	42.00	16.00
Latency $\downarrow$	6.32	10.07	12.40	229.93	23.87	40.14	266.04	216.20
ECE $\downarrow$	0.203	0.314	0.625	0.542	0.046	0.001	0.406	0.125
Brier _{raw} $\downarrow$	1.004	0.376	1.387	0.521	0.094	1.000	0.980	0.977
ai2-arc-challenge $N = 200$								
ACC $\uparrow$	75.50	93.00	33.50	93.50	96.50	27.00	29.00	30.50
Latency $\downarrow$	6.43	9.82	12.88	233.40	23.72	37.46	16.97	221.52
ECE $\downarrow$	0.065	0.108	0.150	0.152	0.021	0.080	0.137	0.067
Brier _{raw} $\downarrow$	0.359	0.128	0.816	0.155	0.065	0.766	0.784	0.748
ai2-arc-easy $N = 200$								
ACC $\uparrow$	91.00	94.50	37.50	98.50	97.00	30.50	48.00	40.50
Latency $\downarrow$	6.30	9.91	12.35	228.80	23.79	39.24	17.44	220.54
ECE $\downarrow$	0.154	0.104	0.100	0.131	0.020	0.062	0.100	0.160
Brier _{raw} $\downarrow$	0.162	0.090	0.740	0.084	0.053	0.752	0.663	0.699
boolq $N = 200$								
ACC $\uparrow$	77.00	<u>87.00</u>	79.50	82.50	85.50	61.00	58.00	40.00
Latency $\downarrow$	6.30	9.74	12.50	234.73	24.03	38.43	31.72	221.18
ECE $\downarrow$	0.075	0.035	0.058	0.090	0.070	0.210	0.066	0.527
Brier _{raw} $\downarrow$	0.284	0.205	0.309	0.256	0.216	0.555	0.478	1.033
clinc150 $N = 200$								
ACC $\uparrow$	16.00	73.50	39.00	74.50	83.00	0.50	39.50	5.00
Latency $\downarrow$	6.29	9.81	12.51	227.32	24.14	39.48	17.89	217.82
ECE $\downarrow$	0.179	0.221	0.556	0.413	0.104	0.012	0.310	0.074
Brier _{raw} $\downarrow$	0.948	0.467	1.125	0.619	0.275	0.994	0.904	0.984
fever $N = 200$								
ACC $\uparrow$	92.00	94.50	78.50	95.00	95.50	52.50	62.50	48.50
Latency $\downarrow$	6.31	9.51	11.53	234.55	23.51	38.56	170.98	220.62
ECE $\downarrow$	0.062	0.019	0.049	0.037	0.036	0.268	0.085	0.475
Brier _{raw} $\downarrow$	0.120	0.091	0.285	0.088	0.081	0.656	0.448	0.944
gpqa $N = 198$								
ACC $\uparrow$	30.30	40.40	23.23	55.05	40.40	21.72	22.73	29.80
Latency $\downarrow$	5.86	9.79	11.35	235.77	23.43	36.56	38.86	222.54
ECE $\downarrow$	0.163	0.122	0.169	0.167	0.208	0.156	0.081	0.044
Brier _{raw} $\downarrow$	0.771	0.719	0.816	0.620	0.740	0.821	0.773	0.764
go-emotions $N = 200$								
ACC $\uparrow$	15.50	37.50	45.00	32.00	29.00	5.00	<u>39.00</u>	30.00
Latency $\downarrow$	5.97	9.44	10.83	237.41	23.39	38.31	18.03	221.61
ECE $\downarrow$	0.111	0.186	0.458	0.130	0.389	0.052	0.106	0.099
Brier _{raw} $\downarrow$	0.931	0.840	0.971	0.860	1.022	0.948	0.830	0.908
mmlu-pro $N = 200$								
ACC $\uparrow$	34.50	48.50	15.00	52.00	55.00	11.00	14.00	11.00
Latency $\downarrow$	8.07	12.31	11.60	226.92	26.87	41.16	30.59	214.15
ECE $\downarrow$	0.099	0.105	0.181	0.180	0.127	0.078	0.069	0.054
Brier _{raw} $\downarrow$	0.745	0.621	0.967	0.669	0.598	0.922	0.910	0.901
nlu-eval-68 $N = 200$								
ACC $\uparrow$	26.50	75.00	27.00	70.00	72.00	6.00	36.50	2.00
Latency $\downarrow$	8.07	12.34	11.98	228.39	26.50	38.16	18.50	209.01
ECE $\downarrow$	0.123	0.144	0.667	0.258	0.141	0.028	0.238	0.152
Brier _{raw} $\downarrow$	0.819	0.369	1.363	0.570	0.420	0.982	0.890	1.008

Continued on next page

Table 11 continued; model band 2/3

N: non-thinking; T: thinking. Parentheses specify reasoning effort.

Metric	Intern- Decision 2B	Intern- Decision 4B	laya	openjev-4b-v5	nimble-9b	nanojev	lightjev	jevforge-0.8b
scifact								$N = 200$
ACC $\uparrow$	87.00	89.50	73.00	89.00	92.00	64.50	65.50	35.00
Latency $\downarrow$	7.68	11.75	10.97	227.67	26.11	39.37	203.38	216.50
ECE $\downarrow$	0.053	0.021	0.060	0.057	0.047	0.149	0.053	0.615
Brier _{raw} $\downarrow$	0.191	0.168	0.400	0.164	0.138	0.505	0.445	1.209
c-eval								$N = 200$
ACC $\uparrow$	52.00	66.50	18.50	66.00	76.50	29.00	27.50	28.50
Latency $\downarrow$	6.95	11.31	10.43	228.40	24.85	39.64	18.34	224.10
ECE $\downarrow$	0.080	0.070	0.230	0.101	0.102	0.064	0.124	0.079
Brier _{raw} $\downarrow$	0.585	0.443	0.842	0.494	0.342	0.770	0.806	0.754
cmmlu								$N = 200$
ACC $\uparrow$	58.50	74.00	20.00	73.00	83.00	31.00	36.50	26.00
Latency $\downarrow$	6.98	11.46	9.45	227.39	24.49	38.97	17.55	223.96
ECE $\downarrow$	0.046	0.079	0.252	0.158	0.064	0.035	0.100	0.126
Brier _{raw} $\downarrow$	0.551	0.361	0.839	0.419	0.244	0.741	0.753	0.768
Domain summary								$N = 2,598$
ACC $\uparrow$	51.46	73.67	39.15	74.52	76.91	26.25	40.07	26.37
Latency $\downarrow$	6.73	10.56	11.62	230.82	24.52	38.89	66.66	219.21
ECE $\downarrow$	0.075	0.042	0.263	0.166	0.097	0.082	0.109	0.152
Brier _{raw} $\downarrow$	0.574	0.375	0.835	0.424	0.330	0.801	0.743	0.900

Table 12 General knowledge and reasoning: model band 3/3. ACC (%) $\uparrow$; benchmark-run latency (s), ECE, and Brier_{raw} $\downarrow$. Rankings span all bands.

N: non-thinking; T: thinking. Parentheses specify reasoning effort.

Metric	Qwen3.5-4B N	Qwen3.8- 27B N	DeepSeek V4-Flash N	GPT-6-Sol N	GPT-6- Luna N	Qwen3.5-4B T	Qwen3.8- 27B T (xhigh)	DeepSeek V4-Flash T (high)	GLM-5.3- Flash T (high)
aida-testc									$N = 200$
ACC $\uparrow$	77.00	96.50	91.00	99.50	96.50	89.00	97.50	98.50	<u>99.00</u>
Latency $\downarrow$	25.21	90.29	54.52	45.27	13.75	74.46	248.88	206.46	211.86
ECE $\downarrow$	0.204	0.024	0.087	<u>0.005</u>	0.035	0.035	0.015	0.020	0.008
Brier _{raw} $\downarrow$	4.291	0.064	0.180	0.010	0.070	0.099	0.028	0.035	<u>0.022</u>
ai2-arc-challenge									$N = 200$
ACC $\uparrow$	83.50	93.00	94.00	97.50	99.00	95.00	<u>98.50</u>	97.00	98.00
Latency $\downarrow$	<u>1.66</u>	3.18	3.30	10.20	8.85	43.20	96.70	61.12	8.76
ECE $\downarrow$	0.101	0.033	0.037	0.018	0.005	0.050	0.010	0.027	0.012
Brier _{raw} $\downarrow$	0.259	0.119	0.110	0.049	0.021	0.103	<u>0.032</u>	0.061	0.038
ai2-arc-easy									$N = 200$
ACC $\uparrow$	91.00	97.00	98.00	<u>99.50</u>	100.00	96.50	<u>99.50</u>	99.00	<u>99.50</u>
Latency $\downarrow$	<u>2.42</u>	3.12	3.84	10.60	18.95	42.54	82.27	58.78	8.10
ECE $\downarrow$	0.065	0.013	0.014	0.001	0.008	0.013	0.016	<u>0.004</u>	0.012
Brier _{raw} $\downarrow$	0.189	0.061	0.031	0.010	0.001	0.039	0.011	0.015	0.010
boolq									$N = 200$
ACC $\uparrow$	68.00	82.00	85.00	79.50	84.50	73.00	83.00	82.00	82.50
Latency $\downarrow$	2.03	3.38	3.00	8.27	4.87	40.08	89.15	51.43	4.93
ECE $\downarrow$	0.256	0.163	0.145	0.189	0.143	0.265	0.150	0.178	0.134
Brier _{raw} $\downarrow$	0.507	0.338	0.289	0.386	0.300	0.529	0.315	0.357	0.297
clinc150									$N = 200$
ACC $\uparrow$	60.00	83.00	73.50	92.50	<u>92.00</u>	81.50	88.50	88.00	85.00
Latency $\downarrow$	12.75	45.02	26.49	22.94	16.48	38.21	199.95	143.83	108.35
ECE $\downarrow$	0.290	0.124	0.225	0.057	0.069	0.160	0.088	0.120	0.070
Brier _{raw} $\downarrow$	2.188	0.299	0.499	0.124	<u>0.150</u>	1.136	0.212	0.243	0.244
fever									$N = 200$
ACC $\uparrow$	81.50	95.50	93.00	89.00	<u>96.50</u>	75.50	85.50	91.00	90.50
Latency $\downarrow$	2.22	3.65	3.10	8.35	5.43	45.34	72.98	44.96	4.20
ECE $\downarrow$	0.176	0.038	0.063	0.098	0.031	0.217	0.137	0.092	0.087
Brier _{raw} $\downarrow$	0.354	0.087	0.130	0.199	0.069	0.459	0.285	0.182	0.179
gpqa									$N = 198$
ACC $\uparrow$	31.31	46.46	44.95	<u>83.84</u>	79.29	60.10	80.30	73.23	85.35
Latency $\downarrow$	<u>1.67</u>	3.44	3.28	32.59	6.01	47.53	147.71	127.61	59.33
ECE $\downarrow$	0.483	0.458	0.481	0.102	0.071	0.216	0.108	0.204	0.064
Brier _{raw} $\downarrow$	1.111	0.965	0.997	0.251	0.167	0.640	0.312	0.455	<u>0.237</u>
go-emotions									$N = 200$
ACC $\uparrow$	30.50	27.00	23.00	28.00	34.50	37.00	35.00	34.50	28.50
Latency $\downarrow$	<u>2.94</u>	9.65	7.37	14.98	4.54	43.80	113.00	85.40	28.83
ECE $\downarrow$	0.478	0.559	0.587	0.487	0.395	0.428	0.405	0.605	0.324
Brier _{raw} $\downarrow$	1.155	1.208	1.268	1.080	0.978	1.064	0.999	1.248	0.923

Continued on next page

Table 12 continued; model band 3/3

N: non-thinking; T: thinking. Parentheses specify reasoning effort.

Metric	Qwen3.5-4B N	Qwen3.8- 27B N	DeepSeek V4-Flash N	GPT-6-Sol N	GPT-6- Luna N	Qwen3.5-4B T	Qwen3.8- 27B T (xhigh)	DeepSeek V4-Flash T (high)	GLM-5.3- Flash T (high)	
mmlu-pro										$N = 200$
ACC $\uparrow$	38.00	56.50	62.50	<u>83.50</u>	84.50	71.50	83.00	82.50		82.50
Latency $\downarrow$	2.36	4.50	5.22	17.79	6.15	43.61	118.42	85.76		24.79
ECE $\downarrow$	0.466	0.358	0.327	0.122	0.102	0.206	0.105	0.158		0.105
Brier _{raw} $\downarrow$	1.080	0.788	0.708	0.296	0.271	0.552	<u>0.284</u>	0.337		0.300
nlu-eval-68										$N = 200$
ACC $\uparrow$	55.00	75.00	72.00	81.50	<u>84.50</u>	75.00	82.50	81.50		81.50
Latency $\downarrow$	<u>5.69</u>	21.74	14.30	17.40	11.17	23.14	142.89	117.57		60.05
ECE $\downarrow$	0.308	0.201	0.248	0.143	0.112	0.157	0.139	0.182		0.081
Brier _{raw} $\downarrow$	0.784	0.446	0.535	0.323	<u>0.266</u>	0.402	0.318	0.364		0.280
scifact										$N = 200$
ACC $\uparrow$	78.50	85.00	85.00	88.50	86.00	82.00	86.50	88.50		85.00
Latency $\downarrow$	<u>0.85</u>	2.21	2.61	6.69	9.52	42.52	97.76	39.25		6.01
ECE $\downarrow$	0.154	0.122	0.121	0.100	0.123	0.145	0.089	0.110		0.106
Brier _{raw} $\downarrow$	0.299	0.272	0.271	0.215	0.250	0.337	0.226	0.225		0.262
c-eval										$N = 200$
ACC $\uparrow$	62.00	74.50	83.00	85.50	87.50	77.00	87.00	91.50		<u>90.00</u>
Latency $\downarrow$	8.16	15.26	97.52	11.31	16.79	42.36	114.36	153.86		17.76
ECE $\downarrow$	0.318	0.199	0.133	0.107	0.092	0.143	0.110	0.075		0.044
Brier _{raw} $\downarrow$	0.706	0.478	0.317	0.248	<u>0.209</u>	0.401	0.241	0.166		0.166
cmmlu										$N = 200$
ACC $\uparrow$	63.50	81.00	83.50	90.50	90.00	79.50	89.50	<u>91.00</u>		94.00
Latency $\downarrow$	8.16	15.30	97.45	13.54	<u>5.16</u>	42.08	117.25	165.39		15.75
ECE $\downarrow$	0.248	0.116	0.120	0.066	0.063	0.094	0.057	0.085		0.026
Brier _{raw} $\downarrow$	0.649	0.343	0.296	0.166	<u>0.157</u>	0.337	0.178	0.179		0.121
Domain summary										$N = 2,598$
ACC $\uparrow$	63.09	76.37	76.06	84.53	85.76	76.37	84.33	84.49		84.72
Latency $\downarrow$	5.88	17.01	24.79	16.90	9.86	43.76	126.24	103.17		42.97
ECE $\downarrow$	0.254	0.177	0.189	0.110	0.093	0.151	0.099	0.141		0.071
Brier _{raw} $\downarrow$	1.046	0.420	0.433	0.257	0.224	0.469	0.265	0.297		0.237

D.2 Medicine

Table 13 Medicine: model band 1/3. ACC (%) $\uparrow$; benchmark-run latency (s), ECE, and Brier_{raw} $\downarrow$. Rankings span all bands.

N: non-thinking; T: thinking. Parentheses specify reasoning effort.

Metric	Jev	InnerJev-4B (ours)	InnerJev-27B (ours)	kev-0.8b	kev-4b	kev-9b	kev-27b	Intern- Decision 0.8B	
cmb									$N = 200$
ACC $\uparrow$	84.50	67.50	80.50	33.50	63.50	68.00	76.00		36.50
Latency $\downarrow$	0.82	5.63	12.45	17.62	57.19	94.69	54.87		6.35
ECE $\downarrow$	0.056	0.095	<u>0.048</u>	0.066	0.076	0.104	0.145		0.073
Brier _{raw} $\downarrow$	0.198	0.425	0.269	0.740	0.484	0.424	0.366		0.714
cmexam									$N = 200$
ACC $\uparrow$	86.50	75.00	84.00	41.00	66.00	68.50	81.00		40.50
Latency $\downarrow$	0.65	5.64	12.38	17.51	56.83	93.76	55.06		6.37
ECE $\downarrow$	<u>0.044</u>	<u>0.044</u>	0.062	0.030	0.079	0.102	0.159		0.064
Brier _{raw} $\downarrow$	0.197	0.345	0.232	0.699	0.441	0.426	0.317		0.690
med-qa-en									$N = 200$
ACC $\uparrow$	88.00	67.00	85.50	42.00	62.50	68.50	82.50		42.00
Latency $\downarrow$	0.53	5.64	13.27	20.16	64.84	111.13	63.63		8.42
ECE $\downarrow$	<u>0.020</u>	0.094	0.044	0.080	0.063	0.064	0.078		0.067
Brier _{raw} $\downarrow$	0.182	0.457	0.233	0.720	0.479	0.433	0.248		0.686
med-qa									$N = 200$
ACC $\uparrow$	92.50	79.50	91.00	41.00	76.50	82.50	86.00		53.50
Latency $\downarrow$	0.56	5.66	12.51	20.42	64.81	112.64	64.89		8.48
ECE $\downarrow$	0.028	0.071	0.057	0.065	0.105	0.152	0.126		0.112
Brier _{raw} $\downarrow$	0.094	0.260	0.129	0.671	0.324	0.281	0.207		0.613
Domain summary									$N = 800$
ACC $\uparrow$	87.88	72.25	85.25	39.38	67.13	71.88	81.38		43.13
Latency $\downarrow$	0.64	5.64	12.66	18.93	60.92	103.06	59.61		7.40
ECE $\downarrow$	0.018	0.061	0.038	0.036	0.046	0.094	0.123		0.033
Brier _{raw} $\downarrow$	0.167	0.372	0.216	0.708	0.432	0.391	0.284		0.676

Table 14 Medicine: model band 2/3. ACC (%) ↑; benchmark-run latency (s), ECE, and Brier_{raw} ↓. Rankings span all bands.

N: non-thinking; T: thinking. Parentheses specify reasoning effort.

Metric	Intern- Decision 2B	Intern- Decision 4B	laya	openjev-4b-v5	nimble-9b	nanojev	lightjev	jevforge-0.8b
cmb								<i>N</i> = 200
ACC ↑	45.00	67.00	26.00	63.00	75.50	20.00	33.50	24.00
Latency ↓	6.35	9.72	12.62	233.39	23.79	39.32	19.22	221.44
ECE ↓	0.112	0.056	0.133	0.140	0.112	0.092	0.090	0.078
Brier _{raw} ↓	0.672	0.460	0.838	0.529	0.342	0.816	0.814	0.810
cmexam								<i>N</i> = 200
ACC ↑	56.00	67.00	20.50	66.50	81.50	16.50	16.50	19.50
Latency ↓	6.31	9.77	11.89	229.51	23.44	39.29	16.92	217.34
ECE ↓	0.066	0.081	0.189	0.107	<u>0.044</u>	0.133	0.247	0.121
Brier _{raw} ↓	0.551	0.398	0.889	0.443	0.299	0.842	0.872	0.820
med-qa-en								<i>N</i> = 200
ACC ↑	51.50	63.50	26.00	64.00	73.00	27.50	28.50	27.50
Latency ↓	8.21	12.46	11.59	216.20	27.02	37.72	80.83	209.76
ECE ↓	0.096	0.063	0.236	0.051	0.089	0.140	0.124	0.094
Brier _{raw} ↓	0.621	0.482	0.851	0.489	0.401	0.813	0.799	0.772
med-qa								<i>N</i> = 200
ACC ↑	63.00	79.50	28.50	76.00	84.50	28.00	34.50	32.00
Latency ↓	8.28	12.88	10.80	204.34	27.38	40.75	17.21	205.13
ECE ↓	0.054	0.099	0.168	0.106	0.067	0.084	0.119	0.053
Brier _{raw} ↓	0.481	0.279	0.811	0.371	0.219	0.773	0.770	0.746
Domain summary								<i>N</i> = 800
ACC ↑	53.88	69.25	25.25	67.38	78.63	23.00	28.25	25.75
Latency ↓	7.29	11.21	11.72	220.86	25.41	39.27	33.54	213.42
ECE ↓	0.052	0.043	0.176	0.084	0.070	0.110	0.141	0.083
Brier _{raw} ↓	0.581	0.405	0.847	0.458	0.315	0.811	0.814	0.787

Table 15 Medicine: model band 3/3. ACC (%) ↑; benchmark-run latency (s), ECE, and Brier_{raw} ↓. Rankings span all bands.

N: non-thinking; T: thinking. Parentheses specify reasoning effort.

Metric	Qwen3.5-4B N	Qwen3.8- 27B N	DeepSeek V4-Flash N	GPT-6-Sol N	GPT-6- Luna N	Qwen3.5-4B T	Qwen3.8- 27B T (xhigh)	DeepSeek V4-Flash T (high)	GLM-5.3- Flash T (high)
cmb									<i>N</i> = 200
ACC ↑	60.50	72.50	82.00	<u>90.50</u>	<u>90.50</u>	67.00	87.00	<u>90.50</u>	91.50
Latency ↓	<u>0.98</u>	3.09	3.23	12.59	5.63	30.37	96.36	63.81	11.64
ECE ↓	0.273	0.184	0.106	0.077	0.075	0.170	0.066	0.088	0.031
Brier _{raw} ↓	0.698	0.496	0.295	0.181	<u>0.179</u>	0.488	0.216	0.180	0.142
cmexam									<i>N</i> = 200
ACC ↑	64.50	74.50	82.00	90.50	<u>93.00</u>	78.50	88.00	91.00	95.00
Latency ↓	<u>2.45</u>	4.81	4.10	12.47	5.16	32.01	93.65	60.85	11.32
ECE ↓	0.253	0.182	0.135	0.055	<u>0.044</u>	0.122	0.072	0.080	0.048
Brier _{raw} ↓	0.616	0.456	0.331	0.140	<u>0.119</u>	0.368	0.217	0.176	0.086
med-qa-en									<i>N</i> = 200
ACC ↑	56.50	80.00	76.50	96.00	<u>95.00</u>	82.00	96.00	93.50	92.50
Latency ↓	<u>2.06</u>	4.45	3.78	12.79	6.21	47.13	114.12	66.49	11.15
ECE ↓	0.310	0.121	0.162	0.023	0.028	0.075	0.033	0.036	0.011
Brier _{raw} ↓	0.735	0.354	0.415	0.072	0.087	0.233	<u>0.074</u>	0.112	0.124
med-qa									<i>N</i> = 200
ACC ↑	76.50	88.00	88.50	93.50	93.50	81.50	<u>94.50</u>	93.00	98.00
Latency ↓	<u>0.95</u>	2.84	3.12	11.12	8.39	29.85	80.45	57.26	10.03
ECE ↓	0.141	0.085	0.059	0.050	0.048	0.071	<u>0.033</u>	0.062	0.028
Brier _{raw} ↓	0.406	0.229	0.203	0.125	0.118	0.255	<u>0.084</u>	0.131	0.040
Domain summary									<i>N</i> = 800
ACC ↑	64.50	78.75	82.25	92.63	<u>93.00</u>	77.25	91.38	92.00	94.25
Latency ↓	<u>1.61</u>	3.80	3.56	12.24	6.35	34.84	96.14	62.10	11.03
ECE ↓	0.229	0.129	0.113	0.050	0.048	0.085	0.028	0.065	<u>0.021</u>
Brier _{raw} ↓	0.615	0.384	0.311	0.130	<u>0.126</u>	0.336	0.147	0.150	0.098

D.3 Law

Table 16 Law: model band 1/3. ACC (%) $\uparrow$; benchmark-run latency (s), ECE, and Brier_{raw} $\downarrow$. Rankings span all bands.

N: non-thinking; T: thinking. Parentheses specify reasoning effort.

Metric	Jev	InnerJev-4B (ours)	InnerJev-27B (ours)	kev-0.8b	kev-4b	kev-9b	kev-27b	Intern- Decision 0.8B
cuad								$N = 200$
ACC $\uparrow$	55.50	81.00	83.50	77.00	83.50	83.00	81.50	69.00
Latency $\downarrow$	1.27	6.08	14.37	17.70	58.57	95.20	56.49	6.58
ECE $\downarrow$	0.035	0.046	0.052	0.181	0.123	0.079	<u>0.030</u>	0.143
Brier _{raw} $\downarrow$	0.122	0.153	0.108	0.283	0.159	0.146	<u>0.118</u>	0.350
disc-law-multi								$N = 200$
ACC $\uparrow$	47.50	19.00	45.00	5.00	12.00	28.50	42.50	2.50
Latency $\downarrow$	0.89	5.64	12.24	16.68	57.83	93.77	54.37	6.38
ECE $\downarrow$	0.089	0.164	0.068	0.080	0.058	0.074	0.160	0.102
Brier _{raw} $\downarrow$	0.675	0.903	0.679	0.944	0.910	0.842	0.764	0.959
disc-law-single								$N = 200$
ACC $\uparrow$	83.50	72.00	85.50	39.00	64.50	68.00	77.50	42.00
Latency $\downarrow$	0.85	5.62	12.39	17.05	57.27	94.14	54.37	6.33
ECE $\downarrow$	<u>0.034</u>	0.084	0.098	0.067	0.076	0.072	0.140	0.059
Brier _{raw} $\downarrow$	0.240	0.383	0.245	0.716	0.483	0.431	0.325	0.676
lex-glue-case-hold								$N = 200$
ACC $\uparrow$	78.50	74.50	83.00	62.00	70.50	70.00	78.50	64.00
Latency $\downarrow$	0.76	5.66	12.39	17.92	56.81	95.64	56.66	6.96
ECE $\downarrow$	0.064	0.093	0.059	0.132	0.117	0.076	0.079	0.102
Brier _{raw} $\downarrow$	0.296	0.352	0.255	0.523	0.426	0.396	<u>0.283</u>	0.510
lex-glue-ecthr-a								$N = 200$
ACC $\uparrow$	71.50	65.50	75.00	50.00	60.50	67.00	68.50	53.50
Latency $\downarrow$	0.66	5.91	14.43	17.82	57.03	96.04	57.06	7.10
ECE $\downarrow$	0.057	0.146	0.077	0.121	0.119	0.169	<u>0.068</u>	0.236
Brier _{raw} $\downarrow$	0.403	0.513	0.381	0.668	0.550	0.514	0.428	0.734
lex-glue-ecthr-b								$N = 200$
ACC $\uparrow$	91.00	87.00	93.00	66.00	79.50	86.00	90.00	69.00
Latency $\downarrow$	0.59	5.87	14.20	17.78	56.96	96.28	57.48	7.23
ECE $\downarrow$	0.093	0.054	0.082	0.248	0.221	0.149	0.177	0.407
Brier _{raw} $\downarrow$	0.166	0.181	0.122	0.571	0.330	0.217	0.200	0.634
lex-glue-ledgar								$N = 200$
ACC $\uparrow$	74.00	68.00	73.50	56.00	63.00	63.00	72.50	13.00
Latency $\downarrow$	0.84	<u>5.77</u>	14.39	17.84	56.86	96.07	57.73	7.10
ECE $\downarrow$	0.148	0.143	0.093	0.239	0.211	0.122	0.113	0.074
Brier _{raw} $\downarrow$	0.422	0.447	0.367	0.676	0.543	0.520	<u>0.395</u>	0.957
lex-glue-scotus								$N = 200$
ACC $\uparrow$	61.00	62.50	55.00	46.50	52.50	60.00	61.00	51.50
Latency $\downarrow$	0.56	6.34	15.29	19.10	60.36	101.50	61.48	7.66
ECE $\downarrow$	0.193	0.157	0.147	0.137	0.126	0.125	0.092	0.362
Brier _{raw} $\downarrow$	<u>0.424</u>	0.454	0.498	0.587	0.527	0.451	0.405	0.726
lex-glue-unfair-tos								$N = 200$
ACC $\uparrow$	79.00	71.00	80.50	75.00	85.50	82.00	89.00	72.50
Latency $\downarrow$	0.53	5.63	12.08	18.52	58.73	99.24	58.40	7.38
ECE $\downarrow$	0.124	0.145	0.058	0.275	0.213	0.196	0.073	0.434
Brier _{raw} $\downarrow$	0.318	0.449	0.306	0.493	0.293	0.326	0.156	0.619
Domain summary								$N = 1, 800$
ACC $\uparrow$	71.28	66.72	74.89	52.94	63.50	67.50	73.44	48.56
Latency $\downarrow$	0.75	5.82	13.49	17.80	57.77	96.36	57.05	6.96
ECE $\downarrow$	0.049	0.082	0.031	0.130	0.121	0.110	0.087	0.181
Brier _{raw} $\downarrow$	0.349	0.429	0.329	0.611	0.472	0.430	<u>0.343</u>	0.689

Table 17 Law: model band 2/3. ACC (%) $\uparrow$; benchmark-run latency (s), ECE, and Brier_{raw} $\downarrow$. Rankings span all bands.

N: non-thinking; T: thinking. Parentheses specify reasoning effort.

Metric	Intern- Decision 2B	Intern- Decision 4B	laya	openjev-4b-v5	nimble-9b	nanojev	lightjev	jevforge-0.8b
cuad								$N = 200$
ACC $\uparrow$	75.00	81.00	18.50	88.00	83.00	22.00	22.00	67.50
Latency $\downarrow$	6.55	9.77	12.22	232.10	24.25	70.99	229.12	217.61
ECE $\downarrow$	0.035	0.019	0.395	0.125	0.036	0.518	0.340	0.182
Brier _{raw} $\downarrow$	0.233	0.158	0.731	0.246	0.133	0.912	0.606	0.433
disc-law-multi								$N = 200$
ACC $\uparrow$	5.00	18.00	1.50	18.50	33.50	9.00	15.00	0.00
Latency $\downarrow$	6.32	9.34	11.47	229.32	23.61	36.99	39.31	218.35
ECE $\downarrow$	0.154	0.127	0.615	<u>0.056</u>	0.209	0.042	0.080	0.151
Brier _{raw} $\downarrow$	0.995	0.919	1.449	0.887	0.862	0.930	0.928	0.965

Continued on next page

Table 17 continued; model band 2/3

N: non-thinking; T: thinking. Parentheses specify reasoning effort.

Metric	Intern- Decision 2B	Intern- Decision 4B	laya	openjev-4b-v5	nimble-9b	nanojev	lightjev	jevforge-0.8b
disc-law-single								$N = 200$
ACC $\uparrow$	53.50	67.50	22.00	62.50	79.50	27.50	26.50	27.50
Latency $\downarrow$	6.26	9.52	11.20	234.81	23.50	37.74	18.72	216.69
ECE $\downarrow$	0.097	0.089	0.172	0.081	0.051	0.083	0.129	0.095
Brier _{raw} $\downarrow$	0.585	0.440	0.804	0.492	0.293	0.775	0.764	0.781
lex-glue-case-hold								$N = 200$
ACC $\uparrow$	68.00	74.00	32.00	59.00	75.50	22.00	38.00	30.00
Latency $\downarrow$	6.50	10.24	13.09	221.82	24.07	39.37	218.21	216.69
ECE $\downarrow$	<u>0.058</u>	0.067	0.167	0.156	0.102	0.079	0.151	0.037
Brier _{raw} $\downarrow$	0.426	0.351	0.694	0.577	0.362	0.805	0.784	0.782
lex-glue-ecthr-a								$N = 200$
ACC $\uparrow$	57.00	67.00	21.50	64.50	64.50	14.50	21.50	7.00
Latency $\downarrow$	6.63	10.30	11.58	218.28	24.32	41.09	226.13	215.26
ECE $\downarrow$	0.099	0.075	0.463	0.057	0.112	0.189	0.112	0.132
Brier _{raw} $\downarrow$	0.606	0.496	1.122	0.486	0.510	0.952	0.901	0.914
lex-glue-ecthr-b								$N = 200$
ACC $\uparrow$	76.00	89.50	20.50	82.00	89.50	13.00	22.50	8.00
Latency $\downarrow$	6.80	10.40	11.49	221.12	24.48	43.15	227.82	211.15
ECE $\downarrow$	0.228	0.216	0.289	0.121	0.036	0.210	0.129	0.116
Brier _{raw} $\downarrow$	0.406	0.236	0.969	0.294	0.172	0.967	0.903	0.927
lex-glue-ledgar								$N = 200$
ACC $\uparrow$	26.00	68.50	20.50	64.50	70.00	7.00	39.00	8.50
Latency $\downarrow$	6.93	10.49	10.28	224.28	24.54	39.59	52.87	212.58
ECE $\downarrow$	0.159	0.060	0.567	0.419	0.206	<u>0.041</u>	0.363	0.038
Brier _{raw} $\downarrow$	0.883	0.441	1.283	0.713	0.473	0.988	0.968	0.980
lex-glue-scotus								$N = 200$
ACC $\uparrow$	58.00	53.00	9.50	69.00	61.00	10.50	33.50	26.50
Latency $\downarrow$	7.58	11.40	11.84	230.08	25.36	130.41	272.80	203.92
ECE $\downarrow$	0.170	<u>0.055</u>	0.443	0.219	0.162	0.032	0.307	0.106
Brier _{raw} $\downarrow$	0.504	0.511	1.053	0.534	0.446	0.916	0.910	0.831
lex-glue-unfair-tos								$N = 200$
ACC $\uparrow$	73.50	85.50	83.00	77.50	76.00	7.00	76.00	<u>87.50</u>
Latency $\downarrow$	7.34	10.78	11.27	218.19	25.56	37.82	20.30	210.75
ECE $\downarrow$	0.246	0.105	<u>0.070</u>	0.225	0.149	0.135	0.554	0.589
Brier _{raw} $\downarrow$	0.502	<u>0.260</u>	0.276	0.376	0.416	0.836	0.736	0.633
Domain summary								$N = 1,800$
ACC $\uparrow$	54.67	67.11	25.44	65.06	70.28	14.72	32.67	29.17
Latency $\downarrow$	6.76	10.23	11.49	225.56	24.39	51.56	141.83	213.78
ECE $\downarrow$	0.072	<u>0.048</u>	0.328	0.154	0.099	0.144	0.233	0.132
Brier _{raw} $\downarrow$	0.576	0.425	0.945	0.512	0.410	0.897	0.835	0.809

Table 18 Law: model band 3/3. ACC (%) $\uparrow$; benchmark-run latency (s), ECE, and Brier_{raw} $\downarrow$. Rankings span all bands.

N: non-thinking; T: thinking. Parentheses specify reasoning effort.

Metric	Qwen3.5-4B N	Qwen3.8- 27B N	DeepSeek V4-Flash N	GPT-6-Sol N	GPT-6- Luna N	Qwen3.5-4B T	Qwen3.8- 27B T (xhigh)	DeepSeek V4-Flash T (high)	GLM-5.3- Flash T (high)
cuad									$N = 200$
ACC $\uparrow$	70.00	77.00	76.00	63.00	80.50	62.50	73.50	80.50	76.00
Latency $\downarrow$	<u>2.56</u>	13.36	15.48	13.76	4.72	43.12	106.46	50.38	8.51
ECE $\downarrow$	0.196	0.114	0.142	0.282	0.092	0.271	0.146	0.099	0.099
Brier _{raw} $\downarrow$	0.379	0.262	0.287	0.566	0.196	0.548	0.322	0.201	0.234
disc-law-multi									$N = 200$
ACC $\uparrow$	15.00	35.50	44.50	<u>70.00</u>	68.50	34.00	61.50	73.00	62.00
Latency $\downarrow$	<u>3.15</u>	6.35	5.61	19.04	10.05	38.88	159.62	78.53	38.92
ECE $\downarrow$	0.648	0.539	0.496	0.266	0.267	0.464	0.291	0.356	0.335
Brier _{raw} $\downarrow$	3.491	2.313	3.083	<u>0.618</u>	0.574	2.038	1.383	2.907	2.231
disc-law-single									$N = 200$
ACC $\uparrow$	60.50	76.50	72.00	90.50	92.50	71.50	92.50	96.00	<u>93.50</u>
Latency $\downarrow$	<u>2.23</u>	2.93	3.68	15.24	5.36	30.63	112.30	68.35	20.75
ECE $\downarrow$	0.270	0.164	0.208	0.073	<u>0.034</u>	0.178	0.036	0.037	0.025
Brier _{raw} $\downarrow$	0.662	0.423	0.464	0.194	<u>0.094</u>	0.449	0.134	0.079	0.123
lex-glue-case-hold									$N = 200$
ACC $\uparrow$	72.00	81.00	71.00	<u>81.50</u>	<u>81.50</u>	70.50	77.00	78.50	75.50
Latency $\downarrow$	1.78	4.65	4.08	9.86	6.08	45.77	146.47	80.77	19.99
ECE $\downarrow$	0.183	0.127	0.227	0.152	0.155	0.248	0.158	0.177	0.137
Brier _{raw} $\downarrow$	0.484	0.334	0.520	0.331	0.338	0.567	0.385	0.389	0.411

Continued on next page

Table 18 continued; model band 3/3

N: non-thinking; T: thinking. Parentheses specify reasoning effort.

Metric	Qwen3.5-4B N	Qwen3.8- 27B N	DeepSeek V4-Flash N	GPT-6-Sol N	GPT-6- Luna N	Qwen3.5-4B T	Qwen3.8- 27B T (xhigh)	DeepSeek V4-Flash T (high)	GLM-5.3- Flash T (high)
lex-glue-ecthr-a									<i>N</i> = 200
ACC ↑	57.00	71.00	67.00	<u>79.50</u>	81.50	66.00	74.50	70.50	72.00
Latency ↓	<u>2.40</u>	11.74	13.23	14.09	6.86	48.65	163.69	112.88	33.74
ECE ↓	0.374	0.240	0.281	0.131	0.140	0.227	0.171	0.213	0.176
Brier _{raw} ↓	0.804	0.528	0.609	0.323	<u>0.337</u>	0.602	0.436	0.523	0.471
lex-glue-ecthr-b									<i>N</i> = 200
ACC ↑	78.00	89.50	82.50	91.50	<u>92.00</u>	85.00	86.00	91.50	91.00
Latency ↓	<u>2.46</u>	10.66	12.19	8.60	8.05	48.70	159.90	76.88	28.44
ECE ↓	0.179	0.064	0.142	<u>0.015</u>	0.025	0.066	0.061	0.054	0.006
Brier _{raw} ↓	0.486	0.181	0.333	0.070	<u>0.089</u>	0.199	0.205	0.150	0.156
lex-glue-ledgar									<i>N</i> = 200
ACC ↑	69.00	72.50	73.00	78.50	<u>76.00</u>	73.50	<u>76.00</u>	72.00	73.50
Latency ↓	7.92	31.29	17.68	15.46	6.98	53.07	174.92	117.35	82.60
ECE ↓	0.238	0.233	0.233	0.179	0.231	0.240	0.193	0.258	0.152
Brier _{raw} ↓	0.591	0.509	0.499	0.401	0.465	0.518	0.435	0.536	0.432
lex-glue-scotus									<i>N</i> = 200
ACC ↑	59.50	59.50	56.50	58.50	56.50	61.00	61.50	<u>63.50</u>	60.50
Latency ↓	<u>5.37</u>	25.89	45.81	8.68	7.74	47.00	138.61	36.16	19.30
ECE ↓	0.224	0.230	0.332	0.272	0.285	0.235	0.208	0.252	0.234
Brier _{raw} ↓	0.551	0.530	0.682	0.566	0.594	0.529	0.486	0.532	0.543
lex-glue-unfair-tos									<i>N</i> = 200
ACC ↑	45.00	75.00	49.50	76.00	75.00	66.50	74.00	74.50	66.00
Latency ↓	<u>2.61</u>	4.82	3.92	6.34	4.38	39.21	102.61	68.67	16.25
ECE ↓	0.401	0.205	0.449	0.219	0.209	0.303	0.214	0.249	0.231
Brier _{raw} ↓	0.881	0.461	0.880	0.448	0.437	0.608	0.466	0.502	0.523
Domain summary									<i>N</i> = 1,800
ACC ↑	58.44	70.83	65.78	76.56	78.22	65.61	75.17	<u>77.78</u>	74.44
Latency ↓	<u>3.37</u>	12.17	13.09	12.40	6.69	43.85	140.95	<u>77.51</u>	30.23
ECE ↓	0.286	0.203	0.276	0.172	0.157	0.237	0.159	0.187	0.150
Brier _{raw} ↓	0.924	0.619	0.826	0.387	0.346	0.677	0.474	0.653	0.574

D.4 Finance

Table 19 Finance: model band 1/3. ACC (%) ↑; benchmark-run latency (s), ECE, and Brier_{raw} ↓. Rankings span all bands.

N: non-thinking; T: thinking. Parentheses specify reasoning effort.

Metric	Jev	InnerJev-4B (ours)	InnerJev-27B (ours)	kev-0.8b	kev-4b	kev-9b	kev-27b	Intern- Decision 0.8B
finqa-yesno								<i>N</i> = 22
ACC ↑	95.45	72.73	86.36	68.18	81.82	86.36	<u>90.91</u>	54.55
Latency ↓	0.59	4.94	11.34	11.29	42.06	62.85	36.49	6.61
ECE ↓	0.051	0.194	0.098	0.113	0.082	0.079	0.056	0.120
Brier _{raw} ↓	<u>0.102</u>	0.263	0.193	0.336	0.227	0.162	0.122	0.439
financebench-yesno								<i>N</i> = 37
ACC ↑	<u>86.49</u>	75.68	81.08	64.86	78.38	81.08	<u>86.49</u>	59.46
Latency ↓	0.68	5.18	9.39	12.18	37.46	71.96	44.21	6.58
ECE ↓	0.105	0.091	0.111	0.158	0.087	0.127	0.061	0.059
Brier _{raw} ↓	0.179	0.294	0.195	0.453	0.295	0.285	0.161	0.448
Domain summary								<i>N</i> = 59
ACC ↑	89.83	74.58	83.05	66.10	79.66	83.05	88.14	57.63
Latency ↓	0.64	5.09	10.12	11.85	39.18	68.56	<u>41.33</u>	6.59
ECE ↓	0.085	0.087	0.066	0.093	0.052	0.075	0.038	0.056
Brier _{raw} ↓	0.150	0.283	0.194	0.409	0.270	0.239	0.147	0.445

Table 20 Finance: model band 2/3. ACC (%) ↑; benchmark-run latency (s), ECE, and Brier_{raw} ↓. Rankings span all bands.

N: non-thinking; T: thinking. Parentheses specify reasoning effort.

Metric	Intern- Decision 2B	Intern- Decision 4B	laya	openjev-4b-v5	nimble-9b	nanojev	lightjev	jevforge-0.8b
finqa-yesno								<i>N</i> = 22
ACC ↑	81.82	81.82	68.18	<u>90.91</u>	86.36	59.09	54.55	36.36
Latency ↓	5.37	8.31	12.49	99.97	16.88	25.08	44.75	68.59
ECE ↓	0.236	0.135	0.275	0.156	0.069	0.103	0.023	0.589
Brier _{raw} ↓	0.400	0.267	0.433	0.178	0.202	0.509	0.489	1.147

Continued on next page

Table 20 continued; model band 2/3

N: non-thinking; T: thinking. Parentheses specify reasoning effort.

Metric	Intern- Decision 2B	Intern- Decision 4B	laya	openjev-4b-v5	nimble-9b	nanojev	lightjev	jevforge-0.8b
financebench-yesno								
ACC ↑	62.16	78.38	56.76	83.78	81.08	72.97	43.24	<i>N</i> = 37
Latency ↓	5.28	9.11	14.07	157.43	18.99	33.57	44.62	29.73
ECE ↓	0.127	0.086	0.196	0.152	0.063	0.105	0.128	133.02
Brier _{raw} ↓	0.390	0.287	0.555	0.270	0.249	0.403	0.522	0.648
Domain summary								
ACC ↑	69.49	79.66	61.02	86.44	83.05	67.80	47.46	1.265
Latency ↓	5.32	8.81	13.48	136.01	18.20	30.40	44.67	<i>N</i> = 59
ECE ↓	0.073	0.104	0.138	0.065	<u>0.043</u>	<u>0.043</u>	0.084	32.20
Brier _{raw} ↓	0.394	0.280	0.510	0.236	0.231	0.442	0.510	109.00

Table 21 Finance: model band 3/3. ACC (%) ↑; benchmark-run latency (s), ECE, and Brier_{raw} ↓. Rankings span all bands.

N: non-thinking; T: thinking. Parentheses specify reasoning effort.

Metric	Qwen3.5-4B N	Qwen3.8- 27B N	DeepSeek V4-Flash N	GPT-6-Sol N	GPT-6- Luna N	Qwen3.5-4B T	Qwen3.8- 27B T (xhigh)	DeepSeek V4-Flash T (high)	GLM-5.3- Flash T (high)
finqa-yesno									
ACC ↑	59.09	86.36	81.82	81.82	95.45	77.27	81.82	95.45	<i>N</i> = 22
Latency ↓	<u>2.67</u>	6.25	68.36	19.01	5.09	58.64	72.74	182.33	<u>90.91</u>
ECE ↓	0.417	0.126	0.190	0.177	<u>0.045</u>	0.210	0.193	<u>0.045</u>	11.23
Brier _{raw} ↓	0.773	0.261	0.366	0.358	0.091	0.439	0.362	0.091	0.091
financebench-yesno									
ACC ↑	72.97	83.78	67.57	83.78	94.59	78.38	83.78	81.08	<i>N</i> = 37
Latency ↓	<u>2.73</u>	6.10	67.78	11.22	8.08	50.30	97.53	175.09	94.59
ECE ↓	0.275	0.152	0.259	0.127	0.034	0.191	0.108	0.173	12.11
Brier _{raw} ↓	0.519	0.316	0.522	0.283	<u>0.090</u>	0.354	0.254	0.359	<u>0.053</u>
Domain summary									
ACC ↑	67.80	84.75	72.88	83.05	94.92	77.97	83.05	86.44	<i>N</i> = 59
Latency ↓	<u>2.71</u>	6.16	68.00	14.13	6.96	53.41	88.29	177.79	<u>93.22</u>
ECE ↓	0.328	0.142	0.227	0.146	0.038	0.198	0.128	0.125	11.78
Brier _{raw} ↓	0.614	0.295	0.464	0.311	0.090	0.385	0.294	0.259	0.051

D.5 Commonsense

Table 22 Commonsense: model band 1/3. ACC (%) ↑; benchmark-run latency (s), ECE, and Brier_{raw} ↓. Rankings span all bands.

N: non-thinking; T: thinking. Parentheses specify reasoning effort.

Metric	Jev	InnerJev-4B (ours)	InnerJev-27B (ours)	kev-0.8b	kev-4b	kev-9b	kev-27b	Intern- Decision 0.8B
commonsense-qa								
ACC ↑	89.00	82.50	91.00	56.50	83.50	80.50	88.50	<i>N</i> = 200
Latency ↓	0.49	5.61	12.55	17.44	57.00	93.18	54.62	54.50
ECE ↓	0.043	0.060	0.072	<u>0.038</u>	0.161	0.145	0.157	6.54
Brier _{raw} ↓	0.172	0.244	0.160	0.560	0.295	0.287	0.214	0.133
hellaswag								
ACC ↑	93.50	85.00	93.50	54.00	83.00	85.00	91.50	<i>N</i> = 200
Latency ↓	0.69	5.65	11.70	16.91	55.51	92.12	54.15	51.00
ECE ↓	<u>0.036</u>	0.068	0.077	0.096	0.223	0.148	0.156	6.59
Brier _{raw} ↓	<u>0.104</u>	0.212	0.129	0.627	0.313	0.262	0.184	0.056
winogrande								
ACC ↑	89.50	70.50	83.50	48.50	69.50	74.50	87.50	<i>N</i> = 200
Latency ↓	0.47	5.63	12.89	18.69	61.11	102.50	59.57	52.50
ECE ↓	<u>0.037</u>	0.162	0.035	0.160	0.137	0.140	0.040	7.26
Brier _{raw} ↓	<u>0.150</u>	0.452	0.214	0.576	0.436	0.404	0.202	0.065
Domain summary								
ACC ↑	<u>90.67</u>	79.33	89.33	53.00	78.67	80.00	89.17	<i>N</i> = 600
Latency ↓	0.55	5.63	12.38	17.68	57.87	95.93	56.11	52.67
ECE ↓	0.010	0.066	0.050	0.093	0.118	0.107	0.103	6.80
Brier _{raw} ↓	0.142	0.303	0.167	0.588	0.348	0.317	0.200	0.041

Table 23 Commonsense: model band 2/3. ACC (%) ↑; benchmark-run latency (s), ECE, and Brier_{raw} ↓. Rankings span all bands.

N: non-thinking; T: thinking. Parentheses specify reasoning effort.								
Metric	Intern-Decision 2B	Intern-Decision 4B	laya	openjev-4b-v5	nimble-9b	nanojev	lightjev	jevforge-0.8b
commonsense-qa N = 200								
ACC ↑	63.50	78.00	44.50	72.00	82.00	17.00	29.50	24.50
Latency ↓	6.23	9.59	12.10	231.86	23.43	37.33	17.60	217.77
ECE ↓	0.066	0.098	0.115	0.195	0.070	0.120	0.267	0.136
Brier _{raw} ↓	0.500	0.332	0.709	0.441	0.255	0.815	0.951	0.824
hellaswag N = 200								
ACC ↑	61.50	79.00	19.50	95.50	89.50	25.50	33.00	30.50
Latency ↓	6.16	9.87	11.45	231.78	23.75	37.17	18.20	219.47
ECE ↓	0.079	0.132	0.182	0.132	0.042	0.088	0.113	0.105
Brier _{raw} ↓	0.507	0.264	0.780	0.122	0.154	0.757	0.732	0.766
winogrande N = 200								
ACC ↑	49.50	70.00	47.50	88.00	75.00	41.50	41.50	51.50
Latency ↓	7.12	11.65	11.97	226.46	25.36	38.11	16.78	220.38
ECE ↓	0.211	0.076	0.307	0.059	0.196	0.359	0.248	0.133
Brier _{raw} ↓	0.603	0.417	0.744	0.152	0.443	0.758	0.634	0.571
Domain summary N = 600								
ACC ↑	58.17	75.67	37.17	85.17	82.17	28.00	34.67	35.50
Latency ↓	6.50	10.37	11.91	230.03	24.18	37.54	17.53	219.21
ECE ↓	0.076	0.047	0.198	0.110	0.080	0.186	0.193	0.122
Brier _{raw} ↓	0.537	0.338	0.738	0.238	0.284	0.777	0.772	0.720

Table 24 Commonsense: model band 3/3. ACC (%) ↑; benchmark-run latency (s), ECE, and Brier_{raw} ↓. Rankings span all bands.

N: non-thinking; T: thinking. Parentheses specify reasoning effort.									
Metric	Qwen3.5-4B N	Qwen3.8-27B N	DeepSeek V4-Flash N	GPT-6-Sol N	GPT-6- Luna N	Qwen3.5-4B T	Qwen3.8-27B T (xhigh)	DeepSeek V4-Flash T (high)	GLM-5.3-Flash T (high)
commonsense-qa N = 200									
ACC ↑	75.00	84.50	75.00	86.50	86.50	78.50	<u>90.50</u>	85.50	85.00
Latency ↓	<u>2.61</u>	3.21	3.97	11.44	4.11	41.66	<u>79.32</u>	65.53	9.54
ECE ↓	0.163	0.093	0.173	0.087	0.071	0.143	0.030	0.120	0.048
Brier _{raw} ↓	0.429	0.275	0.438	0.227	0.208	0.393	<u>0.161</u>	0.267	0.226
hellaswag N = 200									
ACC ↑	72.50	89.00	80.50	<u>94.00</u>	95.50	79.50	89.50	87.00	90.00
Latency ↓	<u>1.73</u>	5.11	3.49	8.26	7.50	38.20	110.93	78.65	9.51
ECE ↓	0.100	0.044	0.098	0.042	0.018	0.078	0.051	0.106	0.070
Brier _{raw} ↓	0.424	0.200	0.307	0.113	0.079	0.343	0.169	0.253	0.135
winogrande N = 200									
ACC ↑	56.00	73.50	68.50	82.00	89.50	80.50	92.50	<u>91.00</u>	88.50
Latency ↓	<u>1.05</u>	2.66	3.40	6.40	4.01	43.28	111.71	61.73	11.17
ECE ↓	0.325	0.209	0.251	0.166	0.074	0.151	0.035	0.083	0.057
Brier _{raw} ↓	0.674	0.460	0.544	0.350	0.192	0.334	0.131	0.172	0.208
Domain summary N = 600									
ACC ↑	67.83	82.33	74.67	87.50	90.50	79.50	90.83	87.83	87.83
Latency ↓	<u>1.80</u>	3.66	3.63	8.70	5.21	41.05	100.65	68.63	10.07
ECE ↓	0.180	0.104	0.168	0.092	0.048	0.118	<u>0.011</u>	0.103	0.021
Brier _{raw} ↓	0.508	0.312	0.430	0.230	0.160	0.356	<u>0.154</u>	0.231	0.190

D.6 Agent and tool use

Table 25 Agent and tool use: model band 1/3. ACC (%) $\uparrow$; benchmark-run latency (s), ECE, and Brier_{raw} $\downarrow$. Rankings span all bands.

N: non-thinking; T: thinking. Parentheses specify reasoning effort.

Metric	Jev	InnerJev-4B (ours)	InnerJev-27B (ours)	kev-0.8b	kev-4b	kev-9b	kev-27b	Intern- Decision 0.8B
agentharm								$N = 97, N_{\text{score}} = 103$
ACC $\uparrow$	84.54	87.63	87.63	77.32	86.60	87.63	89.69	75.26
Latency $\downarrow$	0.79	5.61	12.26	18.20	56.38	95.20	56.37	6.54
ECE $\downarrow$	0.068	0.077	0.069	0.054	0.105	0.072	0.050	0.075
Brier _{raw} $\downarrow$	0.201	0.215	0.200	0.328	0.239	0.213	0.158	0.315
MAE $\downarrow$	0.672	0.853	0.700	0.919	0.760	0.742	0.650	0.988
Valid score	103/103	103/103	103/103	103/103	103/103	103/103	103/103	103/103
decider								$N = 200$
ACC $\uparrow$	79.00	74.00	89.50	51.00	67.00	67.50	70.50	49.50
Latency $\downarrow$	0.47	5.62	12.29	16.94	57.95	93.79	54.48	6.45
ECE $\downarrow$	0.100	0.133	0.035	0.132	0.062	0.062	0.072	0.049
Brier _{raw} $\downarrow$	0.307	0.395	0.168	0.599	0.433	0.432	0.383	0.597
jevbench								$N = 184, N_{\text{score}} = 16$
ACC $\uparrow$	87.50	79.35	86.96	61.41	71.74	73.91	87.50	70.11
Latency $\downarrow$	0.84	5.71	12.70	18.07	57.66	94.71	56.74	6.83
ECE $\downarrow$	0.032	0.065	0.038	0.067	0.043	0.095	0.068	0.105
Brier _{raw} $\downarrow$	0.176	0.274	0.169	0.463	0.356	0.333	0.188	0.379
MAE $\downarrow$	0.159	0.222	0.140	0.437	0.297	0.287	0.188	0.535
Valid score	16/16	16/16	16/16	16/16	16/16	16/16	16/16	16/16
jevforge								$N = 200$
ACC $\uparrow$	66.50	67.50	70.50	56.50	63.00	64.00	64.00	38.50
Latency $\downarrow$	0.74	5.58	12.29	17.82	56.85	95.67	56.88	6.92
ECE $\downarrow$	0.058	0.123	0.076	0.139	0.133	0.128	0.091	0.180
Brier _{raw} $\downarrow$	0.441	0.463	0.354	0.604	0.509	0.525	0.452	0.719
Domain summary								$N = 681, N_{\text{score}} = 119$
ACC $\uparrow$	78.41	75.48	82.97	59.18	69.90	71.07	75.92	55.51
Latency $\downarrow$	0.71	5.63	12.38	17.76	57.21	94.84	56.12	6.69
ECE $\downarrow$	0.044	0.085	0.030	0.059	0.059	0.060	0.023	0.048
Brier _{raw} $\downarrow$	0.296	0.356	0.227	0.525	0.407	0.401	0.319	0.534
MAE $\downarrow$	0.603	0.768	0.625	0.854	0.698	0.681	0.588	0.927
Valid score	119/119	119/119	119/119	119/119	119/119	119/119	119/119	119/119

Table 26 Agent and tool use: model band 2/3. ACC (%) $\uparrow$; benchmark-run latency (s), ECE, and Brier_{raw} $\downarrow$. Rankings span all bands.

N: non-thinking; T: thinking. Parentheses specify reasoning effort.

Metric	Intern- Decision 2B	Intern- Decision 4B	laya	openjev-4b-v5	nimble-9b	nanojev	lightjev	jevforge-0.8b
agentharm								$N = 97, N_{\text{score}} = 103$
ACC $\uparrow$	85.57	73.20	42.27	82.47	90.72	52.58	48.45	47.42
Latency $\downarrow$	6.38	9.94	12.12	230.13	23.67	39.07	19.53	224.29
ECE $\downarrow$	0.074	0.113	0.328	0.112	0.078	0.330	0.111	0.480
Brier _{raw} $\downarrow$	0.243	0.331	0.765	0.306	0.194	0.711	0.522	0.956
MAE $\downarrow$	0.770	0.963	1.308	0.631	0.510	1.558	1.384	1.423
Valid score	103/103	103/103	103/103	103/103	103/103	103/103	103/103	103/103
decider								$N = 200$
ACC $\uparrow$	66.50	72.50	56.00	71.00	76.00	39.50	47.00	33.50
Latency $\downarrow$	6.45	9.24	10.92	232.32	23.39	37.33	16.82	218.69
ECE $\downarrow$	0.054	0.025	0.129	0.068	0.111	0.135	0.073	0.201
Brier _{raw} $\downarrow$	0.443	0.389	0.557	0.408	0.356	0.726	0.649	0.827
jevbench								$N = 184, N_{\text{score}} = 16$
ACC $\uparrow$	77.17	88.04	52.72	81.52	78.26	41.30	47.83	40.76
Latency $\downarrow$	6.40	10.07	12.09	228.00	24.14	38.12	95.96	221.40
ECE $\downarrow$	0.045	0.077	0.139	0.080	0.115	0.163	0.072	0.212
Brier _{raw} $\downarrow$	0.290	0.173	0.566	0.258	0.301	0.726	0.566	0.797
MAE $\downarrow$	0.320	0.304	0.478	0.330	0.202	0.840	0.793	0.844
Valid score	16/16	16/16	16/16	16/16	16/16	16/16	16/16	16/16
jevforge								$N = 200$
ACC $\uparrow$	57.00	56.00	22.50	56.00	66.50	22.50	42.50	70.50
Latency $\downarrow$	6.47	10.10	11.45	226.37	24.05	36.05	216.97	218.71
ECE $\downarrow$	0.073	0.043	0.399	0.064	0.077	0.305	0.115	0.075
Brier _{raw} $\downarrow$	0.552	0.559	0.949	0.562	0.453	0.967	0.693	0.354

Continued on next page

Table 26 continued; model band 2/3

N: non-thinking; T: thinking. Parentheses specify reasoning effort.

Metric	Intern- Decision 2B	Intern- Decision 4B	laya	openjev-4b-v5	nimble-9b	nanojev	lightjev	jevforge-0.8b
Domain summary								$N = 681, N_{\text{score}} = 119$
ACC $\uparrow$	69.31	71.95	43.32	71.07	75.92	36.86	46.11	48.31
Latency $\downarrow$	6.42	9.84	11.65	229.21	23.81	37.64	87.32	220.77
ECE $\downarrow$	0.039	<u>0.030</u>	0.223	0.060	0.083	0.211	0.081	0.199
Brier _{raw} $\downarrow$	0.405	0.372	0.704	0.398	0.346	0.795	0.621	0.698
MAE $\downarrow$	0.710	0.874	1.196	0.590	0.469	1.461	1.304	1.345
Valid score	119/119	119/119	119/119	119/119	119/119	119/119	119/119	119/119

Table 27 Agent and tool use: model band 3/3. ACC (%) $\uparrow$; benchmark-run latency (s), ECE, and Brier_{raw} $\downarrow$. Rankings span all bands.

N: non-thinking; T: thinking. Parentheses specify reasoning effort.

Metric	Qwen3.5-4B N	Qwen3.8- 27B N	DeepSeek V4-Flash N	GPT-6-Sol N	GPT-6- Luna N	Qwen3.5-4B T	Qwen3.8- 27B T (xhigh)	DeepSeek V4-Flash T (high)	GLM-5.3- Flash T (high)
agentharm									$N = 97, N_{\text{score}} = 103$
ACC $\uparrow$	61.86	74.23	78.35	76.29	71.13	70.10	84.54	77.32	85.57
Latency $\downarrow$	1.09	3.62	3.74	10.19	20.26	27.16	98.40	58.86	16.14
ECE $\downarrow$	0.349	0.242	0.201	0.204	0.243	0.256	0.114	0.220	0.089
Brier _{raw} $\downarrow$	0.688	0.484	0.409	0.417	0.485	0.501	0.261	0.437	0.246
MAE $\downarrow$	0.900	0.960	0.780	<u>0.422</u>	0.594	0.680	0.641	0.515	0.418
Valid score	70/103	101/103	101/103	102/103	101/103	103/103	103/103	103/103	103/103
decider									$N = 200$
ACC $\uparrow$	64.50	74.00	73.50	83.00	82.50	75.00	<u>83.50</u>	79.50	80.00
Latency $\downarrow$	2.46	3.88	4.68	20.16	5.71	29.87	84.83	66.37	14.69
ECE $\downarrow$	0.248	0.193	0.203	0.149	0.149	0.178	0.104	0.187	0.113
Brier _{raw} $\downarrow$	0.609	0.445	0.487	0.317	0.318	0.441	<u>0.291</u>	0.390	0.300
jevbench									$N = 184, N_{\text{score}} = 16$
ACC $\uparrow$	65.22	83.70	79.35	92.39	<u>97.28</u>	82.07	94.57	96.20	98.37
Latency $\downarrow$	0.92	3.74	5.73	14.59	7.01	37.67	102.97	75.68	15.38
ECE $\downarrow$	0.294	0.130	0.181	0.080	0.036	0.150	0.045	<u>0.032</u>	0.015
Brier _{raw} $\downarrow$	0.605	0.285	0.388	0.154	0.060	0.285	0.097	<u>0.056</u>	0.040
MAE $\downarrow$	0.417	0.286	0.406	0.000	0.000	0.125	0.000	0.078	<u>0.063</u>
Valid score	12/16	14/16	16/16	16/16	16/16	16/16	16/16	16/16	16/16
jevforge									$N = 200$
ACC $\uparrow$	47.00	60.00	58.50	63.00	<u>70.50</u>	56.00	72.50	67.50	69.00
Latency $\downarrow$	1.45	3.77	4.77	11.09	4.86	17.84	128.55	83.68	17.31
ECE $\downarrow$	0.404	0.229	0.318	0.246	0.114	0.342	0.121	0.219	0.116
Brier _{raw} $\downarrow$	0.889	0.622	0.711	0.580	0.416	0.722	<u>0.415</u>	0.526	0.445
Domain summary									$N = 681, N_{\text{score}} = 119$
ACC $\uparrow$	59.18	72.54	71.37	78.71	81.35	70.63	83.41	80.18	82.53
Latency $\downarrow$	<u>1.50</u>	3.76	4.73	14.02	9.38	28.13	103.69	71.15	15.88
ECE $\downarrow$	0.315	0.191	0.228	0.165	0.114	0.214	0.090	0.151	0.074
Brier _{raw} $\downarrow$	0.702	0.460	0.515	0.364	0.300	0.490	0.271	0.346	<u>0.264</u>
MAE $\downarrow$	0.829	0.878	0.729	0.364	0.513	0.605	0.555	0.456	<u>0.370</u>
Valid score	82/119	115/119	117/119	118/119	117/119	119/119	119/119	119/119	119/119

D.7 Factuality and hallucination

Table 28 Factuality and hallucination: model band 1/3. ACC (%) $\uparrow$; benchmark-run latency (s), ECE, and Brier_{raw} $\downarrow$. Rankings span all bands.

N: non-thinking; T: thinking. Parentheses specify reasoning effort.

Metric	Jev	InnerJev-4B (ours)	InnerJev-27B (ours)	kev-0.8b	kev-4b	kev-9b	kev-27b	Intern- Decision 0.8B
halueval-dialogue								$N = 200$
ACC $\uparrow$	92.00	87.00	92.00	64.00	81.50	85.00	91.50	71.00
Latency $\downarrow$	1.17	5.63	12.08	16.83	56.94	92.24	54.17	6.28
ECE $\downarrow$	0.031	0.050	0.067	0.041	0.063	0.114	0.083	0.070
Brier _{raw} $\downarrow$	0.093	0.169	0.136	0.422	0.274	0.227	0.124	0.411
halueval-general								$N = 200$
ACC $\uparrow$	71.00	80.00	77.50	78.50	77.00	79.50	80.00	73.50
Latency $\downarrow$	1.18	5.62	12.00	17.02	57.01	92.21	53.91	6.33
ECE $\downarrow$	0.039	0.117	0.051	<u>0.041</u>	0.069	0.071	0.078	0.139
Brier _{raw} $\downarrow$	0.396	0.315	0.312	0.331	0.343	0.311	<u>0.305</u>	0.419

Continued on next page

Table 28 continued; model band 1/3

N: non-thinking; T: thinking. Parentheses specify reasoning effort.

Metric	Jev	InnerJev-4B (ours)	InnerJev-27B (ours)	kev-0.8b	kev-4b	kev-9b	kev-27b	Intern- Decision 0.8B
halueval-qa								$N = 200$
ACC $\uparrow$	94.50	94.00	<u>96.00</u>	78.00	91.50	91.00	94.50	79.50
Latency $\downarrow$	1.30	5.62	11.74	17.07	56.11	92.19	53.74	6.31
ECE $\downarrow$	0.039	0.045	0.056	0.047	0.028	0.038	0.028	0.082
Brier _{raw} $\downarrow$	0.091	0.118	0.072	0.309	0.113	0.116	0.089	0.281
halueval-summarization								$N = 200$
ACC $\uparrow$	78.50	74.00	82.00	50.50	77.50	71.00	74.00	49.00
Latency $\downarrow$	1.36	5.80	13.50	16.98	56.02	91.89	54.08	6.37
ECE $\downarrow$	0.095	0.123	<u>0.050</u>	0.133	0.043	0.141	0.077	0.172
Brier _{raw} $\downarrow$	0.283	0.406	0.268	0.517	0.312	0.434	0.370	0.563
truthfulqa								$N = 200$
ACC $\uparrow$	88.50	63.50	71.00	34.50	59.50	58.00	68.50	35.50
Latency $\downarrow$	0.74	5.63	13.01	18.99	61.66	105.10	60.30	7.31
ECE $\downarrow$	0.035	0.136	0.059	0.106	0.061	0.079	0.067	0.090
Brier _{raw} $\downarrow$	0.190	0.499	0.405	0.799	0.543	0.594	0.428	0.755
Domain summary								$N = 1,000$
ACC $\uparrow$	84.90	79.70	83.70	61.10	77.40	76.90	81.70	61.70
Latency $\downarrow$	1.15	5.66	12.47	17.38	57.55	94.73	55.24	6.52
ECE $\downarrow$	0.034	0.066	0.025	0.046	0.032	0.042	<u>0.026</u>	0.051
Brier _{raw} $\downarrow$	0.210	0.301	0.239	0.475	0.317	0.336	0.263	0.486

Table 29 Factuality and hallucination: model band 2/3. ACC (%) $\uparrow$; benchmark-run latency (s), ECE, and Brier_{raw} $\downarrow$. Rankings span all bands.

N: non-thinking; T: thinking. Parentheses specify reasoning effort.

Metric	Intern- Decision 2B	Intern- Decision 4B	laya	openjev-4b-v5	nimble-9b	nanojev	lightjev	jevforge-0.8b
halueval-dialogue								$N = 200$
ACC $\uparrow$	70.00	87.00	59.00	84.50	92.00	47.50	84.00	38.00
Latency $\downarrow$	5.74	9.79	12.83	237.61	23.58	36.33	23.26	223.89
ECE $\downarrow$	0.052	0.043	0.031	0.034	0.059	0.151	0.236	0.246
Brier _{raw} $\downarrow$	0.365	0.186	0.432	0.216	0.149	0.537	0.359	0.652
halueval-general								$N = 200$
ACC $\uparrow$	74.50	81.00	40.00	79.50	<u>80.50</u>	20.00	47.50	80.00
Latency $\downarrow$	5.76	9.85	11.85	238.89	23.16	36.92	45.94	221.26
ECE $\downarrow$	0.048	0.088	0.244	0.165	0.101	0.642	0.155	0.177
Brier _{raw} $\downarrow$	0.333	0.300	0.624	0.353	0.313	1.140	0.503	0.383
halueval-qa								$N = 200$
ACC $\uparrow$	79.50	91.50	66.50	91.00	92.00	41.00	70.50	19.00
Latency $\downarrow$	5.81	9.93	12.18	235.90	23.31	36.23	37.79	218.24
ECE $\downarrow$	0.062	0.043	0.061	0.081	0.035	0.298	0.088	0.529
Brier _{raw} $\downarrow$	0.266	0.133	0.403	0.173	0.114	0.683	0.404	0.940
halueval-summarization								$N = 200$
ACC $\uparrow$	62.00	68.00	1.00	70.50	71.50	43.00	72.50	39.00
Latency $\downarrow$	5.91	9.94	7.04	235.93	23.61	38.14	221.50	219.40
ECE $\downarrow$	0.217	0.178	0.494	0.085	0.177	0.173	0.198	0.233
Brier _{raw} $\downarrow$	0.552	0.475	0.534	0.393	0.430	0.560	0.465	0.625
truthfulqa								$N = 200$
ACC $\uparrow$	57.00	63.50	15.00	51.50	59.50	17.00	37.00	47.00
Latency $\downarrow$	7.38	11.82	12.57	229.56	25.58	39.69	16.46	215.32
ECE $\downarrow$	0.096	0.114	0.341	0.069	0.159	0.174	<u>0.053</u>	0.130
Brier _{raw} $\downarrow$	0.577	0.482	1.011	0.652	0.592	0.872	0.753	0.693
Domain summary								$N = 1,000$
ACC $\uparrow$	68.60	78.20	36.30	75.40	79.10	33.70	62.30	44.60
Latency $\downarrow$	6.12	10.26	12.32	235.58	23.85	37.46	68.99	219.62
ECE $\downarrow$	0.070	0.054	0.156	0.066	0.096	0.286	0.084	0.254
Brier _{raw} $\downarrow$	0.419	0.315	0.622	0.357	0.320	0.758	0.497	0.659

Table 30 Factuality and hallucination: model band 3/3. ACC (%) ↑; benchmark-run latency (s), ECE, and Brier_{raw} ↓. Rankings span all bands.

N: non-thinking; T: thinking. Parentheses specify reasoning effort.									
Metric	Qwen3.5-4B N	Qwen3.8- 27B N	DeepSeek V4-Flash N	GPT-6-Sol N	GPT-6- Luna N	Qwen3.5-4B T	Qwen3.8- 27B T (xhigh)	DeepSeek V4-Flash T (high)	GLM-5.3- Flash T (high)
halueval-dialogue									<i>N</i> = 200
ACC ↑	77.50	83.50	89.50	95.50	<u>95.00</u>	90.50	93.50	90.00	94.50
Latency ↓	<u>2.21</u>	4.46	3.25	15.06	5.50	43.90	130.57	113.87	20.57
ECE ↓	0.144	0.079	0.098	<u>0.030</u>	0.028	0.137	0.090	0.133	0.047
Brier _{raw} ↓	0.346	0.231	0.185	0.083	<u>0.086</u>	0.241	0.164	0.238	0.097
halueval-general									<i>N</i> = 200
ACC ↑	42.00	72.50	77.00	69.00	68.00	67.00	73.50	80.00	75.50
Latency ↓	<u>2.20</u>	3.31	2.64	11.79	5.69	36.31	98.98	48.09	8.38
ECE ↓	0.563	0.223	0.222	0.275	0.289	0.299	0.205	0.195	0.163
Brier _{raw} ↓	1.117	0.494	0.452	0.573	0.604	0.602	0.455	0.390	0.392
halueval-qa									<i>N</i> = 200
ACC ↑	80.00	89.00	87.00	<u>96.00</u>	94.50	94.00	96.50	94.00	92.00
Latency ↓	<u>2.26</u>	4.30	2.91	9.19	3.74	44.67	103.92	71.02	14.35
ECE ↓	0.181	0.082	0.126	0.032	<u>0.027</u>	0.073	0.021	0.056	0.044
Brier _{raw} ↓	0.350	0.187	0.244	<u>0.071</u>	0.086	0.125	0.058	0.116	0.130
halueval-summarization									<i>N</i> = 200
ACC ↑	66.00	79.00	78.50	84.00	87.00	73.00	84.50	<u>85.00</u>	<u>85.00</u>
Latency ↓	<u>2.32</u>	5.58	5.95	11.47	5.34	46.56	132.54	115.66	24.21
ECE ↓	0.280	0.170	0.202	0.122	0.085	0.216	0.099	0.146	0.060
Brier _{raw} ↓	0.596	0.374	0.407	0.266	0.206	0.487	0.261	0.294	<u>0.238</u>
truthfulqa									<i>N</i> = 200
ACC ↑	48.50	78.50	77.00	76.00	80.00	62.50	78.50	78.00	<u>85.00</u>
Latency ↓	<u>1.69</u>	5.06	4.91	7.00	3.53	40.63	96.18	62.14	12.49
ECE ↓	0.332	0.148	0.175	0.206	0.167	0.262	0.157	0.210	0.075
Brier _{raw} ↓	0.814	0.384	0.407	0.440	0.359	0.618	0.370	0.434	<u>0.252</u>
Domain summary									<i>N</i> = 1,000
ACC ↑	62.80	80.50	81.80	84.10	84.90	77.40	85.30	<u>85.40</u>	86.40
Latency ↓	<u>2.14</u>	4.54	3.93	10.90	4.76	42.41	112.44	82.16	16.00
ECE ↓	0.289	0.139	0.161	0.121	0.106	0.184	0.096	0.146	0.056
Brier _{raw} ↓	0.644	0.334	0.339	0.287	0.268	0.415	0.262	0.294	<u>0.222</u>

D.8 Long-context and document understanding

Table 31 Long-context and document understanding: model band 1/3. ACC (%) ↑; benchmark-run latency (s), ECE, and Brier_{raw} ↓. Rankings span all bands.

N: non-thinking; T: thinking. Parentheses specify reasoning effort.								
Metric	Jev	InnerJev-4B (ours)	InnerJev-27B (ours)	kev-0.8b	kev-4b	kev-9b	kev-27b	Intern- Decision 0.8B
helmet								
ACC ↑	74.50	95.00	100.00	32.50	92.50	93.00	100.00	<i>N</i> = 200
Latency ↓	0.71	<u>6.56</u>	15.56	17.89	58.22	96.69	56.96	20.50
ECE ↓	<u>0.001</u>	0.139	0.007	0.283	0.787	0.584	0.027	7.17
Brier _{raw} ↓	0.000	0.098	0.000	0.944	0.771	0.493	<u>0.002</u>	0.154
longbench-lsht								
ACC ↑	61.50	60.50	69.50	46.50	57.50	64.00	64.50	<i>N</i> = 200
Latency ↓	0.60	<u>7.05</u>	15.81	20.62	65.28	110.93	63.91	48.50
ECE ↓	0.178	0.187	0.116	0.132	0.089	0.092	0.059	8.47
Brier _{raw} ↓	0.501	0.572	0.462	0.680	0.538	0.523	<u>0.486</u>	0.258
longbench-retrieval-en								
ACC ↑	100.00	100.00	100.00	28.50	99.00	99.00	100.00	<i>N</i> = 200
Latency ↓	1.12	<u>6.21</u>	15.57	19.79	63.50	108.24	62.79	58.00
ECE ↓	0.000	0.007	0.003	0.194	0.184	0.055	0.003	8.23
Brier _{raw} ↓	0.000	0.000	0.000	0.910	0.062	0.035	0.000	0.384
longbench-retrieval-en-e								
ACC ↑	100.00	100.00	100.00	38.00	99.00	<u>99.50</u>	100.00	<i>N</i> = 200
Latency ↓	0.59	5.98	15.64	20.40	64.90	111.17	65.65	71.00
ECE ↓	0.000	0.003	0.004	0.205	0.107	0.041	0.002	8.41
Brier _{raw} ↓	0.000	0.000	0.000	0.826	0.034	0.017	0.000	0.478
longbench-retrieval-zh								
ACC ↑	100.00	100.00	100.00	40.00	91.50	68.50	100.00	<i>N</i> = 200
Latency ↓	0.98	5.89	15.00	20.29	65.78	111.85	64.74	71.00
ECE ↓	0.000	0.009	0.003	0.268	0.214	0.144	0.003	8.43
Brier _{raw} ↓	0.000	0.002	0.000	0.858	0.183	0.500	0.000	0.505

Continued on next page

Table 31 continued; model band 1/3

N: non-thinking; T: thinking. Parentheses specify reasoning effort.

Metric	Jev	InnerJev-4B (ours)	InnerJev-27B (ours)	kev-0.8b	kev-4b	kev-9b	kev-27b	Intern- Decision 0.8B
longbench-trec								
ACC ↑	<u>86.50</u>	69.50	86.00	13.00	67.00	78.50	86.00	$N = 200$ 4.00
Latency ↓	0.53	<u>6.02</u>	15.14	20.78	66.29	112.70	65.85	8.59
ECE ↓	<u>0.042</u>	0.122	0.049	0.049	0.387	0.277	0.045	0.072
Brier _{raw} ↓	0.199	0.462	0.178	0.921	0.640	0.429	<u>0.187</u>	0.984
longbench-trec-e								
ACC ↑	85.00	65.50	<u>85.50</u>	15.50	67.50	77.00	<u>85.50</u>	$N = 200$ 3.50
Latency ↓	0.63	<u>6.17</u>	15.88	20.81	66.34	112.40	65.57	8.70
ECE ↓	0.047	0.142	<u>0.033</u>	0.041	0.392	0.264	0.065	0.079
Brier _{raw} ↓	0.226	0.514	<u>0.203</u>	0.929	0.636	0.403	0.201	0.981
Domain summary								
ACC ↑	86.79	84.36	91.57	30.57	82.00	82.79	<u>90.86</u>	$N = 1,400$ 39.50
Latency ↓	0.74	<u>6.27</u>	15.51	20.08	64.33	109.14	63.64	8.28
ECE ↓	0.028	0.048	<u>0.023</u>	0.144	0.302	0.201	0.016	0.232
Brier _{raw} ↓	0.130	0.235	0.120	0.867	0.409	0.343	<u>0.125</u>	0.836

Table 32 Long-context and document understanding: model band 2/3. ACC (%) ↑; benchmark-run latency (s), ECE, and Brier_{raw} ↓. Rankings span all bands.

N: non-thinking; T: thinking. Parentheses specify reasoning effort.

Metric	Intern- Decision 2B	Intern- Decision 4B	laya	openjev-4b-v5	nimble-9b	nanojev	lightjev	jevforge-0.8b
helmet								
ACC ↑	33.50	82.00	4.00	<u>99.50</u>	73.00	0.50	19.50	$N = 200$ 1.00
Latency ↓	6.76	10.70	11.83	357.11	24.34	123.96	437.34	277.30
ECE ↓	0.197	0.323	0.107	0.797	0.034	0.014	0.182	0.019
Brier _{raw} ↓	0.775	0.407	0.971	0.729	0.030	0.990	0.988	0.991
longbench-lsht								
ACC ↑	50.00	63.50	0.00	50.00	64.50	10.00	33.50	$N = 200$ 19.00
Latency ↓	8.44	12.22	8.79	265.66	27.98	155.32	352.19	223.77
ECE ↓	0.068	0.064	0.501	0.072	0.213	<u>0.063</u>	0.282	0.071
Brier _{raw} ↓	0.648	0.537	1.290	0.614	0.545	0.969	0.935	0.921
longbench-retrieval-en								
ACC ↑	79.50	98.00	0.00	97.00	100.00	3.00	17.50	$N = 200$ 6.50
Latency ↓	8.00	12.25	—	273.27	27.37	178.62	374.41	218.43
ECE ↓	0.247	0.135	—	0.872	0.002	0.074	0.138	0.022
Brier _{raw} ↓	0.418	0.057	—	0.852	0.000	0.978	0.964	0.965
longbench-retrieval-en-e								
ACC ↑	95.00	98.00	7.50	99.00	100.00	5.00	25.50	$N = 200$ 6.50
Latency ↓	8.36	12.86	14.59	249.20	28.04	101.89	305.62	200.92
ECE ↓	0.334	0.104	0.492	0.603	<u>0.001</u>	0.109	0.189	0.009
Brier _{raw} ↓	0.218	0.040	1.223	0.557	0.000	0.961	0.933	0.941
longbench-retrieval-zh								
ACC ↑	84.50	<u>99.50</u>	2.00	<u>99.50</u>	100.00	5.50	15.00	$N = 200$ 3.00
Latency ↓	8.44	13.08	11.06	220.74	28.12	41.09	300.92	211.68
ECE ↓	0.245	0.150	0.042	0.313	<u>0.001</u>	0.069	0.112	0.011
Brier _{raw} ↓	0.294	0.038	0.962	0.231	0.000	0.980	0.963	0.967
longbench-trec								
ACC ↑	26.00	77.50	1.00	71.00	85.00	0.50	3.50	$N = 200$ 6.50
Latency ↓	8.59	13.37	11.00	263.57	28.15	89.11	340.33	214.78
ECE ↓	0.109	0.237	0.653	0.444	0.097	0.045	0.008	0.056
Brier _{raw} ↓	0.867	0.392	1.532	0.673	0.260	0.986	0.975	0.984
longbench-trec-e								
ACC ↑	29.50	72.50	2.50	71.00	83.50	0.00	6.00	$N = 200$ 6.00
Latency ↓	8.47	13.18	13.54	310.37	28.01	114.14	355.78	215.60
ECE ↓	0.107	0.187	0.672	0.452	0.085	0.049	0.032	0.049
Brier _{raw} ↓	0.848	0.432	1.517	0.698	0.268	0.985	0.974	0.985
Domain summary								
ACC ↑	56.86	84.43	2.43	83.86	86.57	3.50	17.21	$N = 1,400$ 6.93
Latency ↓	8.15	12.52	12.60	277.13	27.54	114.73	352.37	223.01
ECE ↓	0.142	0.161	0.479	0.490	0.051	0.060	0.134	0.024
Brier _{raw} ↓	0.581	0.272	1.320	0.622	0.162	0.978	0.962	0.965

Table 33 Long-context and document understanding: model band 3/3. ACC (%) ↑; benchmark-run latency (s), ECE, and Brier_{raw} ↓. Rankings span all bands.

N: non-thinking; T: thinking. Parentheses specify reasoning effort.									
Metric	Qwen3.5-4B N	Qwen3.8- 27B N	DeepSeek V4-Flash N	GPT-6-Sol N	GPT-6- Luna N	Qwen3.5-4B T	Qwen3.8- 27B T (xhigh)	DeepSeek V4-Flash T (high)	GLM-5.3- Flash T (high)
helmet									<i>N</i> = 200
ACC ↑	40.00	74.50	98.00	73.00	74.00	20.50	46.00	64.00	99.00
Latency ↓	36.02	130.46	73.71	47.77	17.22	62.96	228.30	206.40	183.52
ECE ↓	0.391	0.000	0.030	0.000	0.007	0.087	0.077	0.000	0.000
Brier _{raw} ↓	0.840	0.000	0.055	0.000	0.013	0.232	0.155	0.015	0.000
longbench-lsht									<i>N</i> = 200
ACC ↑	57.50	61.50	61.00	68.00	66.00	<u>68.50</u>	68.00	67.50	67.50
Latency ↓	8.52	39.60	59.74	26.24	8.75	41.37	129.55	52.02	29.01
ECE ↓	0.360	0.288	0.340	0.231	0.272	0.308	0.238	0.303	0.171
Brier _{raw} ↓	0.934	0.650	0.725	0.522	0.593	0.634	0.562	0.628	0.512
longbench-retrieval-en									<i>N</i> = 200
ACC ↑	93.00	<u>99.50</u>	99.00	99.00	<u>99.50</u>	98.00	100.00	100.00	99.00
Latency ↓	8.87	44.52	58.01	15.63	7.80	51.29	125.58	59.50	30.75
ECE ↓	0.018	<u>0.001</u>	0.003	0.000	0.000	0.009	0.004	0.000	0.009
Brier _{raw} ↓	0.025	0.000	0.000	0.000	0.000	0.014	<u>0.001</u>	0.010	0.020
longbench-retrieval-en-e									<i>N</i> = 200
ACC ↑	89.50	100.00	<u>99.50</u>	<u>99.50</u>	99.00	99.00	100.00	<u>99.50</u>	<u>99.50</u>
Latency ↓	<u>5.35</u>	26.14	37.99	7.46	6.89	49.64	112.51	52.78	21.52
ECE ↓	0.035	0.004	0.007	0.000	0.000	0.000	0.004	0.005	0.004
Brier _{raw} ↓	0.027	0.000	0.003	0.000	0.000	0.005	<u>0.001</u>	0.010	0.010
longbench-retrieval-zh									<i>N</i> = 200
ACC ↑	96.50	100.00	97.50	100.00	100.00	97.50	100.00	100.00	100.00
Latency ↓	<u>4.55</u>	19.98	23.15	12.30	7.10	46.53	116.36	63.80	27.71
ECE ↓	0.029	0.009	0.006	0.000	0.000	0.008	0.002	0.000	0.003
Brier _{raw} ↓	0.190	<u>0.001</u>	0.000	0.000	0.000	0.012	0.000	0.000	0.000
longbench-trec									<i>N</i> = 200
ACC ↑	70.00	84.50	72.50	84.50	79.50	74.00	87.00	85.50	81.00
Latency ↓	8.01	33.59	37.85	19.54	6.65	47.62	134.46	38.92	48.49
ECE ↓	0.335	0.130	0.239	0.135	0.181	0.152	0.103	0.139	0.127
Brier _{raw} ↓	0.951	0.286	0.503	0.291	0.373	0.387	0.240	0.284	0.322
longbench-trec-e									<i>N</i> = 200
ACC ↑	69.50	80.50	69.50	83.00	80.00	73.00	86.00	79.00	82.00
Latency ↓	8.92	40.29	42.54	13.38	7.63	50.45	137.31	34.86	48.40
ECE ↓	0.247	0.171	0.271	0.140	0.182	0.236	0.111	0.201	0.121
Brier _{raw} ↓	0.573	0.369	0.566	0.314	0.375	0.524	0.252	0.414	0.314
Domain summary									<i>N</i> = 1,400
ACC ↑	73.71	85.79	85.29	86.71	85.43	75.79	83.86	85.07	89.71
Latency ↓	10.37	44.68	47.64	19.22	8.55	48.65	133.99	65.69	55.50
ECE ↓	0.186	0.084	0.124	0.074	0.095	0.112	0.073	0.097	0.058
Brier _{raw} ↓	0.502	0.194	0.264	0.167	0.200	0.261	0.174	0.203	0.168

D.9 Social science

Table 34 Social science: model band 1/3. ACC (%) ↑; benchmark-run latency (s), ECE, and Brier_{raw} ↓. Rankings span all bands.

N: non-thinking; T: thinking. Parentheses specify reasoning effort.								
Metric	Jev	InnerJev-4B (ours)	InnerJev-27B (ours)	kev-0.8b	kev-4b	kev-9b	kev-27b	Intern- Decision 0.8B
bbq								
ACC ↑	98.00	98.00	96.50	68.50	86.50	73.00	94.50	66.50
Latency ↓	<u>1.03</u>	5.62	12.66	17.93	56.97	95.56	55.72	6.41
ECE ↓	0.012	0.061	0.062	0.154	0.160	0.065	0.154	0.138
Brier _{raw} ↓	<u>0.036</u>	0.052	0.067	0.472	0.235	0.343	0.157	0.454
global-opinions								
ACC ↑	48.50	45.00	51.00	34.50	41.50	49.50	55.00	35.00
Latency ↓	0.85	5.65	11.69	17.06	56.45	93.61	54.29	6.56
ECE ↓	0.151	0.146	0.073	0.088	0.101	0.110	0.137	0.058
Brier _{raw} ↓	0.712	0.696	0.610	0.753	0.715	0.649	0.613	0.742
communitybench-compred								
ACC ↑	88.50	70.50	87.50	47.50	61.00	65.50	88.50	38.50
Latency ↓	0.88	<u>5.86</u>	12.84	18.49	59.42	99.44	58.09	7.20
ECE ↓	0.069	0.061	0.094	0.048	0.094	0.118	0.205	0.042
Brier _{raw} ↓	0.220	0.408	0.198	0.627	0.517	0.469	0.246	0.721

Continued on next page

Table 34 continued; model band 1/3

N: non-thinking; T: thinking. Parentheses specify reasoning effort.

Metric	Jev	InnerJev-4B (ours)	InnerJev-27B (ours)	kev-0.8b	kev-4b	kev-9b	kev-27b	Intern- Decision 0.8B
communitybench-prefid N = 200								
ACC ↑	37.00	<u>41.50</u>	42.00	25.00	34.00	31.50	39.00	26.50
Latency ↓	0.88	5.70	12.74	18.70	58.62	98.61	58.08	7.01
ECE ↓	0.295	0.266	0.157	0.212	0.089	0.169	<u>0.092</u>	0.128
Brier _{raw} ↓	0.843	0.839	0.776	0.824	<u>0.746</u>	0.775	0.724	0.765
compred-finance N = 200								
ACC ↑	61.00	55.00	59.00	54.00	57.00	60.50	60.50	59.00
Latency ↓	0.86	5.69	12.62	18.31	58.39	97.82	58.25	6.92
ECE ↓	0.201	0.237	0.143	0.085	0.101	0.149	<u>0.061</u>	0.075
Brier _{raw} ↓	0.561	0.579	0.496	0.508	0.497	0.520	0.458	0.499
compred-gendersexuality N = 200								
ACC ↑	50.00	50.00	54.00	51.00	44.50	52.00	56.50	49.00
Latency ↓	0.88	5.65	12.50	18.47	58.17	98.20	58.17	6.84
ECE ↓	0.279	0.299	0.188	0.109	0.212	0.169	0.146	0.091
Brier _{raw} ↓	0.683	0.697	0.584	0.532	0.607	0.595	0.540	0.504
compred-history N = 200								
ACC ↑	56.50	55.00	63.00	55.50	52.00	55.00	58.00	53.50
Latency ↓	0.88	<u>5.64</u>	12.71	18.24	57.18	97.57	57.43	6.72
ECE ↓	0.217	0.254	0.123	0.102	0.161	0.153	<u>0.091</u>	0.044
Brier _{raw} ↓	0.590	0.635	0.495	0.533	0.525	0.527	<u>0.482</u>	0.501
compred-politics N = 200								
ACC ↑	58.00	52.00	57.50	53.00	56.50	57.00	56.00	57.50
Latency ↓	0.91	<u>5.66</u>	12.55	18.17	56.17	96.87	57.16	6.64
ECE ↓	0.171	0.260	0.120	0.093	0.173	0.134	0.092	0.064
Brier _{raw} ↓	0.580	0.634	0.522	0.513	0.544	0.529	0.514	0.503
compred-science N = 200								
ACC ↑	57.00	57.50	57.50	51.50	53.50	57.00	57.00	44.00
Latency ↓	0.91	<u>5.73</u>	12.99	18.21	56.44	97.15	56.73	6.52
ECE ↓	0.245	0.226	0.167	0.117	0.142	0.167	<u>0.112</u>	0.151
Brier _{raw} ↓	0.610	0.638	0.552	0.541	0.558	0.584	<u>0.525</u>	0.547
Domain summary N = 1, 800								
ACC ↑	61.61	58.28	63.11	48.94	54.06	55.67	62.78	47.72
Latency ↓	0.90	<u>5.69</u>	12.59	18.17	57.53	97.20	57.10	6.76
ECE ↓	0.162	0.186	0.088	0.067	0.071	0.103	<u>0.051</u>	0.028
Brier _{raw} ↓	0.537	0.575	0.478	0.589	0.549	0.555	0.473	0.582

Table 35 Social science: model band 2/3. ACC (%) ↑; benchmark-run latency (s), ECE, and Brier_{raw} ↓. Rankings span all bands.

N: non-thinking; T: thinking. Parentheses specify reasoning effort.

Metric	Intern- Decision 2B	Intern- Decision 4B	laya	openjev-4b-v5	nimble-9b	nanojev	lightjev	jevforge-0.8b
bbq N = 200								
ACC ↑	85.50	96.50	30.00	76.50	73.50	28.50	24.50	48.00
Latency ↓	6.23	9.93	12.09	233.57	23.90	39.77	18.52	218.71
ECE ↓	0.129	0.097	0.393	0.093	0.104	0.333	0.239	0.111
Brier _{raw} ↓	0.261	0.076	0.944	0.339	0.355	0.837	0.877	0.626
global-opinions N = 200								
ACC ↑	33.50	41.00	40.50	41.00	49.50	27.00	35.50	15.50
Latency ↓	6.11	9.42	10.63	237.10	23.60	37.69	16.53	222.19
ECE ↓	0.096	0.077	0.114	0.038	0.186	0.185	<u>0.043</u>	0.276
Brier _{raw} ↓	0.718	0.701	0.736	0.677	0.664	0.814	0.717	0.937
communitybench-compred N = 200								
ACC ↑	45.00	66.00	0.00	63.50	73.00	23.00	17.50	32.50
Latency ↓	6.72	11.35	—	230.77	24.45	38.98	159.13	226.69
ECE ↓	0.114	0.052	—	0.063	0.062	0.139	0.156	0.076
Brier _{raw} ↓	0.680	0.465	—	0.501	0.368	0.789	0.798	0.731
communitybench-prefid N = 200								
ACC ↑	31.50	35.50	7.00	31.50	32.50	26.50	25.00	31.50
Latency ↓	6.70	11.15	12.35	231.83	24.01	37.29	146.52	226.48
ECE ↓	0.201	0.250	0.125	0.245	0.346	0.126	0.105	0.125
Brier _{raw} ↓	0.837	0.817	0.771	0.869	0.914	0.799	0.772	0.793
compred-finance N = 200								
ACC ↑	52.50	53.50	10.00	59.50	60.00	50.00	52.00	49.50
Latency ↓	6.70	10.78	9.64	236.57	23.99	39.00	165.75	229.11
ECE ↓	0.161	0.206	0.029	0.217	0.216	0.122	0.086	0.151
Brier _{raw} ↓	0.559	0.583	<u>0.474</u>	0.612	0.613	0.575	0.509	0.564

Continued on next page

Table 35 continued; model band 2/3

N: non-thinking; T: thinking. Parentheses specify reasoning effort.

Metric	Intern- Decision 2B	Intern- Decision 4B	laya	openjev-4b-v5	nimble-9b	nanojev	lightjev	jevforge-0.8b
compred-gendersexuality								
ACC ↑	46.50	51.00	8.00	53.50	50.00	55.00	<u>57.50</u>	$N = 200$ 50.00
Latency ↓	6.68	10.89	12.36	234.54	23.85	38.48	131.59	225.73
ECE ↓	0.215	0.264	0.129	0.179	0.286	<u>0.055</u>	0.049	0.141
Brier _{raw} ↓	0.630	0.651	0.420	0.599	0.705	0.504	<u>0.503</u>	0.538
compred-history								
ACC ↑	56.50	50.00	6.50	48.50	61.00	50.50	43.50	$N = 200$ 52.50
Latency ↓	6.38	10.48	8.00	233.46	23.66	39.14	147.13	226.69
ECE ↓	0.114	0.224	0.151	0.243	0.206	0.100	0.171	0.125
Brier _{raw} ↓	0.532	0.595	0.531	0.667	0.550	0.537	0.581	0.518
compred-politics								
ACC ↑	53.00	54.50	10.50	58.00	53.00	52.00	51.00	$N = 200$ 52.00
Latency ↓	6.32	10.15	13.43	232.88	23.71	40.06	137.48	224.06
ECE ↓	0.131	0.204	0.130	0.195	0.260	0.113	<u>0.080</u>	0.116
Brier _{raw} ↓	0.561	0.585	<u>0.509</u>	0.606	0.656	0.545	0.515	0.546
compred-science								
ACC ↑	51.50	55.00	9.50	53.50	<u>59.50</u>	39.50	46.00	$N = 200$ 55.00
Latency ↓	6.18	10.14	14.67	232.39	<u>23.79</u>	38.97	139.63	224.43
ECE ↓	0.171	0.196	0.166	0.170	0.240	0.213	0.165	0.111
Brier _{raw} ↓	0.571	0.590	0.583	0.623	0.638	0.602	0.574	0.528
Domain summary								
ACC ↑	50.61	55.89	13.56	53.94	56.89	39.11	39.17	$N = 1,800$ 42.94
Latency ↓	6.45	10.48	11.59	233.68	23.88	38.82	118.03	224.90
ECE ↓	0.104	0.138	0.190	0.125	0.200	0.150	0.120	0.135
Brier _{raw} ↓	0.594	0.563	0.746	0.610	0.607	0.667	0.650	0.642

Table 36 Social science: model band 3/3. ACC (%) ↑; benchmark-run latency (s), ECE, and Brier_{raw} ↓. Rankings span all bands.

N: non-thinking; T: thinking. Parentheses specify reasoning effort.

Metric	Qwen3.5-4B N	Qwen3.8- 27B N	DeepSeek V4-Flash N	GPT-6-Sol N	GPT-6- Luna N	Qwen3.5-4B T	Qwen3.8- 27B T (xhigh)	DeepSeek V4-Flash T (high)	GLM-5.3- Flash T (high)
bbq									
ACC ↑	67.50	90.50	86.50	96.00	94.00	<u>97.00</u>	96.00	96.50	$N = 200$ 91.00
Latency ↓	0.88	2.86	2.68	8.62	6.36	41.40	100.79	53.93	12.73
ECE ↓	0.160	0.049	0.080	0.034	0.057	0.023	<u>0.020</u>	0.034	0.029
Brier _{raw} ↓	0.445	0.162	0.222	0.079	0.117	0.018	0.071	0.069	0.138
global-opinions									
ACC ↑	41.00	40.00	50.50	60.00	<u>57.00</u>	40.00	49.50	50.50	$N = 200$ 51.50
Latency ↓	2.66	5.59	5.10	8.19	5.87	39.82	136.90	82.49	16.50
ECE ↓	0.234	0.187	0.138	0.094	0.068	0.129	0.075	0.072	0.086
Brier _{raw} ↓	0.789	0.715	0.663	<u>0.544</u>	0.571	0.695	0.625	0.585	0.533
communitybench-compred									
ACC ↑	50.00	79.00	75.00	92.50	<u>89.50</u>	69.50	87.00	85.00	$N = 200$ 88.50
Latency ↓	7.56	16.07	93.57	11.96	21.99	39.04	117.26	163.88	14.09
ECE ↓	0.347	0.111	0.137	0.055	0.061	0.164	<u>0.045</u>	0.105	0.055
Brier _{raw} ↓	0.831	0.359	0.418	0.130	0.175	0.496	0.215	0.274	<u>0.171</u>
communitybench-prefid									
ACC ↑	38.50	32.00	38.50	38.00	39.50	36.00	36.00	38.00	$N = 200$ 39.50
Latency ↓	8.26	15.87	95.39	13.29	<u>5.14</u>	38.99	131.69	150.26	22.39
ECE ↓	0.408	0.447	0.481	0.495	0.462	0.425	0.344	0.515	0.260
Brier _{raw} ↓	1.008	1.013	1.034	1.049	0.989	0.997	0.909	1.086	0.807
compred-finance									
ACC ↑	51.50	56.50	56.00	<u>65.50</u>	66.00	58.00	60.00	58.00	$N = 200$ 61.00
Latency ↓	6.19	13.09	84.12	12.01	<u>5.05</u>	32.76	118.14	170.22	15.74
ECE ↓	0.329	0.277	0.245	0.237	0.164	0.276	0.165	0.221	0.140
Brier _{raw} ↓	0.738	0.635	0.610	0.554	0.499	0.639	0.519	0.577	0.502
compred-gendersexuality									
ACC ↑	48.50	55.00	54.50	55.50	63.00	51.50	54.00	<u>57.50</u>	$N = 200$ 57.00
Latency ↓	6.64	13.05	83.76	11.32	<u>4.91</u>	33.13	118.76	169.96	15.63
ECE ↓	0.350	0.280	0.254	0.252	0.191	0.338	0.213	0.181	0.167
Brier _{raw} ↓	0.754	0.672	0.662	0.635	0.551	0.734	0.595	0.579	0.574
compred-history									
ACC ↑	54.00	60.50	60.50	59.50	59.50	46.00	<u>64.50</u>	65.50	$N = 200$ 60.50
Latency ↓	6.39	13.17	84.90	12.84	6.28	30.12	117.04	183.58	19.81
ECE ↓	0.331	0.265	0.195	0.244	0.208	0.364	0.183	0.136	0.137
Brier _{raw} ↓	0.731	0.611	0.519	0.609	0.566	0.759	0.520	0.497	0.481

Continued on next page

Table 36 continued; model band 3/3

N: non-thinking; T: thinking. Parentheses specify reasoning effort.

Metric	Qwen3.5-4B N	Qwen3.8- 27B N	DeepSeek V4-Flash N	GPT-6-Sol N	GPT-6- Luna N	Qwen3.5-4B T	Qwen3.8- 27B T (xhigh)	DeepSeek V4-Flash T (high)	GLM-5.3- Flash T (high)
compred-politics									$N = 200$
ACC $\uparrow$	54.50	56.00	54.50	57.50	62.00	50.50	<u>58.50</u>	54.00	54.50
Latency $\downarrow$	6.62	12.66	82.21	12.36	6.02	36.32	125.42	183.14	18.02
ECE $\downarrow$	0.272	0.272	0.222	0.228	0.166	0.306	0.153	0.238	0.192
Brier _{raw} $\downarrow$	0.658	0.643	0.571	0.585	0.544	0.690	0.548	0.595	0.571
compred-science									$N = 200$
ACC $\uparrow$	53.00	56.00	53.00	54.50	56.00	<u>59.50</u>	57.50	55.00	62.50
Latency $\downarrow$	6.64	13.00	84.04	12.08	6.40	32.53	113.01	173.65	19.02
ECE $\downarrow$	0.284	0.288	0.261	0.302	0.269	0.235	0.199	0.245	0.142
Brier _{raw} $\downarrow$	0.700	0.668	0.643	0.682	0.652	0.618	0.554	0.620	0.518
Domain summary									$N = 1,800$
ACC $\uparrow$	50.94	58.39	58.78	<u>64.33</u>	65.17	56.44	62.56	62.22	62.89
Latency $\downarrow$	5.80	11.71	68.49	11.41	7.55	36.01	119.89	147.90	17.10
ECE $\downarrow$	0.287	0.237	0.216	0.205	0.171	0.238	0.141	0.187	0.110
Brier _{raw} $\downarrow$	0.740	0.609	0.594	0.541	0.518	0.627	0.506	0.542	<u>0.477</u>

D.10 Personalization and preferences

Table 37 Personalization and preferences: model band 1/3. ACC (%) $\uparrow$; benchmark-run latency (s), ECE, and Brier_{raw} $\downarrow$. Rankings span all bands.

N: non-thinking; T: thinking. Parentheses specify reasoning effort.

Metric	Jev	InnerJev-4B (ours)	InnerJev-27B (ours)	kev-0.8b	kev-4b	kev-9b	kev-27b	Intern- Decision 0.8B
prism-alignment								$N = 200$
ACC $\uparrow$	50.50	50.00	54.00	41.00	45.00	47.00	49.50	43.50
Latency $\downarrow$	0.84	5.88	14.00	19.46	63.08	105.86	62.30	7.82
ECE $\downarrow$	0.174	0.183	0.106	0.169	0.127	0.147	0.098	0.100
Brier _{raw} $\downarrow$	0.607	0.642	<u>0.562</u>	0.638	0.603	0.621	0.572	0.577
prism-alignment-score								$N_{\text{score}} = 200$
Latency $\downarrow$	0.45	5.85	13.59	19.33	61.97	105.54	61.33	7.72
MAE $\downarrow$	1.058	0.970	1.038	1.281	1.043	1.007	1.018	1.300
Valid score	200/200	200/200	200/200	200/200	200/200	200/200	200/200	200/200
Domain summary								$N = 200, N_{\text{score}} = 200$
ACC $\uparrow$	50.50	50.00	54.00	41.00	45.00	47.00	49.50	43.50
Latency $\downarrow$	0.65	5.87	13.80	19.39	62.52	105.70	61.82	7.77
ECE $\downarrow$	0.174	0.183	0.106	0.169	0.127	0.147	0.098	0.100
Brier _{raw} $\downarrow$	0.607	0.642	<u>0.562</u>	0.638	0.603	0.621	0.572	0.577
MAE $\downarrow$	1.058	0.970	1.038	1.281	1.043	1.007	1.018	1.300
Valid score	200/200	200/200	200/200	200/200	200/200	200/200	200/200	200/200

Table 38 Personalization and preferences: model band 2/3. ACC (%) $\uparrow$; benchmark-run latency (s), ECE, and Brier_{raw} $\downarrow$. Rankings span all bands.

N: non-thinking; T: thinking. Parentheses specify reasoning effort.

Metric	Intern- Decision 2B	Intern- Decision 4B	laya	openjev-4b-v5	nimble-9b	nanojev	lightjev	jevforge-0.8b
prism-alignment								
ACC $\uparrow$	44.00	50.50	2.50	43.50	51.00	46.00	47.50	$N = 200$ 51.00
Latency $\downarrow$	8.04	12.27	4.31	228.04	26.07	38.84	219.78	211.15
ECE $\downarrow$	0.158	0.144	0.472	0.068	0.199	<u>0.062</u>	0.033	0.083
Brier _{raw} $\downarrow$	0.632	0.611	0.455	0.573	0.618	0.577	0.568	0.587
prism-alignment-score								
Latency $\downarrow$	8.03	12.20	12.30	227.53	25.99	39.81	222.55	$N_{\text{score}} = 200$ 214.57
MAE $\downarrow$	1.139	1.048	1.087	1.040	<u>1.002</u>	1.252	1.376	1.339
Valid score	200/200	200/200	200/200	200/200	200/200	200/200	200/200	200/200
Domain summary								
ACC $\uparrow$	44.00	50.50	2.50	43.50	51.00	46.00	47.50	$N = 200, N_{\text{score}} = 200$ 51.00
Latency $\downarrow$	8.03	12.23	12.07	227.78	26.03	39.32	221.17	212.86
ECE $\downarrow$	0.158	0.144	0.472	0.068	0.199	<u>0.062</u>	0.033	0.083
Brier _{raw} $\downarrow$	0.632	0.611	0.455	0.573	0.618	0.577	0.568	0.587
MAE $\downarrow$	1.139	1.048	1.087	1.040	<u>1.002</u>	1.252	1.376	1.339
Valid score	200/200	200/200	200/200	200/200	200/200	200/200	200/200	200/200

Table 39 Personalization and preferences: model band 3/3. ACC (%) ↑; benchmark-run latency (s), ECE, and Brier_{raw} ↓. Rankings span all bands.

N: non-thinking; T: thinking. Parentheses specify reasoning effort.									
Metric	Qwen3.5-4B N	Qwen3.8- 27B N	DeepSeek V4-Flash N	GPT-6-Sol N	GPT-6- Luna N	Qwen3.5-4B T	Qwen3.8- 27B T (xhigh)	DeepSeek V4-Flash T (high)	GLM-5.3- Flash T (high)
prism-alignment $N = 200$									
ACC ↑	39.50	44.50	50.50	52.00	50.50	49.00	<u>52.50</u>	48.00	50.00
Latency ↓	<u>1.09</u>	4.15	5.25	9.41	26.85	37.44	126.47	127.13	16.12
ECE ↓	0.301	0.311	0.284	0.300	0.331	0.282	0.129	0.348	0.128
Brier _{raw} ↓	0.757	0.728	0.713	0.729	0.777	0.692	0.574	0.818	0.599
prism-alignment-score $N_{\text{score}} = 200$									
Latency ↓	<u>2.45</u>	7.37	5.80	11.59	9.86	33.85	129.81	127.46	18.84
MAE ↓	1.267	1.364	1.379	1.798	2.174	1.351	1.377	1.676	1.100
Valid score	187/200	198/200	192/200	196/200	200/200	200/200	200/200	200/200	200/200
Domain summary $N = 200, N_{\text{score}} = 200$									
ACC ↑	39.50	44.50	50.50	52.00	50.50	49.00	<u>52.50</u>	48.00	50.00
Latency ↓	<u>1.78</u>	5.75	5.52	10.49	18.34	35.64	128.14	127.29	17.48
ECE ↓	0.301	0.311	0.284	0.300	0.331	0.282	0.129	0.348	0.128
Brier _{raw} ↓	0.757	0.728	0.713	0.729	0.777	0.692	0.574	0.818	0.599
MAE ↓	1.267	1.364	1.379	1.798	2.174	1.351	1.377	1.676	1.100
Valid score	187/200	198/200	192/200	196/200	200/200	200/200	200/200	200/200	200/200

E More Analysis Details

E.1 Mathematical and Probability Task Prompts

Each row presents a complete task template for JEV (left) and LLMs (right). Braced placeholders are filled per item. Numerical options expand to all 101 integers; each probability map contains every candidate key. Field headings are presentation aids. JEV receives structured fields; LLMs receive the displayed system and user messages with a strict response JSON Schema. Area variants are alternatives within one task; posterior observations likewise select one of the two stated outcomes. Figure 13 shows one example of each task: the numerical examples identify the nearest-integer gold choice, while the prior and posterior examples show the gold choice together with the analytically determined probabilities of all candidate labels.

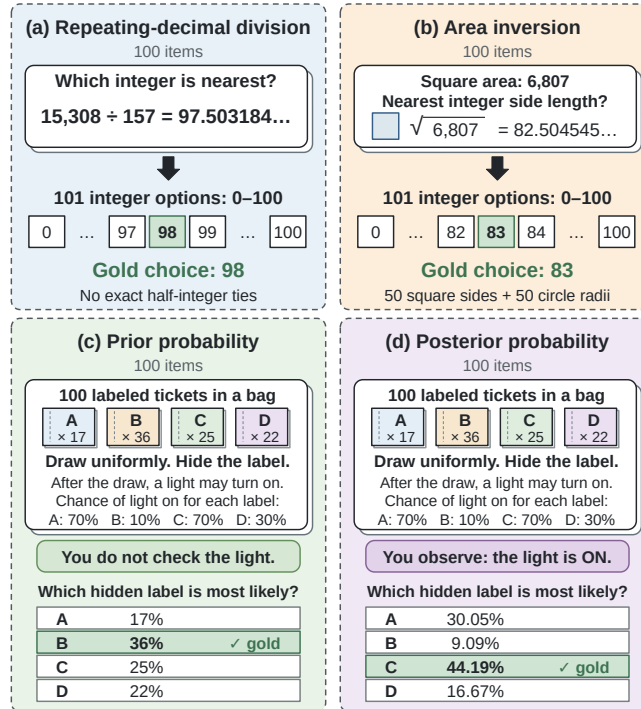


Figure 13 Examples of the four tasks, with gold choices in green and true probabilities shown. Numerical options are abbreviated.

1. Repeating-decimal division — JEV

Request fields

Question ID: q; type: choice.

State:

```
{"numerator": {numerator}, "denominator": {denominator}}
```

Task instruction:

Divide numerator by denominator. Select the integer from 0 through 100 nearest to the exact quotient. No case is exactly halfway between two integers.

Criteria:

n_0: 0

n_1: 1

...

n_100: 100

Response structure:

```
{"answers":{"q":{"type":"choice", "choice":{"selected_key"}, "confidence":{"confidence"}, "probabilities":{"probability_for_every_key}}}}
```

1. Repeating-decimal division — LLM

System message

Return only one JSON object conforming to the response JSON Schema.

Use the original question ID under "answers".

Include every field required for this question type.

For choice and score questions, include every option or level exactly once in "probabilities", including entries with value 0.

Output confidence and probability values as JSON numbers between 0 and 1.

For noul questions, "noul" is the probability that the answer is true.

Do not output Markdown, comments, explanations, or additional fields.

User message

State:

```
{"numerator": {numerator}, "denominator": {denominator}}
```

Task instruction:

Divide numerator by denominator. Select the integer from 0 through 100 nearest to the exact quotient. No case is exactly halfway between two integers.

Options:

- n_0: 0

- n_1: 1

...

- n_100: 100

Response structure:

```
{"answers":{"q":{"type":"choice", "choice":{"selected_key"}, "confidence":{"confidence"}, "probabilities":{"probability_for_every_key}}}}
```

2. Area inversion — JEV

Request fields

Question ID: q; type: choice.

State:

Square: {"shape":"square", "area_square_units":{"area}}

Circle: {"shape":"circle", "area_square_units":{"area"}, "pi":"3.1415926535897932384626433832795028841971693993751"}

Task instruction:

[Square]

A square has the area supplied in state. Its area equals side length squared. Select the integer from 0 through 100 nearest to its side length. No case is exactly halfway between two integers.

[Circle]

A circle has the area supplied in state. Its area equals pi times radius squared. Use the decimal pi supplied in state as the exact constant for this problem. Select the integer from 0 through 100 nearest to its radius. No case is exactly

halfway between two integers.

Criteria:

n_0: 0
n_1: 1
...
n_100: 100

Response structure:

```
{"answers":{"q":{"type":"choice", "choice":{"selected_key"},"confidence":{"confidence}, "probabilities":{"probability_for_every_key}}}}
```

2. Area inversion — LLM

System message

Return only one JSON object conforming to the response JSON Schema.
Use the original question ID under "answers".
Include every field required for this question type.
For choice and score questions, include every option or level exactly once in "probabilities", including entries with value 0.
Output confidence and probability values as JSON numbers between 0 and 1.
For noul questions, "noul" is the probability that the answer is true.
Do not output Markdown, comments, explanations, or additional fields.

User message

State:

Square: {"shape":"square", "area_square_units":{"area}}
Circle: {"shape":"circle", "area_square_units":{"area}, "pi":"3.1415926535897932384626433832795028841971693993751"}

Task instruction:

[Square]

A square has the area supplied in state. Its area equals side length squared. Select the integer from 0 through 100 nearest to its side length. No case is exactly halfway between two integers.

[Circle]

A circle has the area supplied in state. Its area equals pi times radius squared. Use the decimal pi supplied in state as the exact constant for this problem. Select the integer from 0 through 100 nearest to its radius. No case is exactly halfway between two integers.

Options:

- n_0: 0
- n_1: 1
...
- n_100: 100

Response structure:

```
{"answers":{"q":{"type":"choice", "choice":{"selected_key"},"confidence":{"confidence}, "probabilities":{"probability_for_every_key}}}}
```

3. Prior probability — JEV

Request fields

Question ID: q; type: choice.

State:

```
{"sampling":"A ticket is drawn uniformly at random from the bag. Its label is hidden.", "ticket_counts":{"A":{"n_A}, "B":{"n_B}, "C":{"n_C}, "D":{"n_D}}}, "clue_likelihood":{"P(X|label=A)":{"likelihood_A"}, "P(X|label=B)":{"likelihood_B"}, "P(X|label=C)":{"likelihood_C"}, "P(X|label=D)":{"likelihood_D"}}, "observation":{"observation}}}
```

Observation value:

The clue was not checked. No result was observed; whether X occurred is unknown. The ticket label was not revealed.

Task instruction:

Which label is on the drawn ticket?

For each option, report the probability that it is the drawn ticket's label, given the available information.

Criteria:

A: The hidden ticket has label A.
B: The hidden ticket has label B.
C: The hidden ticket has label C.
D: The hidden ticket has label D.

Response structure:

```
{"answers":{"q":{"type":"choice", "choice":{"selected_key"},"confidence":{"confidence"}, "probabilities":{"probability_for_every_key}}}}
```

3. Prior probability — LLM

System message

Return only one JSON object conforming to the response JSON Schema.
Use the original question ID under "answers".
Include every field required for this question type.
For choice and score questions, include every option or level exactly once in "probabilities", including entries with value 0.
Output confidence and probability values as JSON numbers between 0 and 1.
For noul questions, "noul" is the probability that the answer is true.
Do not output Markdown, comments, explanations, or additional fields.

User message

State:

```
{"sampling":"A ticket is drawn uniformly at random from the bag. Its label is hidden.", "ticket_counts":{"A":{"n_A"}, "B":{"n_B"}, "C":{"n_C"}, "D":{"n_D"}}, "clue_likelihood":{"P(X|label=A)":{"likelihood_A"}, "P(X|label=B)":{"likelihood_B"}, "P(X|label=C)":{"likelihood_C"}, "P(X|label=D)":{"likelihood_D"}}, "observation":{"observation"}}
```

Observation value:

The clue was not checked. No result was observed; whether X occurred is unknown. The ticket label was not revealed.

Task instruction:

A ticket label was randomly selected and is hidden. Given all information in state, estimate the conditional probability that the hidden ticket has each label A, B, C, or D. The four events are mutually exclusive and exhaustive. For each label k, probabilities[k] must mean $P(\text{hidden label} = k \mid \text{information provided in state})$, not the probability that k is the most likely label. Select the label with the largest estimated probability.

Options:

- A: The hidden ticket has label A.
- B: The hidden ticket has label B.
- C: The hidden ticket has label C.
- D: The hidden ticket has label D.

Response structure:

```
{"answers":{"q":{"type":"choice", "choice":{"selected_key"},"confidence":{"confidence"}, "probabilities":{"probability_for_every_key}}}}
```

4. Posterior probability — JEV

Request fields

Question ID: q; type: choice.

State:

```
{"sampling":"A ticket is drawn uniformly at random from the bag. Its label is hidden.", "ticket_counts":{"A":{"n_A"}, "B":{"n_B"}, "C":{"n_C"}, "D":{"n_D"}}, "clue_likelihood":{"P(X|label=A)":{"likelihood_A"}, "P(X|label=B)":{"likelihood_B"}, "P(X|label=C)":{"likelihood_C"}, "P(X|label=D)":{"likelihood_D"}}, "observation":{"observation"}}
```

Observation value (choose one):

The clue was checked. The observed result was X ($X=\text{true}$); the ticket label was not revealed.
The clue was checked. The observed result was not X ($X=\text{false}$); the ticket label was not revealed.

Task instruction:

Which label is on the drawn ticket?
For each option, report the probability that it is the drawn ticket's label, given the available information.

Criteria:

A: The hidden ticket has label A.

B: The hidden ticket has label B.
C: The hidden ticket has label C.
D: The hidden ticket has label D.

Response structure:

```
{"answers":{"q":{"type":"choice", "choice":{"selected_key"},"confidence":{"confidence"}, "probabilities":{"probability_for_every_key}}}}
```

4. Posterior probability — LLM

System message

Return only one JSON object conforming to the response JSON Schema.
Use the original question ID under "answers".
Include every field required for this question type.
For choice and score questions, include every option or level exactly once in "probabilities", including entries with value 0.
Output confidence and probability values as JSON numbers between 0 and 1.
For noul questions, "noul" is the probability that the answer is true.
Do not output Markdown, comments, explanations, or additional fields.

User message

State:

```
{"sampling":"A ticket is drawn uniformly at random from the bag. Its label is hidden.", "ticket_counts":{"A":{"n_A"}, "B":{"n_B"}, "C":{"n_C"}, "D":{"n_D"}}, "clue_likelihood":{"P(X|label=A)":{"likelihood_A"}, "P(X|label=B)":{"likelihood_B"}, "P(X|label=C)":{"likelihood_C"}, "P(X|label=D)":{"likelihood_D"}}, "observation":{"observation}}"
```

Observation value (choose one):

The clue was checked. The observed result was X (X=true); the ticket label was not revealed.
The clue was checked. The observed result was not X (X=false); the ticket label was not revealed.

Task instruction:

A ticket label was randomly selected and is hidden. Given all information in state, estimate the conditional probability that the hidden ticket has each label A, B, C, or D. The four events are mutually exclusive and exhaustive. For each label k, probabilities[k] must mean $P(\text{hidden label} = k \mid \text{information provided in state})$, not the probability that k is the most likely label. Select the label with the largest estimated probability.

Options:

- A: The hidden ticket has label A.
- B: The hidden ticket has label B.
- C: The hidden ticket has label C.
- D: The hidden ticket has label D.

Response structure:

```
{"answers":{"q":{"type":"choice", "choice":{"selected_key"},"confidence":{"confidence"}, "probabilities":{"probability_for_every_key}}}}
```

E.2 Agentic Workflow Evaluation

We evaluate the original τ -bench tool-calling agent and its JEV-based variant on the full Retail (115 tasks) and Airline (50 tasks) test sets. Each task is run once under both systems, yielding 330 interaction episodes in total. Both settings use the same version of τ -bench and DeepSeek-V4-Flash as both the base agent model and the user simulator, with temperature set to 0. Task configurations, available tools, user simulation, environment execution, and reward verification are kept identical across the two systems. Each episode is allowed up to 30 interaction turns.

To isolate the effect of JEV on agent decision making, we replace the action-selection component of the original τ -bench agent while leaving the remaining execution pipeline unchanged. In the original agent, the base LLM jointly determines the next action from the dialogue context and tool descriptions and generates the corresponding tool arguments or user-facing response. In the JEV variant, JEV first selects the next action from the currently available tools together with a ‘respond’ option. The same base LLM then generates arguments for the selected tool, or produces the response text when ‘respond’ is chosen. In both systems, the resulting action is executed by the same τ -bench environment, which then updates the interaction state. Figure 14 illustrates the two workflows.

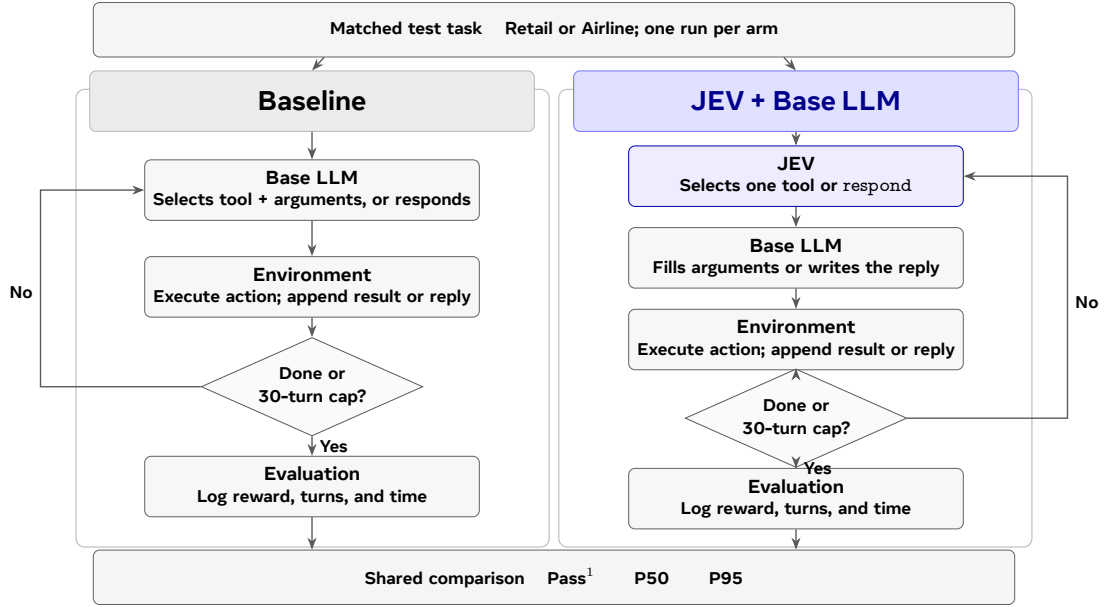


Figure 14 Matched τ -bench workflow for the baseline and JEV arms.

E.3 ElectionSim Results

Table 40 summarizes the simulation of Section 5.4.3. Following ElectionSim, States and Battleground count the correctly called states among all 51 and among the 15 battleground states. MAE, RMSE, and Bias measure the error of the predicted Democratic two-party share across states; the MAE equals the per-state RMSE that ElectionSim reports, averaged over states, and a positive bias overestimates Harris.

Table 40 Summary of the 2024 U.S. presidential election simulated in ElectionSim with voters sampled at a ratio of 1/1,000. Errors are in percentage points. Bold marks the better value in each column, with the bias closer to zero counted as better.

Voter model	States $\uparrow$	Battleground $\uparrow$	MAE $\downarrow$	RMSE $\downarrow$	Bias
ElectionSim (GPT-4o-mini)	46/51	11/15	3.06	3.80	+2.31
InnerJev-27B (decision model)	47/51	12/15	6.52	9.42	+3.73

Table 41 lists the per-state results behind Table 40 and Figure 8. Shares are Harris’s percentage of the two-party vote, and errors are predicted minus actual shares in percentage points. Voters counts the simulated voters of InnerJev-27B; the archived run of the original pipeline covers 331,836 voters with nearly the same per-state counts. Following ElectionSim, the 15 battleground states are Arizona, Colorado, Florida, Georgia, Iowa, Michigan, Minnesota, Nevada, New Hampshire, North Carolina, Ohio, Pennsylvania, Texas, Virginia, and Wisconsin.

Table 41 Per-state results of the 2024 ElectionSim simulation with a 1/1,000 voter sample. BG marks the 15 battleground states, and † marks a state called incorrectly.

State	BG	Voters	Actual	InnerJev-27B		ElectionSim, GPT-4o-mini	
				Share	Error	Share	Error
Alabama		5,031	34.5	32.6	−1.9	34.4	−0.2
Alaska		737	43.2	12.6	−30.5	38.0	−5.2
Arizona	•	7,159	47.2	41.4	−5.8	46.6	−0.6
Arkansas		3,014	34.3	35.5	+1.2	33.6	−0.7
California		39,577	60.6	66.2	+5.6	64.1	+3.4
Colorado	•	5,783	55.6	56.4	+0.8	58.0	+2.4
Connecticut		3,609	57.4	62.4	+5.0	61.6	+4.2
Delaware		991	57.5	71.7	+14.3	64.3	+6.8
District of Columbia		690	93.3	87.6	−5.7	92.9	−0.4
Florida	•	21,571	43.4	43.0	−0.3	46.1	+2.7
Georgia	•	10,726	48.9	45.5	−3.4	48.4	−0.5
Hawaii		1,461	61.8	71.7	+9.9	64.1	+2.3
Idaho		1,842	31.2	38.5	+7.2	36.0	+4.8
Illinois		12,823	55.4	59.1	+3.7	57.9	+2.5
Indiana		6,791	40.3	47.0	+6.7	44.3	+4.0
Iowa	•	3,193	43.3	47.7	+4.4	47.9	+4.6

Continued on next page

Table 41 continued

State	BG	Voters	Actual	InnerJev-27B		ElectionSim, GPT-4o-mini	
				Share	Error	Share	Error
Kansas		2,941	41.8	41.7	0.0	41.1	−0.6
Kentucky		4,510	34.5	42.8	+8.4	36.7	+2.2
Louisiana		4,662	38.8	38.4	−0.5	38.5	−0.3
Maine		1,364	53.6	55.3	+1.7	58.6	+5.0
Maryland		6,186	64.4	65.4	+1.0	64.9	+0.5
Massachusetts		7,034	62.6	74.7	+12.1	70.6	+8.0
Michigan	•	10,085	49.2	54.7 [†]	+5.5	53.4 [†]	+4.1
Minnesota	•	5,710	52.1	56.8	+4.6	54.5	+2.4
Mississippi		2,964	37.8	38.0	+0.1	40.0	+2.2
Missouri		6,161	40.7	47.2	+6.6	43.2	+2.6
Montana		1,086	39.7	43.7	+4.0	41.5	+1.7
Nebraska		1,964	39.6	34.3	−5.3	35.7	−3.9
Nevada	•	3,109	48.4	61.5 [†]	+13.1	53.3 [†]	+4.9
New Hampshire	•	1,380	51.4	68.6	+17.1	60.4	+9.0
New Jersey		9,295	53.0	60.6	+7.5	60.0	+6.9
New Mexico		2,121	53.1	53.4	+0.2	49.6 [†]	−3.6
New York		20,216	55.9	64.1	+8.2	61.9	+6.0
North Carolina	•	10,454	48.3	48.5	+0.2	49.0	+0.7
North Dakota		780	31.3	21.6	−9.6	30.1	−1.2
Ohio	•	11,809	44.3	43.4	−0.9	47.9	+3.6
Oklahoma		3,964	32.5	31.3	−1.2	33.9	+1.3
Oregon		4,242	57.4	67.5	+10.0	60.1	+2.7
Pennsylvania	•	13,012	49.1	53.4 [†]	+4.3	50.8 [†]	+1.7
Rhode Island		1,099	56.9	60.0	+3.0	58.7	+1.8
South Carolina		5,125	41.0	42.8	+1.9	43.5	+2.6
South Dakota		888	35.0	48.5	+13.4	35.0	0.0
Tennessee		6,917	34.9	36.5	+1.6	37.4	+2.4
Texas	•	29,184	43.0	40.2	−2.8	43.9	+0.9
Utah		3,276	38.9	37.4	−1.5	36.9	−2.0
Vermont		644	66.4	80.2	+13.8	68.9	+2.6
Virginia	•	8,655	52.9	61.5	+8.6	56.1	+3.2
Washington		7,716	59.5	68.2	+8.7	62.8	+3.2
West Virginia		1,796	28.6	41.7	+13.0	36.1	+7.5
Wisconsin	•	5,898	49.6	48.2	−1.4	51.0 [†]	+1.4
Wyoming		578	26.5	60.3 [†]	+33.9	34.6	+8.1

F Details of Reasoning-to-Readout Self-Distillation

This appendix gives the input template used by the first-token readout of Section 6, the data points of Figure 10, and the effect of the training set size on Reasoning-to-Readout Self-Distillation.

F.1 Readout Template

All open models in Section 6 share the plain-text template below; no few-shot examples are used. The readout is taken at the token that follows the final answer prefix, over the labels of the valid options only. When the teacher reasons, the same prompt is used with thinking enabled, and the answer prefix is appended after its reasoning segment. Noul questions present the two outcomes as options (A) and (B); score questions present every rating level as an option.

Readout template
State: {state} Question: {instruction} Options: (A) {option_1_key}: {option_1_description} (B) {option_2_key}: {option_2_description} ... The correct answer is (
Example: noul question
State: Order #1182. Status: shipped on 3 September. Carrier: DHL. Question: Has the order been shipped? Answer strictly from the facts stated in the text.

Options:
 (A) yes: The text states that this is so
 (B) no: The text states that this is not so
 The correct answer is (
 The readout returns the probability of (A) as $P(\text{true})$.

Example: score question

State:
 The icon is misaligned. Every function works.
 Question: Rate incident impact using only reported facts. Use the highest fully supported level.
 Options:
 (A) 0: No function impaired; cosmetic only
 (B) 1: One user or a nonessential function impaired, with a workaround
 (C) 2: Many users blocked from a core function, no data loss
 (D) 3: Confirmed irreversible data loss or physical harm
 The correct answer is (
 The readout returns the expected level $0 \cdot p(A) + 1 \cdot p(B) + 2 \cdot p(C) + 3 \cdot p(D)$.

F.2 Training Details

Table 42 lists the training settings of InnerJev-4B and InnerJev-27B. Both scales use the same settings, and the full-rank and LoRA versions differ only in which weights are updated and in the learning rate.

Table 42 Training settings for Reasoning-to-Readout Self-Distillation, shared by both backbones.

Setting	Full-rank (main)	LoRA
Updated weights	attention and MLP projections	LoRA adapters on the same modules
LoRA rank and α	—	64 and 128
Learning rate	2×10^{-6}	5×10^{-5}
Frozen weights	embeddings and output layer	
Optimizer and schedule	AdamW, 3% warmup, cosine decay	
Global batch size and epochs	64, one epoch	
Precision	BF16	
Teacher chains per question	2	
Loss weights	$\lambda = 0.5$; $\kappa = 0.05$, or 0.2 for hard-labeled <code>noul</code>	
Checkpoint selection	frozen development set	

F.3 Accuracy and Latency of Decision Models

This section describes how the data of Figure 10 are measured. Accuracy is the pooled JEVal accuracy of Section 4.2. Latency is the mean latency for inputs of up to 1K tokens, measured for every open model on a single H100 80 GB GPU; Jev’s value is that of its hosted API and includes network transport. On the same GPU, we also time our models with one request at a time over a sample of 232 JEVal questions, each timed three times, including tokenization, the forward pass, and the readout. The median latency is 58.7 ms for InnerJev-4B and 112.2 ms for InnerJev-27B. Intern-Decision-0.8B and Intern-Decision-2B were not part of the latency test and are therefore not plotted.

F.4 Effect of Training Set Size

We measure how development performance grows with the number of training questions under Reasoning-to-Readout Self-Distillation, where each backbone learns from its own post-reasoning distributions. The training sets from 5,000 to 29,326 questions are nested samples of one pool, drawn with a fixed seed and stratified by capability group, so every size keeps the same mixture, and Qwen3.5-4B is also trained on a smaller set of 2,500 questions. The pool cannot grow further with this mixture, so for Qwen3.8-27B we add questions from new sources: science from QASC and ScienceQA, quantitative and logical reasoning from MathQA, AQuA, LogiQA2, and ReClor, medical and legal practice, and factual and conditional boundaries from VitaminC, ConditionalQA, and ContractNLI. This gives a set of 39,279 questions and a larger set of 48,358 questions, which also contains science questions that Qwen3.8-27B wrote itself from paper abstracts and encyclopedia articles. Both sets keep all 29,326 earlier questions but are drawn independently of each other. As before, targets for the added questions average two reasoning chains of the model and never use gold labels, and questions whose chains are truncated or invalid or disagree on the top option

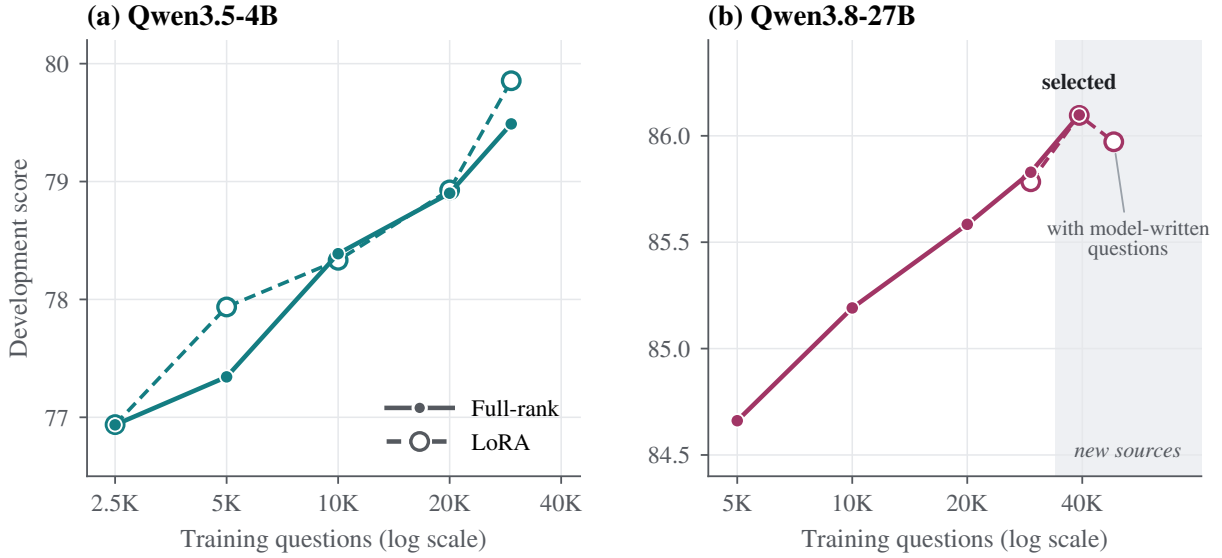


Figure 15 Development score against the number of training questions under Reasoning-to-Readout Self-Distillation. Filled markers with solid lines show full-rank training, and open markers with dashed lines show LoRA. Points for Qwen3.5-4B are means over two seeds. For Qwen3.8-27B, sizes up to 20,000 questions use one seed and larger sizes average two or three seeds; the shaded region marks sets that add new sources, and the largest set also contains questions written by the model itself.

are left out. Table 43 gives the composition of every set. Every run selects its checkpoint on a frozen development set of 7,506 questions, whose score weights nine capability groups equally and sources equally within each group, using accuracy for classification questions and one minus the normalized MAE for soft `noul` and `score` questions.

Table 43 Number of training questions per capability group. The sets of 5,000 to 29,326 questions are nested and keep the same mixture. The sets of 39,279 and 48,358 questions keep all 29,326 questions, add new sources, and are drawn independently of each other; InnerJev-27B is trained on the set of 39,279 questions.

Capability group	5,000	10,000	20,000	29,326	39,279	48,358
Language understanding	1,199	2,398	4,795	7,031	7,031	7,031
General knowledge	938	1,875	3,751	5,500	5,500	5,500
Commonsense	693	1,387	2,774	4,068	4,068	4,068
Logic	520	1,040	2,081	3,051	3,051	3,051
Medicine, English	520	1,040	2,080	3,050	3,050	3,050
Medicine, Chinese	498	996	1,992	2,921	2,921	2,921
Decisions, rules, and computation	473	946	1,891	2,773	2,773	2,773
Safety and subjective judgment	159	318	636	932	932	932
<i>New sources</i>						
Science	—	—	—	—	3,279	6,014
of which written by the model	—	—	—	—	—	2,731
Quantitative and logical reasoning	—	—	—	—	3,000	6,925
Medical and legal practice	—	—	—	—	2,500	4,384
Factual and conditional boundaries	—	—	—	—	1,174	1,709
Total	5,000	10,000	20,000	29,326	39,279	48,358

Figure 15 shows that development performance rises steadily with the amount of training data at both scales. The growth is roughly linear in the logarithm of the training set size, so each doubling of the data buys a similar gain, and the gain per doubling is smaller for Qwen3.8-27B, whose scores are already higher. Full-rank and LoRA training follow nearly the same curve. For Qwen3.8-27B, the new sources carry the trend to 39,279 questions, but the larger set brings no further gain, so we train InnerJev-27B on 39,279 questions.

We attribute the stall mainly to the quality of the added questions rather than to their number. Reasoning-to-Readout Self-Distillation uses neither a stronger model nor gold labels, and this limits how new training questions can be made. First, questions that a model writes itself tend to fall within its comfort zone. The science questions that Qwen3.8-27B wrote turned out to be easy for it: it answered them correctly without thinking, so its teacher and student already agree and the questions carry little training signal. A useful question must be hard enough that the student fails without reasoning. Second, the model cannot reliably check whether such a hard question is itself correct, and a flawed question turns directly into a wrong target. In earlier experiments, questions generated

from programmatic templates failed in yet another way: the student learned the template rather than the skill, and the gain did not transfer to real questions. Scaling Reasoning-to-Readout Self-Distillation further therefore depends on finding hard, well-posed questions rather than on adding more questions.

These curves describe a trend rather than a law. The development set also selects checkpoints, so absolute scores are optimistic. The smaller 27B sizes use a single seed, the expanded sets also change the mixture, and the runs on 48,358 questions stopped at a preset time limit about halfway through their epoch, so they show only that the larger set brings no early gain. The curves therefore support the overall trend but not the gain between any two neighboring sizes, and they do not support fitting a scaling law.